

CAS3230 : Introduction to Mathematics for Deeplearning

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Introduction

0.1 Why Math?

How to use and apply Computer Science or Artificial Intelligence is not enough.

You need to truly understand, ...and it often requires math

Applications:

- Probability, Linear Algebra, and other Mathematical Tools : Rigorous Problem Definition
- Supervised Learning : Image Classification, Next Word Prediction, Translation, Price Prediction

0.2 Supervised Learning

1. Supervised Learning

- Data : $x \in \mathcal{X}$ usually a vector
- Label : $y \in \mathcal{Y}$
- Dataset : $(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(n)}, y^{(n)})$

2. Supervised Learning: Setup

- True function : f^* where $f^*(x^{(i)}) = y^{(i)}$
- Want to find : $g(x) \approx f^*(x)$
- Function class : $\mathcal{G}$
- Goal : $g_\theta(x) \approx f^*(x), \quad g_\theta \in \mathcal{G}$

Find g_θ from function class that approximates the true function **well**

3. Deep Learning

- True function : f^* where $f^*(x^{(i)}) = y^{(i)}$
- Want to find : $g(x) \approx f^*(x)$
- Function class : **Neural Networks**
- Goal : $g_\theta \approx f^*(x), \quad g_\theta \in \mathcal{G}$

Find g_θ from function class that approximates the true function **well**

0.3 Good Generative Model

We are Generating Vectors (Generative Model = Learning Distribution)

Diffusion Models

- Forward SDE (data $\rightarrow$ noise)

$$\mathbf{x}(0) \xrightarrow{\quad} d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w} \rightarrow \mathbf{x}(T)$$

- Reverse SDE (noise $\rightarrow$ data)

$$\mathbf{x}(0) \leftarrow [\mathbf{f}(\mathbf{x}, t) - g^2(t)\nabla_{\mathbf{x}} \log p_t(\mathbf{x})]dt + g(t)d\bar{\mathbf{w}} \xrightarrow{\quad} \mathbf{x}(T)$$

0.4 Optimisation

Learning, Training $\Rightarrow$ Deep Learning : $g(x) \approx f^*(x)$

1 Review on Probability Theory

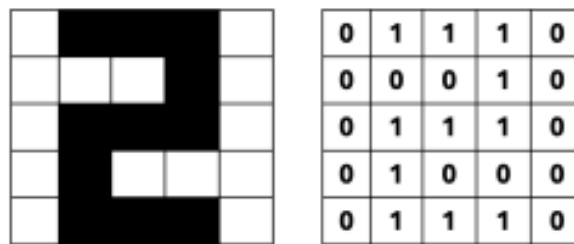
1.1 Probability

- Ω : Sample Space
- E : Event $\subseteq \Omega$ which is a measurable subset
 - **Measurable?** : Satisfying a certain criteria
- $\mathcal{F}$: Set of events $\Rightarrow |\mathcal{F}| = 2^{\# \text{ of events}}$
- P : Probability Measure
 - $P : \mathcal{F} \mapsto [0, 1] = \{x : 0 \leq x \leq 1\}$

Example 01. Fair Coin

$$\Omega = \{H, T\}, \quad \mathcal{F} = \{\{H\}, \{T\}, \{H, T\}, \emptyset\}$$

$$P(\emptyset) = 0, \quad P(\{T\}) = \frac{1}{2}, \quad P(\{H\}) = \frac{1}{2}, \quad P(\{H, T\}) = 1$$

Example 02. 5×5 Binary ImageFigure 1: 5×5 Binary Image

$$\Omega = \{0, 1\}^{5 \times 5} \cong \{0, 1\}^{25}$$

$$|\Omega| = 2^{25}, \quad |\mathcal{F}| = 2^{2^{25}}$$

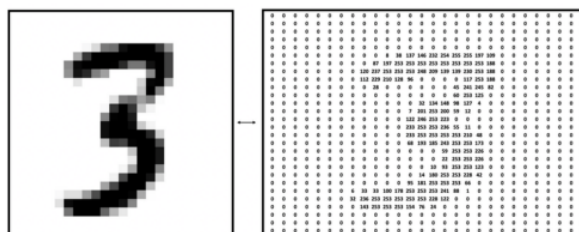
Example 03. MNIST Dataset

Figure 2: MNIST Dataset

$$\Omega = \{0, 1, \dots, 255\}^{28 \times 28}$$

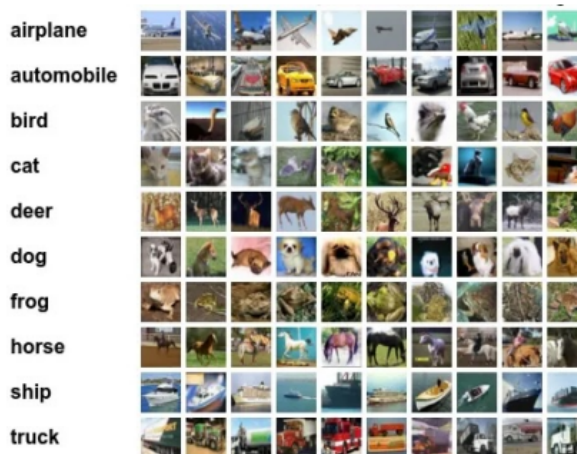
Example 04. CIFAR-10 Dataset

Figure 3: CIFAR-10 Dataset

CIFAR-10 image is 32×32 RGB colour image where the intensity of each pixel ranges from 0 to 255.

$$\Omega = \{0, \dots, 255\}^{32 \times 32 \times 3}$$

Here, $32 \times 32 \times 3$ stands for RGB channels of size 32×32 each.

Example 05. ImageNet Dataset

ImageNet image is 224×224 RGB colour image where the intensity of each pixel ranges from 0 to 255.

$$\Omega = \{0, \dots, 255\}^{224 \times 224 \times 3}$$

1.2 Random Variable

Map from sample space to real number, in other words, converting a sample to a number.

$$X : \Omega \rightarrow \mathbb{R}$$

Example 06. 5×5 Binary image

In case of $\Omega = \{0, 1\}^{5 \times 5}$, a random variable X can be number of 1's.

$$X : \Omega \rightarrow \mathbb{R}$$

$$\begin{pmatrix} 0 & 1 & 0 & \dots \\ \vdots & \ddots & & \vdots \\ \vdots & \dots & \dots & \vdots \end{pmatrix} \mapsto \frac{1}{25}(X_{11} + \dots + X_{55})$$

Example 07. 7-tosses of a fair coin

X : # of heads

$$\Omega = \{(H_i T_j) \mid \text{where } 0 \leq i \leq 7, 0 \leq j \leq 7, (i+j) = 7, (i, j) \in \mathbb{Z}\}$$

$$X(HHHTTHT) = 4$$

***Discrete Random Variable** : where the sample space is finite (*i.e.*, $|\Omega| \leq \infty$) and countable

1.3 Probability Mass Function

$$p_X(x) = \Pr(X = x)$$

Where X denotes a random variable and x denotes a specific value (realisation)

Example 08. Rolling a dice

X : # of outcome

$$p_X(4) = \Pr(X = 4)$$

$$\Pr(X = 1) + \dots + \Pr(X = 6) = 1$$

Note that $\sum_{x \in \mathcal{X}} p_X(x) = 1$ where $\mathcal{X}$ denotes all possible values of X

1.4 Expectation

$$\mathbb{E}[X] = \sum_{x \in \mathcal{X}} x \cdot p_X(x)$$

Example 09. Rolling a dice

X : # of outcome

$$\mathbb{E}[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + \dots + 6 \cdot \frac{1}{6}$$

Lemma

For $f : \mathcal{X} \rightarrow \mathbb{R}$

W.T.S is below

$$\mathbb{E}[f(X)] = \sum_{x \in \mathcal{X}} f(x) \cdot p_X(x)$$

Hint

$$\begin{aligned} \mathbb{E}[f(X)] &= \mathbb{E}[Y] = \sum_{y \in \mathcal{Y}} y \cdot p_Y(y) \\ Y &= f(X), \quad \underbrace{\Omega \xrightarrow{X} \mathbb{R} \xrightarrow{f} \mathbb{R}}_Y \end{aligned}$$

Proof

Let Define $Y = f(X)$, and its p.m.f is as follows:

$$p_Y(y) = \sum_{x \in \mathcal{X}: f(x)=y} p_X(x)$$

Then,

$$\begin{aligned} \mathbb{E}[f(X)] &= \mathbb{E}[Y] = \sum_y y \cdot p_Y(y) \\ &= \sum_y \sum_{x: f(x)=y} y \cdot p_X(x) = \sum_y \sum_{x: f(x)=y} f(x) \cdot p_X(x) = \sum_{x \in \mathcal{X}} f(x) \cdot p_X(x) \\ \therefore \mathbb{E}[f(X)] &= \sum_{x \in \mathcal{X}} f(x) \cdot p_X(x) \end{aligned}$$

For a continuous random variable, the proof follows similarly by using integrals, so it's sufficient to demonstrate the discrete case.

2 Information Theory

2.1 Entropy

Let $\mathcal{X} = \{0, 1, \dots, M-1\}$, and X be a discrete random variable with p.m.f $p_X(x)$.

The "Surprisal" $S(x)$ is defined as:

$$S(x) = \log \frac{1}{p_X(x)}$$

Also, consider the following $H(X)$:

$$H(X) = \mathbb{E}[S(X)] = \sum_{x=0}^{M-1} p_X(x) \log \frac{1}{p_X(x)}$$

Here, the function $H(X)$ is denoted as the "Entropy".

Example 01. Earthquake occurrence

$$X = \begin{cases} 0 & \text{O.W.} \\ 1 & \text{earthquake occur} \end{cases}$$

Of course, $p_X(1) < 1$, but assume $p_X(1) = 0.0001$

$$\Rightarrow H(X) = 0.0001 \log \frac{1}{0.0001} + 0.9999 \log \frac{1}{0.9999} \approx 0 \quad (\text{Since } \lim_{\alpha \rightarrow 0} \alpha \log \frac{1}{\alpha} = 0)$$

Example 02. "Fair-Coin" Toss

$$X = \begin{cases} 1 & H \\ 0 & T \end{cases}$$

$$H(X) = 0.5 \log \frac{1}{0.5} + 0.5 \log \frac{1}{0.5} = 0.5 \cdot 1 + 0.5 \cdot 1 = 1$$

***Note** that $\log$ represented in this course denotes $\log_2$ (it's just a convention... don't take it seriously). Also, keep in mind that the following expressions represent the equivalent meaning.

$$\text{"Randomness"} = \text{"Unpredictability"} = \text{"More Informative"} = \text{"Uncertainty"}$$

Example 03. Consider 2 coins

$$\text{Coin 1} = \begin{cases} 0.6 & H \\ 0.4 & T \end{cases} \quad \text{Coin 2} = \begin{cases} 0.3 & H \\ 0.7 & T \end{cases}$$

$$H(\text{Coin 1}) = 0.6 \log \frac{1}{0.6} + 0.4 \log \frac{1}{0.4}$$

$$H(\text{Coin 2}) = 0.3 \log \frac{1}{0.3} + 0.7 \log \frac{1}{0.7}$$

$$\therefore H(\text{Coin 1}) > H(\text{Coin 2})$$

Example 04. Guessing Game

Let Alice sets a single number from $0, 1, 2, \dots, 7$ and tells Bob to guess by asking some questions.

Let X as the number Alice has selected. Bob's question is as follows:

$$\text{Question 1 : } X \in \{0, 2, 4, 6\} \rightarrow (\text{Success, Fail}) = \left(\frac{1}{2}, \frac{1}{2}\right)$$

$$\text{Question 2 : } X \in \{0, 4\} \rightarrow (\text{Success, Fail}) = \left(\frac{1}{4}, \frac{3}{4}\right)$$

Considering the fact, Question 1 is better than Question 2.

Also, suppose Bob has an another task to solve which either of the dice is less likely to predict.

The data below is the probability the dice has to have the corresponding value, respectively.

$$\text{dice 1 : } \frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{2}{9}, \frac{2}{9}, \frac{2}{9} \Rightarrow H(\text{dice 1}) = \frac{1}{9} \log_2 9 + \dots$$

$$\text{dice 2 : } \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{32} \Rightarrow H(\text{dice 2}) = \frac{1}{2} \log_2 2 + \dots$$

Since, $H(\text{dice 1}) > H(\text{dice 2})$, dice 1 is less likely to predict than dice 2.

***Note** that the following expressions in the examples represents the equivalent meaning.

“Better Question” = More “Informative” Question = Random Variable with “Higher Entropy”

Case Study 01. Image Classification

Suppose there exist a classifier f , which has a task to classify the image from either of the following categories dog/cat/horse.

The classifiers have two different strategies.

- Hard labeling/prediction
Set the classifier as $f(\square) = \text{dog}$, which is quite deterministic
- Soft labeling/prediction
Let there exist two images Image#1 and Image#2, assume soft labeling/prediction is applied.

$$f(\boxed{1}) = \begin{pmatrix} \text{dog} & 70\% \\ \text{cat} & 20\% \\ \text{horse} & 10\% \end{pmatrix}$$

$$f(\boxed{2}) = \begin{pmatrix} \text{dog} & 40\% \\ \text{cat} & 30\% \\ \text{horse} & 30\% \end{pmatrix}$$

Since $H(f(\boxed{1})) < H(f(\boxed{2}))$, Image#2 is harder(= challenging) to classify(= uncertain = random).

Case Study 02. Reinforcement Learning

Suppose you are selecting either one of the stores A, B, C to have lunch during your lunch break.

The data below shows

(# of satisfactory meals)/(# of trials) of the corresponding stores respectively:

$$\begin{array}{ccc} \text{A} & \text{B} & \text{C} \\ 3/5 & 1/2 & 5/10 \end{array}$$

From the naive perspective, selecting store A might be the wisest option out of the three.

However, there are some loopholes you did not consider.

You might consider option B and option C may look as equivalent situations, but it is more likely to occur different consequences.

Note that though they have the equivalent value, the number of trials of B is far less than that of C which can be interpreted as an “*undereexplored*” situation.

This leads to the concepts of “*Exploration*” and “*Exploitation*” when selecting an action in Reinforcement Learning.

“*Exploration*” is the process that selects an action which is *undereexplored* (in this case, selecting option B), whereas “*Exploitation*” is the process that selects the most conservative action which is likely to have the highest probability to success considering the background information (in this case, selecting option A).

You decided to select either one of the strategies which represents the probability to select a certain store, respectively:

strategy 1 : 0.4 0.4 0.2

strategy 2 : 0.6 0.2 0.2

Since $H(\text{strategy 1}) > H(\text{strategy 2})$, strategy 1 is more likely to select as an *exploration* option than strategy 2, this represents a more “*explorative*” strategy, and equivalently a more “*random*” strategy.

2.2 Properties Related to Entropy

1. Non-Negativity

Since $0 \leq p_X(x) \leq 1 \Rightarrow 1 \leq \frac{1}{p_X(x)}$:

This makes $0 \leq \log \frac{1}{p_X(x)}$ and $\sum_{x \in \mathcal{X}} p_X(x) \log \frac{1}{p_X(x)} \geq 0$.

Therefore, $\mathbb{E}[\log \frac{1}{p_X(x)}] = H(X) \geq 0$, which suffices the non-negativity of the entropy.

$H(X) = 0$, if $p_X(x) = 0$ or $p_X(x) = 1$ which makes X deterministic.

2. Convexity

- $f : \mathcal{X} \rightarrow \mathbb{R}$ is convex if:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) \quad \forall x, y \in \mathcal{X}, \quad \lambda \in [0, 1]$$

- $f : \mathcal{X} \rightarrow \mathbb{R}$ is concave if:

$$f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y) \quad \forall x, y \in \mathcal{X}, \quad \lambda \in [0, 1]$$

Lemma

Let X be a Bernoulli random variable with parameter p .

$$X = \begin{cases} 0 & \text{with probability } (1 - p) \\ 1 & \text{with probability } (p) \end{cases}$$

W.T.S is below

$$H(X) = p \log \frac{1}{p} + (1 - p) \log \frac{1}{(1 - p)} \text{ is a concave function.}$$

Proof

Let Define $h_2(p) = H(X) = -p \log p - (1-p) \log(1-p)$, and its derivative is as follows:
 We have $h_2'(p) = \log \frac{(1-p)}{p}$ and $h_2''(p) = -\frac{1}{p} - \frac{1}{(1-p)} < 0$ which implies $h_2'(p)$ is decreasing.
 For $0 \leq x < y \leq 1$, by the **Mean Value Theorem**, there exists $x \leq z \leq y$ such that:

$$\frac{h_2(x) - h_2(y)}{x - y} = h_2'(z)$$

Then, for $0 < p_1 < p_2 \leq 1$ and $0 \leq \lambda \leq 1$, we have:

$$\begin{aligned} & \lambda h_2(p_1) + (1-\lambda)h_2(p_2) - h_2(\lambda p_1 + (1-\lambda)p_2) \\ &= \lambda(h_2(p_1) - h_2(\lambda p_1 + (1-\lambda)p_2)) + (1-\lambda)(h_2(p_2) - h_2(\lambda p_1 + (1-\lambda)p_2)) \\ &= \lambda(p_1 - (\lambda p_1 + (1-\lambda)p_2))h_2'(q_1) + (1-\lambda)(p_2 - (\lambda p_1 + (1-\lambda)p_2))h_2'(q_2) \end{aligned}$$

for some q_1, q_2 such that $p_1 \leq q_1 \leq \lambda p_1 + (1-\lambda)p_2 \leq q_2 \leq p_2$.

Thus we have:

$$\begin{aligned} & \lambda h_2(p_1) + (1-\lambda)h_2(p_2) - h_2(\lambda p_1 + (1-\lambda)p_2) \\ &= \lambda(1-\lambda)(p_1 - p_2)h_2'(q_1) - \lambda(1-\lambda)(p_1 - p_2)h_2'(q_2) \\ &= \lambda(1-\lambda)(p_1 - p_2)(h_2'(q_1) - h_2'(q_2)) \leq 0 \\ \therefore H(X) &= p \log \frac{1}{p} + (1-p) \log \frac{1}{(1-p)} \text{ is a concave function.} \end{aligned}$$

3. Jensen's Inequality

If f is concave:

$$\mathbb{E}[f(X)] \leq f(\mathbb{E}[X])$$

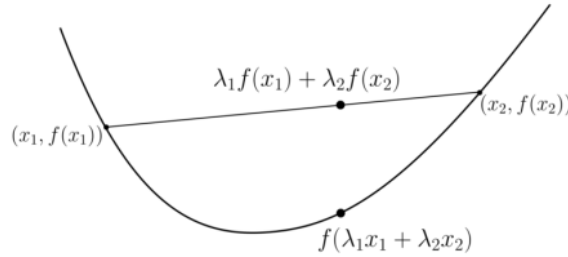


Figure 4: Jensen's Inequality

That is:

$$\sum_{x \in \mathcal{X}} p_X(x) \cdot f(x) \leq f\left(\sum_{x \in \mathcal{X}} p_X(x) \cdot x\right)$$

Jensen's Inequality is a generalised version of the definition of concavity, where $\mathbb{E}[f(X)]$ is weighted average of f and $f(\mathbb{E}[X])$ is a function value at weighted average input.

For strictly concave function f , Jensen's inequality holds with equality if and only if $f(x)$ is constant.

4. Maximum Entropy

For a random variable $X \in \mathcal{X} = \{0, 1, 2, \dots, M-1\}$ we have:

$$H(X) \leq \log_2 |\mathcal{X}| = \log_2 M$$

Proof

Since log is a concave function, by applying Jensen's inequality:

$$\mathbb{E}\left[\log \frac{1}{p_X(x)}\right] \leq \log \mathbb{E}\left[\frac{1}{p_X(x)}\right] = \log \sum_{x \in \mathcal{X}} \left(p_X(x) \cdot \frac{1}{p_X(x)}\right) = \log M$$

Considering this, the entropy is maximised when $p_X(0) = \dots = p_X(M-1) = \frac{1}{M}$.

5. Mismatch

For “wrong” p.m.f $Q(x)$:

$$H(X) = \mathbb{E}_{p_X} \left[\log \frac{1}{p_X(x)} \right] = \sum_x p_X(x) \cdot \log \left(\frac{1}{p_X(x)} \right) \leq \mathbb{E}_{p_X} \left[\log \frac{1}{Q(X)} \right] = \sum_x p_X(x) \cdot \log \left(\frac{1}{Q(x)} \right)$$

Proof

Since log is a concave function, by applying Jensen’s inequality:

$$\begin{aligned} H(X) - \mathbb{E}_{p_X} \left[\log \frac{1}{Q(X)} \right] \\ = \mathbb{E}_{p_X} \left[\log \frac{Q(X)}{p_X(X)} \right] \leq \log \left(\mathbb{E} \left[\frac{Q(X)}{p_X(X)} \right] \right) = \log \left(\sum_x p_X(x) \cdot \frac{Q(x)}{p_X(x)} \right) = \log \left(\sum_x Q(x) \right) = \log 1 = 0 \end{aligned}$$

The equality holds if and only if $\frac{Q(x)}{p_X(x)}$ is constant, which implies $Q = p_X$.

Lemma

The proof presented above is incomplete due to an edge case where $p_X(x) = 0$ for some x . Provide a rigorous proof that the inequality holds even in the edge case.

W.T.S is below

$$\mathbb{E}_{p_X} \left[\log \frac{1}{p_X(X)} \right] \leq \mathbb{E}_{p_X} \left[\log \frac{1}{Q(X)} \right]$$

Rigorous Proof

$$\begin{aligned} H(X) - \mathbb{E}_{p_X} \left[\log \frac{1}{Q(X)} \right] &= \sum_x P_X(x) \cdot \log \left(\frac{Q(x)}{P_X(x)} \right) = \sum_{x: P_X(x) > 0} P_X(x) \cdot \log \left(\frac{Q(x)}{P_X(x)} \right) \\ &\leq \log \left(\sum_{x: P_X(x) > 0} P_X(x) \cdot \frac{Q(x)}{P_X(x)} \right) = \log \left(\sum_{x: P_X(x) > 0} Q(x) \right) \leq \log 1 = 0 \end{aligned}$$

To achieve equality, both the first (Jensen) and the second ($\sum_{x: P_X(x) > 0} Q(x) < 1$)

inequality should hold with equality.

In other words:

- 1) $\frac{Q(x)}{P_X(x)} = c$ is constant for all x and
- 2) $Q(x) = 0$ if $P_X(x) = 0$.

This can only be true if $Q = P_X$.

3 Kullback-Leibler (KL) Divergence

3.1 Relative Entropy

For two distributions P and Q on $\mathcal{X}$,

The KL divergence (or relative entropy) is defined by:

$$D(P||Q) = \sum_{x \in \mathcal{X}} \left(P(x) \cdot \log \left(\frac{P(x)}{Q(x)} \right) \right)$$

Due to the previous proof, we have:

$$\mathbb{E}_{P_X} \left[\log \frac{P_X(X)}{Q(X)} \right] = \sum_x P_X(x) \cdot \log \frac{P_X(x)}{Q(x)} \geq 0$$

This implies that:

$$D(P_X||Q) \geq 0$$

In other words, the KL divergence is non-negative and 0 if and only if $P_X = Q$.

Example 01. KL divergence vs Total Variation

Consider a task rolling two dice.

The data below is the probability the dice has to have the corresponding value, respectively.

$$\text{dice 1} : \left(\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6} \right)$$

$$\text{dice 2} : \left(\frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{2}{9}, \frac{2}{9}, \frac{2}{9} \right)$$

The KL divergence($D(\text{dice 1}||\text{dice 2})$) and Total Variation($TV(\text{dice 1}, \text{dice 2})$) is as follows:

$$D(\text{dice 1}||\text{dice 2}) = \frac{1}{6} \log \left(\frac{\frac{1}{6}}{\frac{1}{9}} \right) \times 3 + \frac{1}{6} \log \left(\frac{\frac{1}{6}}{\frac{2}{9}} \right) \times 3$$

$$TV(\text{dice 1}, \text{dice 2}) = \left| \frac{1}{6} - \frac{1}{9} \right| \times 3 + \left| \frac{1}{6} - \frac{2}{9} \right| \times 3$$

Here, the Total Variation($TV(\text{dice 1}, \text{dice 2})$) is equivalent to the ℓ^1 -distance.

3.2 Cross Entropy

Recall the previous inequality represented in 2.2.5:

$$H(X) = \mathbb{E}_{P_X} \left[\log \frac{1}{P_X(X)} \right] \leq \mathbb{E}_{P_X} \left[\log \frac{1}{Q(X)} \right] = \sum_{x \in \mathcal{X}} P_X(x) \log \left(\frac{1}{Q(x)} \right)$$

Here, the *right-hand side* $\sum_{x \in \mathcal{X}} P_X(x) \log \left(\frac{1}{Q(x)} \right)$ is denoted as “Cross Entropy” where we happen to use the wrong p.m.f $Q(x)$ instead of the true p.m.f $P_X(x)$.

Example 02. Gambling

Consider a game where you can bet a trial on flipping a coin.

The data below is the probability and the odds to have the corresponding value, respectively.

Coin	H	T	
Probability	p	$(1-p)$	(where, $p > \frac{1}{2}$)
Odds	2	2	
Bet	qB	$(1-q)B$	

Here, B denotes the entire budget you have (let's say about \$100.00)

Let the random variable S be your budget after the bet, when you bet qB on ‘ H ’ and $(1-q)B$ on ‘ T ’.

Then we have:

$$\mathbb{E}[S] = p \cdot 2(qB) + (1-p) \cdot 2(1-q)B = q \cdot (2pB) + (1-q) \cdot (2(1-p)B)$$

Here, the budget will be maximised when $q^* = 1$, which is the optimal strategy.

However, if the trial is repeated many times, there will be high chance of the occurrence of ‘ T ’.

(i.e., losing all your money).

One way to prevent this is to maximise $\mathbb{E}[\log S]$ instead since $\log 0 = -\infty$.

This will surely prevent you from losing all your budget:

$$\begin{aligned} \mathbb{E}[\log S] &= p \cdot \log(2qB) + (1-p) \cdot \log(2(1-q)B) \\ &= p \log(2B) + (1-p) \log(2B) - \left(p \log \left(\frac{1}{q} \right) + (1-p) \log \left(\frac{1}{(1-q)} \right) \right) \\ &\leq p \log(2B) + (1-p) \log(2B) - \left(p \log \left(\frac{1}{p} \right) + (1-p) \log \left(\frac{1}{(1-p)} \right) \right) \end{aligned}$$

The following inequality can be derived based on the inequality above:

$$\left(p \log \left(\frac{1}{q}\right) + (1-p) \log \left(\frac{1}{(1-q)}\right)\right) \geq \left(p \log \left(\frac{1}{p}\right) + (1-p) \log \left(\frac{1}{(1-p)}\right)\right)$$

This shows that **Cross Entropy** $\geq$ **Entropy**.

Considering this, the optimal strategy where $\mathbb{E}[\log S]$ is maximised, is when $q^* = p$.

Example 03. Cost of mismatch

From the previous example, the ‘**cost of mismatch**’ is denoted as the difference between your strategy $(q, 1-q)$ and the optimal strategy $(p, 1-p)$:

$$\begin{aligned} & \left(p \log(2B) + (1-p) \log(2B) - \left(p \log \frac{1}{q} + (1-p) \log \left(\frac{1}{(1-q)}\right)\right)\right) \\ & - \left(p \log(2B) + (1-p) \log(2B) - \left(p \log \frac{1}{p} + (1-p) \log \left(\frac{1}{(1-p)}\right)\right)\right) \\ & = p \log \frac{q}{p} + (1-p) \log \left(\frac{(1-q)}{(1-p)}\right) \end{aligned}$$

Here, the term can be interpreted as the KL Divergence (*i.e.*, $D((p, 1-p) || (q, 1-q))$)

Also, consider the following equation again.

$$\mathbb{E}[\log S] = \log(2B) - p \log \frac{1}{q} - (1-p) \log \left(\frac{1}{(1-q)}\right) = \log B + (\log 2 - p \log \frac{1}{q} - (1-p) \log \left(\frac{1}{(1-q)}\right))$$

Let the term, $W = \log 2 - p \log \frac{1}{q} - (1-p) \log \left(\frac{1}{(1-q)}\right)$

Then, $\log S = \log B + W$ and this derives the following equation:

$$S = B \times 2^{nW}$$

Here, the W can be denoted as the “**doubling rate**”

Example 04. Horse Racing

Consider the trial of betting a game of horse racing with M horses.

The data below is the probability and the odds to have the corresponding value, respectively.

Outcomes	$1, 2, \dots, M$
Probability	$p_1, p_2, \dots, p_M$
Odds	$o_1, o_2, \dots, o_M$
Adjusted Probability (for simplicity)	$\frac{1}{M}, \frac{1}{M}, \dots, \frac{1}{M}$
Adjusted Odds (for simplicity)	$\times M, \times M, \dots, \times M$
Bet	$q_1 B, q_2 B, \dots, q_M B$

Here $\sum q_i = 1$, and $o_i = \frac{1}{p_i}$ represents a fair game, and $o_i \neq \frac{1}{p_i}$ denotes an unfair game.

Note that by the previous problem ($\mathbb{E}[\log S]$), we know that the optimal strategy is when $q^* = p$

The cost of mismatch therefore, is denoted as:

$$\text{Cost of Mismatch} = \sum_{i=1}^M p_i \log \left(\frac{p_i}{q_i}\right) = D(p || q)$$

3.3 Cross Entropy Loss in Classification Tasks

Consider a classification problem with the corresponding constraints:

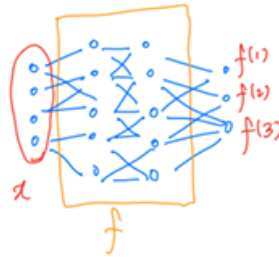
- input : $x \in \mathcal{X}$
- label : $y \in \{1, \dots, M\}$

- classifier $f : \mathcal{X} \rightarrow M$

$$x \xrightarrow{f} \begin{pmatrix} f(x)_1 \\ \vdots \\ f(x)_M \end{pmatrix}, \quad \begin{bmatrix} f(x)_1 \approx \Pr(\text{label of } x = 1) \\ \vdots \\ f(x)_M \approx \Pr(\text{label of } x = M) \end{bmatrix}$$

- Goal: have a classifier (lower loss)
- Here the Cross Entropy Loss is defined as follows

$$\ell(f(x), y) = \log \frac{1}{f(x)_y}$$

Figure 5: Classifier f **Example 05.** Weather Prediction

Consider a task where you need to predict what weather will be in the future.

To make the case simple, let suppose the weather will be either "rain", "cloudy", or "sunny".

Suppose you predict it will be 70% rain, 20% cloudy, and 10% sunny.

This can be interpreted as a classification task with the corresponding labels.

$$\begin{array}{ll} \text{rain} & 70\% \\ \text{cloudy} & 20\% \\ \text{sunny} & 10\% \end{array} \implies f(x) = \begin{pmatrix} 0.7 \\ 0.2 \\ 0.1 \end{pmatrix}$$

Assume we use the Cross Entropy Loss to determine the truth.

$$\text{truth}(y = \text{rain}) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \log \frac{1}{0.7}, \quad \text{truth}(y = \text{sunny}) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \log \frac{1}{0.1}$$

Since the classification task is to have a classifier with lower loss, this situation can be viewed as it is likely to "rain" in the future.

Consider these cases below

$$\begin{array}{ll} \text{rain} & 90\% \\ \text{cloudy} & 5\% \\ \text{sunny} & 5\% \end{array} \implies \text{Case 2}, \quad \begin{array}{ll} \text{rain} & 100\% \\ \text{cloudy} & 0\% \\ \text{sunny} & 0\% \end{array} \implies \text{Case 3}$$

Comparing the both cases of the label "rain":

Case 2 has the truth value of $\log \frac{1}{0.9}$ and Case 3 has the truth value of $\log \frac{1}{1}$

Since Case 3 has the lower loss, Case 3 is more likely to be classified as "rain" in the future.

The application of Cross Entropy Loss in classification tasks is to penalise to training classifier to find out which one is better.

It makes sense though but in one hand, it is weird that why we actually use $\log \frac{1}{f(x)_y}$ as the loss function.

Considering the same classifier, the y -th label can be interpreted as a One-Hot notation, which is a deterministic probability distribution.

$$f(x) = \begin{pmatrix} f(x)_1 \\ f(x)_2 \\ \vdots \\ f(x)_M \end{pmatrix}, \quad y \rightarrow \mathbb{1}_y = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 (= y\text{-th index}) \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

Considering this, the Cross Entropy Loss can be viewed as follows:

$$\ell(f(x), y) = \log \frac{1}{f(x)_y} = 1 \cdot \log \frac{1}{f(x)_y} = 0 \cdot \log \frac{0}{f(x)_1} + \dots + 1 \cdot \log \frac{1}{f(x)_y} + \dots + 0 \cdot \log \frac{0}{f(x)_M} = D(\mathbb{1}_y || f(x))$$

$\therefore$ The Cross Entropy Loss in Classification Tasks can be interpreted as a KL Divergence of $\mathbb{1}_y$ and $f(x)$

Case Study 01. Generative Adversarial Networks (GAN)

For general distribution (other than one-hot distribution), we have

$$D(p_X || Q) = \sum_x p_X(x) \log \frac{p_X(x)}{Q(x)} = - \sum_x p_X(x) \log \frac{1}{p_X(x)} + \sum_x \log \frac{1}{Q(x)} = -H(p_X) + \sum_x \log p_X(x) \frac{1}{Q(x)}.$$

where the second term is called cross entropy which coincides with the cross entropy loss

$$\text{Cross Entropy}(\mathbb{1} || f(x)) = \sum_m \mathbb{1}_m \log \frac{1}{f(x)_m}.$$

This is the reason why we call it "cross entropy loss".

Thus, even when the true label is "soft" distribution (instead of one-hot), we have

$$D(y^{(\text{soft})} || f(x)) = -H(y^{(\text{soft})}) + \text{Cross Entropy}(y^{(\text{soft})} || f(x)).$$

Since the true label is given an fixed, minimising KL divergence is equivalent to minimising cross entropy loss.

Case Study 02. Knowledge Distillation

Consider a teacher model with classifier f_{teacher} and a student model with classifier f_{student} .

This can be interpreted as follows:

$$\begin{aligned} x &\longrightarrow \boxed{\text{f teacher}} \longrightarrow (f_{\text{teacher}})(x) \\ x &\longrightarrow \boxed{\text{f student}} \longrightarrow (f_{\text{student}})(x) \implies \log \frac{1}{f_{\text{student}}(x)} \end{aligned}$$

The task is to train and penalise the student network by comparing the teacher's belief and the students belief which forms a probability distribution.

$\therefore$ From this perspective, Knowledge Distillation can be viewed as a KL divergence of the classifiers between the teacher model and the student model (*i.e.*, $D(f_{\text{teacher}} || f_{\text{student}})$).

4 Joint Random Variables

Let $X \in \mathcal{X}$ and $Y \in \mathcal{Y}$ are random variables. The joint pmf is given by

$$p_{X,Y}(x, y) = \Pr(X = x, Y = y)$$

The *marginal distribution* can be obtained by summing joint pmf with respect to the separate random

variable:

$$p_X(x) = \sum_y p_{X,Y}(x,y) = \sum_y p_{X|Y}(x|y)p_Y(y) = \mathbb{E}_{p_Y}[p_{X|Y}(x|Y)]$$

- **Conditional pmf**

- $p_{Y|X}(y|x) = \Pr(Y = y|X = x)$
- $p_{X,Y}(x,y) = p_X(x)p_{Y|X}(y|x) = p_Y(y)p_{X|Y}(x|y)$
- $\mathbb{E}[f(Y)] = \sum f(y)p_Y(y)$

- **Independence**

- If $X \perp\!\!\!\perp Y \iff p_{X,Y}(x,y) = p_X(x)p_Y(y) \quad \forall x \in \mathcal{X}, y \in \mathcal{Y}$
- Note that $p_{X,Y}(x,y) = p_X(x)p_{Y|X}(y|x)$
- $\therefore$ If $X \perp\!\!\!\perp Y \implies p_Y(y) = p_{Y|X}(y|x)$

Lemma

Consider jointly distributed random variables X and Y .

For any functions $f : \mathcal{X} \rightarrow \mathbb{R}$ and $g : \mathcal{Y} \rightarrow \mathbb{R}$

$$\mathbb{E}[f(X) + g(Y)] = \sum_{x,y} p_{X,Y}(x,y)(f(x) + g(y))$$

W.T.S is below:

$$\mathbb{E}[f(X) + g(Y)] = \mathbb{E}[f(X)] + \mathbb{E}[g(Y)].$$

Proof

We have

$$\begin{aligned} \mathbb{E}[f(X) + g(Y)] &= \sum_{x,y} p_{X,Y}(x,y)(f(x) + g(y)) = \sum_{x,y} p_{X,Y}(x,y)f(x) + \sum_{x,y} p_{X,Y}(x,y)g(y) \\ &= \sum_x p_X(x)f(x) + \sum_y p_Y(y)g(y) = \mathbb{E}[f(X)] + \mathbb{E}[g(Y)]. \\ \therefore \mathbb{E}[f(X) + g(Y)] &= \mathbb{E}[f(X)] + \mathbb{E}[g(Y)]. \end{aligned}$$

Also, let X and Y be independent random variables.

For any functions $f : \mathcal{X} \rightarrow \mathbb{R}$ and $g : \mathcal{Y} \rightarrow \mathbb{R}$

$$\mathbb{E}_{X,Y}[f(X)g(Y)] = \sum_{x,y} p_{X,Y}(x,y)f(x)g(y)$$

W.T.S is below:

$$\mathbb{E}_{X,Y}[f(X)g(Y)] = \mathbb{E}_X[f(X)]\mathbb{E}_Y[g(Y)].$$

Proof

We have

$$\begin{aligned} \mathbb{E}_{X,Y}[f(X)g(Y)] &= \sum_{x,y} p_{X,Y}(x,y)f(x)g(y) = \sum_x \sum_y p_X(x)p_Y(y)f(x)g(y) \\ &= \sum_x \left(p_X(x)f(x) \sum_y p_Y(y)g(y) \right) = \sum_x p_X(x)f(x)\mathbb{E}_Y[g(Y)] = \mathbb{E}_X[f(X)]\mathbb{E}_Y[g(Y)] \\ \therefore \mathbb{E}_{X,Y}[f(X)g(Y)] &= \mathbb{E}_X[f(X)]\mathbb{E}_Y[g(Y)]. \end{aligned}$$

Lemma

Let X and Y be a joint random variable with pmf $p_{X,Y}(x,y)$.

If there exists $\phi_1 : \mathcal{X} \rightarrow \mathbb{R}$ and $\phi_2 : \mathcal{Y} \rightarrow \mathbb{R}$ such that $p_{X,Y}(x,y) = \phi_1(x)\phi_2(y)$ for all x and y .

W.T.S is below:

X and Y are independent.

Proof

If we marginalise $p_{X,Y}$, we have

$$p_X(x) = \sum_y p_{x,y}(x,y) = \sum_y \phi_1(x)\phi_2(y) = C_1\phi_1(x)$$

where $C_1 = \sum_y \phi_2(y)$.

Similarly, we have $p_Y(y) = C_2\phi_2(y)$ where $C_2 = \sum_x \phi_1(x)$.

Thus, $p_{X,Y}(x,y) = C_1C_2p_X(x)p_Y(y)$ where C_1C_2 should be 1

$\therefore$ This implies that X and Y are independent.

For more than two random variables, we can define a random vector.

Let $X^n = (X_1, \dots, X_n) \in \mathcal{X}^n$ be random vector. The joint pmf is given by

$$p_{X^n}(x^n) = \Pr[X_1 = x_1, X_2 = x_2, \dots, X_n = x_n].$$

In Supervised Learning problems, we often have (X, Y) as a data point, where X is often a random vector and Y is random vector (label).

Thus, one way of interpreting classification problem is to learn $p_{Y|X}(y|x)$

Example 01. Image Classification

Recall the 5×5 of 0-1 image. It can be collection of 25-dimensional binary random vector, i.e.,

$$X^{5 \times 5} = \begin{pmatrix} X_{11}, \dots, X_{15} \\ \vdots \\ X_{51}, \dots, X_{55} \end{pmatrix} \in \{0, 1\}^{25}.$$

Let f be a classifier that classifies input images as numbers from $1, \dots, M$ then:

$$x \longrightarrow \boxed{f(x)} \longrightarrow f(x) = \begin{pmatrix} f(x)_1 \\ \vdots \\ f(x)_M \end{pmatrix} = \begin{bmatrix} p_{Y|X}(1|x) \\ p_{Y|X}(2|x) \\ \vdots \\ p_{Y|X}(M|x) \end{bmatrix}$$

Assume the input image x is as follows, note that 0 represents "empty" and 1 represents "coloured".

$$x = \begin{pmatrix} 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

Considering x , some might think the image represents the number "1",

while others might think the image represents the number "7".

Let the classifier's output turned out to be $f(x)_1 = 0.4$, and $f(x)_7 = 0.6$.

Since $f(x)_7 > f(x)_1$, this represents that the classifier classified the image as "7".

4.1 Joint Entropy

For jointly distributed random variables X_1 and X_2 with joint pmf $P_{X_1, X_2}(x_1, x_2)$.

The *joint entropy* is given by

$$H(X_1, X_2) = \mathbb{E} \left[\log \frac{1}{p_{X_1, X_2}(X_1, X_2)} \right] = \sum_{x_1, x_2} \left(p_{X_1, X_2}(x_1, x_2) \log \frac{1}{p_{X_1, X_2}(x_1, x_2)} \right).$$

Property of joint entropy

$$\text{If } X_1 \perp\!\!\!\perp X_2 \implies H(X_1, X_2) = H(X_1) + H(X_2)$$

Proof

$$\begin{aligned} H(X_1, X_2) &= \mathbb{E} \left[\log \frac{1}{p_{X_1, X_2}(X_1, X_2)} \right] = \mathbb{E} \left[\log \frac{1}{p_{X_1}(X_1)p_{X_2}(X_2)} \right] \\ &= \mathbb{E} \left[\log \left(\frac{1}{p_{X_1}(X_1)} \right) + \log \left(\frac{1}{p_{X_2}(X_2)} \right) \right] = \mathbb{E} \left[\log \frac{1}{p_{X_1}(X_1)} \right] + \mathbb{E} \left[\log \frac{1}{p_{X_2}(X_2)} \right] = H(X_1) + H(X_2) \\ \therefore H(X_1, X_2) &= H(X_1) + H(X_2) \end{aligned}$$

The property above can be generalised to multi-variables.

If $X_1, \dots, X_n$ are independent, *i.e.*, $p_{X_1 \dots X_n}(x_1, \dots, x_n) = \prod_{i=1}^n p_{X_i}(x_i)$, then

$$H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i).$$

Example 02. Guessing Game

Let Alice picks number X uniformly at random from $\{0, 1, \dots, 7\}$ and tells Bob to guess.

Bob asks the following questions:

Q1: Is $X \in \{4, 5, 6, 7\}$? The random variable $Y_1 = 1$ if the answer is yes, and $Y_1 = 0$ if not.

Q2: Is $X \in \{2, 3, 6, 7\}$? The random variable $Y_2 = 1$ if the answer is yes, and $Y_2 = 0$ if not.

Q3: Is $X \in \{1, 3, 5, 7\}$? The random variable $Y_3 = 1$ if the answer is yes, and $Y_3 = 0$ if not.

Note that Y_1, Y_2, Y_3 are all Bernoulli $\frac{1}{2}$ random variable, and therefore $H(Y_1) = H(Y_2) = H(Y_3) = 1$.

Also, Y_1, Y_2, Y_3 are independent since

$$p_{Y_1, Y_2, Y_3}(y_1, y_2, y_3) = \frac{1}{8} = p_{Y_1}(y_1)p_{Y_2}(y_2)p_{Y_3}(y_3)$$

for all y_1, y_2, y_3 (*i.e.*, $\Pr(Y_2 = 1 | Y_1 = 1) = \frac{1}{2} = \Pr(Y_2 = 1)$).

$$\text{Thus, } H(Y_1, Y_2, Y_3) = H(Y_1) + H(Y_2) + H(Y_3) = 3 = \left(\frac{1}{8} \log 8 \right) \times 8$$

This implies that each question worth of 1 bit information, and the total information we can obtain after three question is 3 bits (if questions are independent).

Since the amount of uncertainty (amount of information) in X is also 3 bits, three questions is enough to deterministically estimate the random variable X .

This time consider the following case where Bob asks questions as follows:

Q1: Is $X \in \{4, 5, 6, 7\}$? The random variable $Y_1 = 1$ if the answer is yes, and $Y_1 = 0$ if not.

Q2: Is $X \in \{3, 4, 5, 6\}$? The random variable $Y_2 = 1$ if the answer is yes, and $Y_2 = 0$ if not.

Q3: Is $X \in \{2, 3, 4, 5\}$? The random variable $Y_3 = 1$ if the answer is yes, and $Y_3 = 0$ if not.

Though $H(Y_1) = H(Y_2) = H(Y_3) = 1$, Y_1, Y_2, Y_3 are not independent.

This makes $H(Y_1, Y_2, Y_3) < 3$, which implies there is not enough information to determine access.

4.2 Conditional Entropy

We showed that if $X_1 \perp\!\!\!\perp X_2 \Rightarrow p_{X_1, X_2}(x_1, x_2) = p_{X_1}(x_1)p_{X_2}(x_2) \Rightarrow H(X_1, X_2) = H(X_1) + H(X_2)$

This implies $p_{X_1, X_2}(x_1, x_2) = p_{X_1}(x_1)p_{X_2|X_1}(x_2|x_1) \Rightarrow H(X_1, X_2) = H(X_1) + H(X_2|X_1)$

Here, the $H(X_2|X_1)$ is denoted as the *conditional entropy*.

The *conditional entropy* $H(X|Y)$ is an amount of information of X given the information of Y , which can be defined as

$$H(X|Y) = \mathbb{E} \left[\log \frac{1}{p_{X|Y}(X|Y)} \right] = \sum p_{X,Y}(x, y) \log \left(\frac{1}{p_{X|Y}(x|y)} \right).$$

Property of conditional entropy

$$\text{If } X \perp\!\!\!\perp Y \Rightarrow H(X|Y) = H(X)$$

$$\text{Also, if } X = Y \Rightarrow H(X|Y) = 0$$

$$\therefore 0 \leq H(X|Y) \leq H(X)$$

Relation with joint entropy

In our case, the function that we want to compute is $H(X, Y) = \mathbb{E} \left[\log \frac{1}{p_{X,Y}(x, y)} \right]$,

and the probability of x, y can be expressed with joint distribution $P_{X,Y}(x, y)$.

We are considering random variables X and Y to compute the expectation.

The joint entropy is a sum of individual entropies, which becomes apparent when we consider each outcome y and its respective contribution to the overall entropy.

$$H(X|Y = y) = \mathbb{E} \left[\log \frac{1}{p_{X|Y}(X|Y = y)} \right] = \sum p_{X|Y}(x|y) \log \left(\frac{1}{p_{X|Y}(x|y)} \right)$$

If we aggregate all possible instances of conditional entropy given a specific condition y , we can represent the overall conditional entropy of X given Y as:

$$H(X|Y) = \sum_y p_Y(y) H(X|Y = y)$$

Proof

$$\begin{aligned} H(X|Y) &= \mathbb{E} \left[\log \frac{1}{p_{X|Y}(X|Y)} \right] = \sum_{x,y} p_{X,Y}(x, y) \log \left(\frac{1}{p_{X|Y}(x|y)} \right) = \sum_y \sum_x p_Y(y) p_{X|Y}(x|y) \log \left(\frac{1}{p_{X|Y}(x|y)} \right) \\ &= \sum_y p_Y(y) \left(\sum_x p_{X|Y}(x|y) \log \left(\frac{1}{p_{X|Y}(x|y)} \right) \right) = \sum_y p_Y(y) \cdot H(X|Y = y). \end{aligned}$$

We all know that if $X \perp\!\!\!\perp Y \Rightarrow H(X, Y) = H(X) + H(Y)$

What if $X \not\perp\!\!\!\perp Y$? then:

- $H(X, Y) \neq H(X) + H(Y)$
- $H(X, Y) = H(X) + H(Y|X)$

Proof

$$\begin{aligned} H(X) + H(Y|X) &= \mathbb{E} \left[\log \frac{1}{p_X(X)} \right] + \mathbb{E} \left[\log \frac{1}{p_{Y|X}(Y|X)} \right] = \mathbb{E} \left[\log \frac{1}{p_X(X)p_{Y|X}(Y|X)} \right] \\ &= \mathbb{E} \left[\log \frac{1}{p_X(X) \cdot \frac{p_{X,Y}(X,Y)}{p_X(X)}} \right] = \mathbb{E} \left[\log \frac{1}{p_{X,Y}(X,Y)} \right] = H(X, Y). \end{aligned}$$

Example 03. Gambling

Let X be a trial of tossing a fair coin and Y the money you get with the corresponding trial.

The bet is as follows:

$$\text{If } (X = H) \implies Y = \$1.00$$

$$\text{If } (X = T) \implies Y = \begin{cases} \$0.00 & \text{with probability } \frac{1}{2} \\ \$2.00 & \text{with probability } \frac{1}{2} \end{cases}$$

Computing the conditional entropy:

$$H(Y|X = H) = \mathbb{E} \left[\log \frac{1}{p_{Y|X}(Y|X = H)} \right] = \mathbb{E} \left[\log \frac{1}{1} \right] = 0$$

$$\begin{aligned} H(Y|X = T) &= \mathbb{E} \left[\log \frac{1}{p_{Y|X}(Y|X = T)} \right] \\ &= p_{Y|X}(Y = 0|X = T) \times \log \frac{1}{p_{Y|X}(Y = 0|X = T)} + p_{Y|X}(Y = 2|X = T) \times \log \frac{1}{p_{Y|X}(Y = 2|X = T)} \\ &= \frac{1}{2} \times \log \frac{1}{\frac{1}{2}} + \frac{1}{2} \times \log \frac{1}{\frac{1}{2}} = 1 \end{aligned}$$

$$\therefore H(Y|X) = p_X(X = H) \times H(Y|X = H) + p_X(X = T) \times H(Y|X = T) = \frac{1}{2} \times 0 + \frac{1}{2} \times 1 = \frac{1}{2}$$

Example 04. Guessing Game

Recall the guessing game in **Example 02**. I.e.,:

Alice : Pick $X \in \{0, 1, 2, 3, 4, 5, 6, 7\}$

$$\text{Bob : } \begin{cases} Q1 : X \in \{4, 5, 6, 7\} \Rightarrow Y_1 \\ Q2 : X \in \{2, 3, 6, 7\} \Rightarrow Y_2 \\ Q3 : X \in \{1, 3, 5, 7\} \Rightarrow Y_3 \end{cases}$$

$$H(X) = H(Y_1, Y_2, Y_3) = H(Y_1) + H(Y_2) + H(Y_3) = 3$$

This time consider Bob asks the question as follows:

$$\text{Bob : } Q2' : X \in \{3, 4, 5, 6\} \Rightarrow Y_2'$$

This makes $H(Y_2') = 1$, then consider the following joint entropy:

$$H(Y_1, Y_2') = H(Y_1) + H(Y_2'|Y_1)$$

This makes the conditional entropy $H(Y_2'|Y_1)$ as follows:

$$X \begin{cases} Y_1 = 0 & \Rightarrow \{0, 1, 2, 3\} \Rightarrow \begin{cases} \Rightarrow \{0, 1, 2\} \\ \Rightarrow \{3\} \end{cases} \\ Y_1 = 1 & \Rightarrow \{4, 5, 6, 7\} \Rightarrow \begin{cases} \Rightarrow \{4, 5, 6\} \\ \Rightarrow \{7\} \end{cases} \end{cases}$$

$$\begin{aligned} H(Y_2'|Y_1) &= p_{Y_1}(Y_1 = 0) \times H(Y_2'|Y_1 = 0) + p_{Y_1}(Y_1 = 1) \times H(Y_2'|Y_1 = 1) \\ &= \frac{1}{2} \times h_2\left(\frac{1}{4}\right) + \frac{1}{2} \times h_2\left(\frac{3}{4}\right) = \frac{1}{2} \times \left(\frac{1}{4} \log \frac{1}{1/4} + \frac{3}{4} \log \frac{1}{3/4} \right) < 1 \\ \therefore H(Y_2'|Y_1) &< 1 \end{aligned}$$

More aspects of conditional entropy

- $H(Y_1|Y_1) = 0$
- If $Y_1 \perp\!\!\!\perp Y_2$, then $H(Y_2|Y_1) = H(Y_2)$

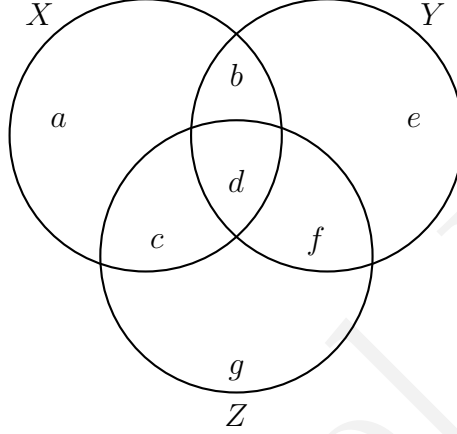
***Note** that there are no representations such as ~~$H(X|Y|Z)$~~ or ~~$\Pr(X = x|Y = y|Z = z)$~~

Conditional entropy of multiple variables

Let three random variables X, Y, Z have the joint probability $P_{X,Y,Z}(X, Y, Z)$

$$H(X, Y, Z) = \mathbb{E} \left[\log \frac{1}{p_{X,Y,Z}(X, Y, Z)} \right] = H(X) + H(Y|X) + H(Z|X, Y)$$

$$H(Z|X, Y) = \mathbb{E}_{p_{X,Y,Z}} \left[\log \frac{1}{p_{Z|X,Y}(Z|X, Y)} \right] = \sum_{x,y} p_{X,Y}(x, y) H(Z|X = x, Y = y)$$



Considering the diagram above, each area corresponds to the following representation.

- $H(X) = a + b + c + d$
- $H(Y|X) = e + f$
- $H(Z|X, Y) = g$

Case Study 01. Data Processing

Let X a random variable $\in \mathcal{X}$, and f be deterministic function (*i.e.*, $f : \mathcal{X} \rightarrow \mathbb{R}$), then:

$$H(X) \geq H(f(X))$$

Proof

$$H(X, f(X)) = H(X) + H(f(X)|X) = H(X)$$

$$H(X, f(X)) = H(f(X)) + H(X|f(X)) \geq H(f(X))$$

Consider the following example below:

$$X = \begin{cases} -2 & \text{with probability } \frac{1}{5} \\ -1 & \text{with probability } \frac{1}{5} \\ 0 & \text{with probability } \frac{1}{5} \\ 1 & \text{with probability } \frac{1}{5} \\ 2 & \text{with probability } \frac{1}{5} \end{cases}, \quad f(X) = X^2 = \begin{cases} 0 & \text{with probability } \frac{1}{5} \\ 1 & \text{with probability } \frac{2}{5} \\ 4 & \text{with probability } \frac{2}{5} \end{cases} \Rightarrow H(X) > H(f(X))$$

If f is one-to-one and onto (therefore there exists an inverse), then $H(X) = H(f(X))$.

The above theorem implies that the amount information may not increase by post processing (applying a deterministic function).

4.3 Mutual Information

Mutual information is defined as the subtraction of the entropy and the conditional entropy.

$$I(X; Y) = H(X) - H(X|Y), \quad I(Y; X) = H(Y) - H(Y|X)$$

Conditional entropy and Mutual information can be expressed in the diagram below:

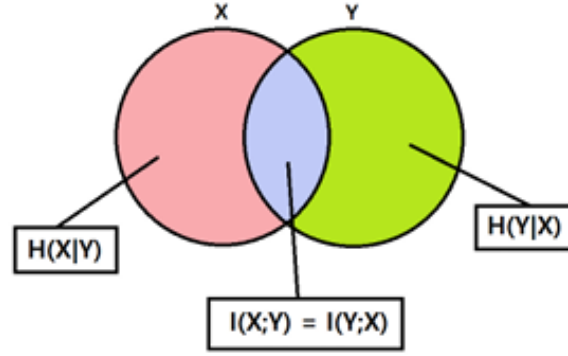


Figure 6: Conditional entropy and Mutual information diagram

Some characteristics of mutual information

- $I(X; X) = H(X) - H(X|X) = H(X) - 0 = H(X)$
- If $X \perp\!\!\!\perp Y$, then $I(X; Y) = H(X) - H(X|Y) = H(X) - H(X) = 0$

4.4 Properties of Mutual Information**Non-negativity**

$$\begin{aligned}
 I(X; Y) &= \mathbb{E} \left[\log \frac{1}{p_X(X)} \right] - \mathbb{E} \left[\log \frac{1}{p_{X|Y}(X|Y)} \right] = \mathbb{E} \left[\log \frac{p_{X|Y}(X|Y)p_Y(Y)}{p_X(X)p_Y(Y)} \right] \\
 &= \mathbb{E} \left[\log \frac{p_{X,Y}(X, Y)}{p_X(X)p_Y(Y)} \right] = D(p_{X,Y} \| p_X p_Y) \geq 0
 \end{aligned}$$

Here, mutual information can be viewed as the KL divergence between two probability distributions.

Symmetric

$$\begin{aligned}
 I(X; Y) &= H(X) - H(X|Y) = H(X) - (H(X, Y) - H(Y)) = H(X) + H(Y) - H(X, Y) = I(Y; X) \\
 \therefore I(X; Y) &= I(Y; X)
 \end{aligned}$$

Example 05. Mutual information with KL divergence

Consider two random variables X, Y having the problem distribution as below:

$$\begin{aligned}
 p_{X,Y}(0, 0) &= 0.45 & p_{X,Y}(0, 1) &= 0.05 \\
 p_{X,Y}(1, 0) &= 0.05 & p_{X,Y}(1, 1) &= 0.45
 \end{aligned}$$

This makes $p_X(0) = p_X(1) = 0.5$ and $p_Y(0) = p_Y(1) = 0.5$.

This time consider a new distribution $Q_{X,Y}$ as $Q_{X,Y}(X, Y) = p_X(X) \cdot p_Y(Y)$ which is as follows:

$$\begin{aligned}
 Q_{X,Y}(0, 0) &= 0.25 & Q_{X,Y}(0, 1) &= 0.25 \\
 Q_{X,Y}(1, 0) &= 0.25 & Q_{X,Y}(1, 1) &= 0.25
 \end{aligned}$$

Note that the KL divergence represents $D(p_{X,Y} \| p_X \cdot p_Y) = 0.45 \log \frac{0.45}{0.25} + \dots$

By using the relation between the mutual information and the KL divergence:

Mutual information $I(X; Y)$ denotes how different between $p_{X,Y}$ vs $p_X \cdot p_Y$ is.

In the independent case, *i.e.* $X \perp\!\!\!\perp Y \Rightarrow p_{X,Y} = p_X \cdot p_Y \Rightarrow I(X; Y) = 0$

Since independence denotes $p_{X,Y}$ and $p_X \cdot p_Y$ has no difference, this makes mutual information $I(X; Y) = 0$

Example 06. Mutual information related to uncertainty

Using the non-negativity property of mutual information $I(X; Y) \geq 0$ we have:

$$I(X; Y) = H(X) - H(X|Y) \geq 0 \implies H(X) \geq H(X|Y)$$

Here, $H(X)$ denotes the amount of uncertainty of X , and $H(X|Y)$ denotes the remaining uncertainty of X correlated with Y which is surely less than the former.

This can be interpreted as "conditioning always decreases (or maintains) the amount of uncertainty."

Case Study 02. Markov Chain

For any function $f : \mathcal{X} \rightarrow \mathbb{R}$, we have

$$I(X; Y) \geq I(f(X); Y).$$

Proof

$$I(X; Y) = H(Y) - H(Y|X) = H(Y) - H(Y|X, f(X)) \geq H(Y) - H(Y|f(X)) = I(f(X); Y)$$

Markov chain is a generalised version, which are not special situations like the above.

Let $X - Y - Z$ form a Markov chain (or X and Z are conditionally independent given Y) then:

$$X - Y - Z \iff X \perp\!\!\!\perp Z|Y : \text{Conditional independence; } X \text{ and } Z \text{ are independent given } Y$$

$$\begin{aligned} &\iff p_{X,Z|Y}(x, z|y) = p_{X|Y}(x|y)p_{Z|Y}(z|y) \\ &\iff p_{Z|Y,X}(z|y, x) = p_{Z|Y}(z|y) \quad \text{for all } x, y, z \\ &\quad \text{if } X \perp\!\!\!\perp Y \iff p_{Y|X}(y|x) = p_Y(y) \end{aligned}$$

This implies that X is useless to estimate Z when Y is given.

Consider $X \perp\!\!\!\perp Z \not\iff X \perp\!\!\!\perp Z|Y$ where X, Y, Z is as follows

$$\left. \begin{array}{l} X = 0 \text{ or } 1 \\ Z = 0 \text{ or } 1 \end{array} \right\} \text{independent, } Y = X + Z$$

$$X \perp\!\!\!\perp Z \not\iff X \perp\!\!\!\perp Z|Y$$

This shows for given Y , the relation between X and Z are not independent.

Case Study 03. Data Processing on Markov Chain conditions

If $X - Y - Z$ forms a Markov chain, then:

$$I(Z; X) \leq I(Z; Y)$$

Proof

$$\begin{aligned} I(Z; Y) &= H(Z) - H(Z|Y) \longrightarrow \mathbb{E} \left[\log \frac{1}{p_{Z|Y}(Z|Y)} \right] \\ &= H(Z) - H(Z|Y, X) \longrightarrow \mathbb{E} \left[\log \frac{1}{p_{Z|Y,X}(Z|Y, X)} \right] \\ &\geq H(Z) - H(Z|X) = I(X; Z) \end{aligned}$$

This implies Y is more correlated to Z .

Here, $X - Y - Z$ can be viewed as $f(X) - X - Y$.

4.5 Conditional Mutual Information

We can define the *conditional mutual information* $I(X; Z|Y)$,

which is the mutual information between X and Z when Y is given.

$$I(X; Z|Y) = H(X|Y) - H(X|Y, Z) = H(Z|Y) - H(Z|X, Y)$$

This makes $I(X; Z|Y) = 0 \iff X \perp\!\!\!\perp Z|Y$ which is the equivalent definition of a Markov chain.

***Note** that there is no general relationship between $I(X; Z|Y)$ and $I(X; Z)$.

This can be examined through the following case:

$$\begin{aligned} X &= 0 \text{ or } 1, & \text{Each probability with } \frac{1}{2} \\ Z &= 0 \text{ or } 1, & \text{Each probability with } \frac{1}{2} \\ Y &= X \oplus Z \end{aligned}$$

Here, $I(X; Z|Y) = 1$, and $I(X; Z) = 0$ which makes $I(X; Z|Y) \geq I(X; Z)$.

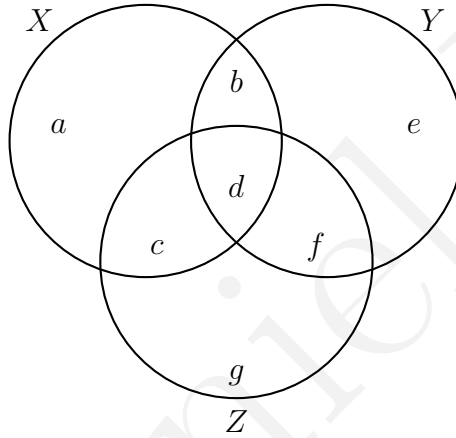
The opposite case exist in the case of a Markov chain where $I(X; Z|Y) = 0$ and $I(X; Z) \geq 0$.

Mutual information can be also defined when we have multiple random variables.

For instance, $I(X, Y; Z)$ means mutual information between (X, Y) and Z .

This can be expressed as follows according to the chain rule:

$$I(X, Y; Z) = I(X; Z) + I(Y; Z|X)$$



From the diagram above, $I(X; Z) = c + d$ and $I(Y; Z|X) = f$ which is equivalent to $I(X, Y; Z) = c + d + f$

Considering this, $I(X^n; Y)$ can be expressed as the equation below:

$$I(X^n; Y) = I(X_1; Y) + I(X_2; Y|X_1) + I(X_3; Y|X_1, X_2) + \dots = \sum_{i=1}^n I(X_i; Y|X^{i-1})$$

Example 06. Image Classification

Let $X^{28 \times 28}$ an input MNIST image which needs to be classified, and Y as the classified value.

(i.e., $Y \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$)

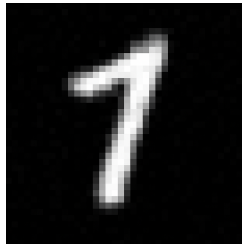


Figure 7: Input image $X^{28 \times 28}$

Consider X as the figure above and $Y = 7$, then:

$$I(X^{28 \times 28}; Y) \rightarrow H(Y) - H(Y|X^{28 \times 28}) = \log_2 10 - H(Y|X^{28 \times 28})$$

Since $H(Y|X^{28 \times 28}) \geq 0$, the greater the value is the harder the image is to classify.

Entropy in Continuous Cases

You might think it might be similar to the discrete case, when it comes to the continuous space however, this quite not makes sense.

Consider the trial X where Alice selects a number from $X \in \{0, 1, 2, \dots, 7\}$, and Y where Bob selects a number from $Y \in [0, 1]$.

We know that $H(X) = \log_2 8 = 3$ which requires 3 bits.

If we apply the same method to Y , then $H(Y) = \infty$ which makes no sense.

This leads to the concept of *Differential Entropy* which will be stated in the future.

***Note** that the entropy between the two should differ since p.d.f and p.m.f has different mechanism.

$$\begin{array}{ll} \text{Probability density} & \text{Probability mass} \\ f_X(x) \neq \Pr(X = x) & p_X(x) = \Pr(X = x) \end{array}$$

5 Continuous Random Variables

5.1 Probability Density Function

Define a cumulative distribution function $F_X(x)$ where

$$F_X(x) = \Pr[X \leq x]$$

If $F_X(x)$ is differentiable, the probability density function is as follows:

$$f_X(x) = \frac{\partial}{\partial x} F_X(x).$$

Unlike p.m.f, the p.d.f requires additional attention during change of variables. If $Y = aX$, we have

$$f_Y(y) = \frac{1}{|a|} f_X(y/a).$$

This is because the probability should be conserved, *i.e.*, $|f_X(x)dx| = |f_Y(y)dy|$ where $|dx/dy| = 1/|a|$.

More generally, let $Y = g(X)$ for a bijective function, we have

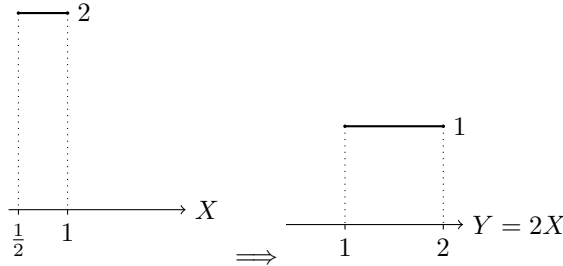
$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

Example 01. Discrete r.v. vs. Continuous r.v.

Let X a discrete random variable and $Y = 2X$, then the result is as follows:

$$X = \begin{cases} 1 & \text{with probability } \frac{1}{2} \\ 2 & \text{with probability } \frac{1}{4} \\ 3 & \text{with probability } \frac{1}{4} \end{cases} \Rightarrow Y = 2X = \begin{cases} 2 & \text{with probability } \frac{1}{2} \\ 4 & \text{with probability } \frac{1}{4} \\ 6 & \text{with probability } \frac{1}{4} \end{cases}$$

This time let X a continuous random variable where $X \sim \text{Unif}[\frac{1}{2}, 1]$ and $Y = 2X$, the result quite differs:



5.2 Gaussian

Let X be a *Gaussian distribution* with mean μ and variance σ^2 ,

then we say $X \sim \mathcal{N}(\mu, \sigma^2)$ and the corresponding p.d.f:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Jointly Gaussian

Let X_1, X_2 be a *jointly Gaussian* with mean μ and covariance matrix

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}$$

We say $(X_1, X_2) \sim \mathcal{N}(\mu, \Sigma)$ and the corresponding joint p.d.f:

$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{\sqrt{(2\pi)^2 |\Sigma|}} \exp\left(-\frac{1}{2}(\mathbb{X} - \mu)^\top \Sigma^{-1}(\mathbb{X} - \mu)\right)$$

where $\mathbb{X} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

For generalisation, let $X^n = (X_1, X_2, \dots, X_n)$ be a jointly Gaussian with mean $\mu = (\mu_1, \mu_2, \dots, \mu_n)$ and covariance matrix Σ . We say $X^n \sim \mathcal{N}(\mu, \Sigma)$ and the corresponding joint p.d.f:

$$f_{X^n}(x^n) = \frac{1}{\sqrt{(2\pi)^n \det(\Sigma)}} \exp\left(-\frac{1}{2}(x^n - \mu)^\top \Sigma^{-1}(x^n - \mu)\right).$$

For independent Gaussian random variable $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$,

the sum of the Gaussian, *i.e.*, $X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$, the generalisation is as follows:

Let X^n be a Gaussian vector, then $Y^m = AX^n$ is also Gaussian vector for any matrix $A \in \mathbb{R}^{m \times n}$.

5.3 Differential Entropy

Mutual Information for Continuous Random Variables

In discrete setting, entropy becomes maximum if the probability distribution is uniform.

The number of events become larger, the entropy increases.

However, there are some drawbacks for the extension when it comes to continuous settings.

This is why we should first define the mutual information for continuous random variables.

Consider the extension of the definition of KL divergence in continuous setting:

$$D(f||g) = \mathbb{E}_f \left[\log \frac{f(X)}{g(X)} \right]$$

As in the discrete case, the KL-divergence is non negative.

Since $-\log$ is a convex function, Jensen's inequality can be applied as follows:

$$D(f||g) = \mathbb{E}_f \left[\log \frac{f(X)}{g(X)} \right] = \int f(x) \cdot \log \frac{f(x)}{g(x)} dx = - \int f(x) \cdot \log \frac{g(x)}{f(x)} dx \geq - \log \left(\int f(x) \cdot \frac{g(x)}{f(x)} dx \right) = 0$$

To extend the definition of mutual information to the continuous scenario,

we see the mutual information in terms of KL-divergence.

The mutual information in the continuous scenario is as follows:

$$I(X; Y) = \mathbb{E} \left[\log \frac{f_{X,Y}(X, Y)}{f_X(X)f_Y(Y)} \right] = D(f_{X,Y}||f_X f_Y) \geq 0$$

Which explains the non-negativity of mutual information.

This implies $h(X) \geq h(X|Y)$ where conditioning also decreases (non-increases) the differential entropy.

Differential Entropy

If we discretise the continuous random variables X and Y to X^Δ and Y^Δ i.e.:

$$X \triangleq i \quad \text{if} \quad i\Delta \leq X < (i+1)\Delta$$

Writing the definition of mutual information, where $P(X^\Delta) = P(i \cdot \Delta \leq X \leq (i+1) \cdot \Delta) = \Delta \cdot f_X(X)$.

$$I(X^\Delta; Y^\Delta) = \mathbb{E} \left[\log \frac{P_{X^\Delta, Y^\Delta}(X^\Delta, Y^\Delta)}{P_{X^\Delta}(X^\Delta) P_{Y^\Delta}(Y^\Delta)} \right] = \mathbb{E} \left[\log \frac{\Delta^2 \cdot f_{X,Y}(X, Y)}{\Delta \cdot f_X(X) \Delta \cdot f_Y(Y)} \right] = \mathbb{E} \left[\log \frac{f_{X,Y}(X, Y)}{f_X(X) \cdot f_Y(Y)} \right]$$

Like this, we can naturally derive continuous version of mutual information from the discrete case.

Differential entropy (h) is the continuous correspondence of entropy for discrete random variable.

We use the same strategy to derive a continuous version of entropy and see the obstacles to be tackled:

As $\Delta \rightarrow 0 \implies X \approx X^\Delta$ and $P_{X^\Delta}(i) = \int_{i\Delta}^{(i+1)\Delta} f_X(x) dx \cong \Delta \cdot f_X(i\Delta)$

$$\begin{aligned} H(X^\Delta) &= \sum P_{X^\Delta}(i) \log \frac{1}{P_{X^\Delta}(i)} \approx \sum \Delta f_X(i\Delta) \log \frac{1}{\Delta f_X(i\Delta)} = \sum \Delta f_X(i\Delta) \log \frac{1}{\Delta} + \sum \Delta f_X(i\Delta) \log \frac{1}{f_X(i\Delta)} \\ &\cong \log \frac{1}{\Delta} + \int f_X(x) \log \frac{1}{f_X(x)} dx = \log \frac{1}{\Delta} + h(X) \end{aligned}$$

where

$$h(X) = \int f_X(x) \log \frac{1}{f_X(x)} dx$$

is a differential entropy, and $\log \frac{1}{\Delta}$ denotes the infinite resolution which functions as a baseline.

This denotes as Δ becomes smaller, the entropy becomes bigger,

which is intuitive since smaller Δ leads to more possibilities, which makes the entropy bigger.

The differential entropy is not equal to discretised version of entropy ($H(X^\Delta)$) with difference of $\log \Delta$.

Note that discrete entropy (H) has *label invariance* which definition is as follows:

Considering the discrete random variable X , if $Y = g(X)$ maps to one-one function then:

$$\text{If } H(Y) = H(X), \quad H(g(X)) \leq H(X) \implies H(Y) \geq H(g^{-1}(Y))$$

However, this does not hold when it comes to the continuous scenario.

Consider the continuous random variable X and $Y = g(X) = aX$ which maps to one-to-one function:

$$h(Y) = \mathbb{E} \left[\log \frac{1}{\frac{1}{|a|} \cdot f_X(X)} \right] = \mathbb{E} \left[\log |a| + \log \frac{1}{f_X(X)} \right] = \log |a| + h(X)$$

This shows $h(X) \neq h(g(X))$ even if $g(X)$ maps to one-to-one function.

Also, note that the differential entropy $h(X)$ may be negative.

For example, if $U \sim \text{Unif}(0, \frac{1}{2})$, then $h(U) = -\log 2 < 0$

This differs due to the fact that differential entropy is "*normalised*" version of entropy with $\log \Delta$ term.

5.4 Properties of Differential Entropy

In discrete case 1 to 1 function $f(x)$ discrete entropy are the same. But differential entropy does not.

$$h(X) = \mathbb{E} \left[\log \frac{1}{f_X(x)} \right] = \int f_X(x) \cdot \log \frac{1}{f_X(x)} dx \neq h(aX) = h(X) + \log |a|$$

Conditional Differential Entropy

$$h(X|Y) = \mathbb{E}_{f_{X,Y}} \left[\log \frac{1}{f_{X|Y}(x|y)} \right] = \int f_{X,Y}(x, y) \log \frac{1}{f_{X|Y}(x|y)} dx dy$$

***Note** that when it comes to differential entropy, $h(X)$ can be negative.

However, mutual information are the same. You can interpret the mutual information in the same way. Amount of information X tells about the Y or the amount of information aX tells about the Y . They should be the same and that's also the case in the continuous case.

Property of Mutual Entropy

$$I(aX; Y) = I(X; Y)$$

Proof

$$\begin{aligned} I(aX; Y) &= h(aX) - h(aX|Y) = h(aX) - \mathbb{E} \left[\log \frac{1}{f_{X|Y}(aX|Y)} \right] \\ &= h(aX) - \left(\mathbb{E} \left[\log \frac{1}{f(X|Y)} \right] + \log |a| \right) = (h(X) + \log |a|) - (h(X|Y) + \log(|a|)) = I(X; Y) \end{aligned}$$

5.5 Joint Differential Entropy

Chain Rule of Differential Entropy

$$h(X_1, X_2) = h(X_1) + h(X_2|X_1)$$

Proof

Let's denote $X_1, X_2, X_3, \dots, X_n$ as X . Then we may need to define a differential entropy for joint distributions.

For example if X_1 and X_2 are continuous random variables,

$$h(X_1|X_2) = \mathbb{E} \left[\log \frac{1}{f_{X_1, X_2}(X_1, X_2)} \right] = \mathbb{E} \left[\log \frac{1}{f_{X_1}(X_1)} \right] + \mathbb{E} \left[\log \frac{1}{f_{X_2|X_1}(X_2|X_1)} \right] = h(X_1) + h(X_2|X_1)$$

This is using the same property of the pmf. Joint is a product between the marginal times the conditional which is also the case on the pmf.

Properties of Mutual Entropy

$$I(X; Y) = \mathbb{E} \left[\log \frac{f_{X|Y}(X|Y)}{f_X(X)} \right] = \mathbb{E} \left[\log \frac{1}{f_X(X)} \right] - \mathbb{E} \left[\log \frac{1}{f_{X|Y}(X|Y)} \right] = h(X) - h(X|Y)$$

$$H(X) \geq H(X|Y), \quad h(X) \geq h(X|Y), \quad H(X) \geq H(f(X)), \quad h(X) \geq h(f(X)) \Rightarrow \text{we cannot determine}$$

X and Y are *independent* if and only if

$$I(X; Y) = 0$$

Data Processing Inequality

If $X - Y - Z$ form a *Markov Chain*, then

$$I(X; Y) \geq I(X; Z)$$

Proof

$$X - Y - Z \text{ Markov Chain} \Leftrightarrow X \perp\!\!\!\perp Z|Y \Leftrightarrow f_{X,Z|Y}(x, z|y) = f_{X|Y}(x|y)f_{Z|Y}(z|y) \Leftrightarrow f_{Z|Y,X}(z|y, x) = f_{Z|Y}(z|y)$$

Since conditioning decreases (non-increases) the entropy.

$$I(X; Y) = h(Z) - h(Z|Y) = h(Z) - h(Z|Y, X) \geq h(Z) - h(Z|X) = I(X; Z)$$

where the first equality is from Markov property $f(Z|Y) = f(Z|Y, X)$.

5.6 Maximum Differential Entropy

For $X = \{1, 2, 3, \dots, K\}$, we showed $H(X) \leq \log_2 K$ where the equality holds for uniform distribution. In other words, the maximum entropy is achieved by an uniform distribution in discrete case.

$$H(X) \leq \log_2 K \quad \Rightarrow \quad \max H(X) \text{ when } p_X(1) = \dots = p_X(K) = \frac{1}{K}$$

Then, what about $\max h(X)$?

If $X \in [a, b]$ with probability 1, then $h(X) \leq \log(b - a)$ where $h(X)$ is maximised when $f_X(x) = \frac{1}{b-a}$, $x \in [a, b]$

Proof

$$h(X) = \mathbb{E} \left[\log \frac{1}{f_X(x)} \right] = \mathbb{E} \left[\log \frac{1}{f_X(x)} \right] \geq \log \mathbb{E} \left[\frac{1}{f_X(x)} \right] = \log \int_1^b \frac{1}{f_X(x)} f_X(x) dx = \log(b - a).$$

Lemma

The Gaussian distribution achieves maximum differential entropy subject to the second moment constraint.

Proof

Let X be a random variable with pdf f_X where $\mathbb{E}_f[X^2] = p$, while X' be a Gaussian random variable $\mathcal{N}(0, p)$ where pdf is $g(x) = \frac{1}{\sqrt{2\pi p}} \exp(-x^2/(2p))$. Then consider the KL divergence between f and g . We will show

$$\begin{aligned} \int f_X(x) \log \frac{1}{f_X(x)} dx &\leq \int g(x) \log \frac{1}{g(x)} dx \\ 0 \leq D(f_X||g) &= \mathbb{E}_f \left[\log \frac{f_X(X)}{g(X)} \right] = \int f_X(x) \log \frac{f_X(x)}{g(x)} dx \\ &= \underbrace{\int f_X(x) \log \frac{1}{g(x)} dx}_{\mathbb{E}_{f_X} \left[\log \frac{1}{g(X)} \right]} - \underbrace{\int f_X(x) \log \frac{1}{f_X(x)} dx}_{h(X) = \mathbb{E}_{f_X} \left[\log \frac{1}{f_X(X)} \right]} = \mathbb{E}_f \left[\log \frac{1}{g(X)} \right] - \mathbb{E}_f \left[\log \frac{1}{f_X(X)} \right] \\ &= \mathbb{E}_f \left[\log \frac{1}{g(X)} \right] - h(f_X) = \log \sqrt{2\pi p} + \underbrace{\mathbb{E}_{f_X} \left[\frac{X^2}{2p} \right]}_{\leq \frac{1}{2}} \leq \log \sqrt{2\pi p} + \underbrace{\mathbb{E}_g \left[\frac{X^2}{2p} \right]}_{=\frac{1}{2}} \end{aligned}$$

where $h(f) = h(X)$ with $X \sim f_X$. The first term is simply

$$\mathbb{E}_f \left[\log \frac{1}{g(X)} \right] = \log \sqrt{2\pi p} + \mathbb{E}_f \left[\frac{X^2}{2p} \right] = \log \sqrt{2\pi p} + \mathbb{E}_g \left[\frac{X^2}{2p} \right] = \mathbb{E}_g \left[\log \frac{1}{g(X)} \right] = h(g).$$

Therefore, we see that

$$D(f_X||g) = h(g) - h(f_X) \geq 0$$

by the non-negativity of divergence. Therefore,

$$h(g) \geq h(f)$$

Thus, under the second moment constraint, Gaussian distribution is "maximally" random.

6 Random Process

We define random process $\mathcal{X} = \{X_n\}_{n=1}^\infty$ as a sequence of random variables. The most simple random process is an independent and identically distributed (i.i.d.) process. For i.i.d. process, we have

$$P_{X^n}(x^n) = \prod_{i=1}^n P_{X_i}(x_i), \quad (P_{X_i}(x_i) = P_X(x_i))$$

In practice, for example, in text symbols are highly correlated (e.g., one can easily deduce the missing letter is m). Therefore, such sequence does not follow an i.i.d. distribution.

my na_e is Albert

6.1 What is Markovian?

iid <----|-----> Practical
1st order Markov

The concept of a first-order Markov chain serves as a bridge between the i.i.d. assumption and practical scenarios. Markov means there is first order correlation (1st order markov).

Example 01. Random Walk

$$X_0 = 0, \quad X_n = \begin{cases} X_{n-1} + 1 & \text{with prob. of } 1/2 \\ X_{n-1} - 1 & \text{with prob. of } 1/2 \end{cases}$$

$$\text{---} * \xleftarrow{1/2} \text{---} | \text{---} \xrightarrow{1/2} * \text{---}$$

0

Given $X_{101} = 51$, X_{102} could be either 50 or 52. The outcome is unaffected by the additional information that $X_{100} = 50$, aligning with the property of a first-order Markov chain.

6.2 First Order Markov Process

Consider a random process $\mathbb{X}$. If the probability of the i^{th} step $P(X_i | X_{i-1}, \dots, X_1) = P(X_i | X_{i-1})$, the sequence is said to follow a 1st order Markov process.

$$P_{X_n | X^{n-1}}(x_n | x^{n-1}) = P_{X_n | X_{n-1}}(x_n | x_{n-1})$$

$$\underbrace{X_1 - X_2 - X_3}_{X_3 \perp\!\!\!\perp X_1 | X_2} - X_4 - \dots - X_{n-1} - \underbrace{X_n}_{X_n \perp\!\!\!\perp X_{n-1} | X_{n-2}}$$

$$P_{X_3 | X_2, X_1} = P_{X_3 | X_2}$$

Remark

Consider the joint probability distribution of all possible tuples in the sequence denoted as $P_{X^n}(x^n) = P_{X_1, \dots, X_n}(x_1, \dots, x_n)$. Then, the first order Markov process satisfies

$$X_{n+2} \perp\!\!\!\perp X_n | X_{n+1} \not\Rightarrow X_{n+2} \perp\!\!\!\perp X_n$$

$$P_{X^n}(x^n) = P_{X_1}(x_1)P_{X_2|X_1}(x_2|x_1) \cdots = P_{X_1}(x_1) \prod_{i=2}^n P_{X_i|X_{i-1}}(x_i|x_{i-1}) = P_{X_1}(x_1) \prod_{i=2}^n P_{X_i|X^{i-1}}(x_i|X^{i-1})$$

Consider a random process $\mathbb{X}$ where $X_i \in \mathcal{X} = \{1, 2, \dots, n\}$ for all i . Suppose that the transition probability $P_{X_i|X_{i-1}}(x_i|x_{i-1})$ remains the same for all i . Then, we can define a $n \times n$ transition matrix P where the (u, v) -th element is $P_{u,v} = \Pr[X_i = u | X_{i-1} = v]$. All elements in P are nonnegative and the sum of elements of each column in the transition matrix is 1.

$$P(X_i = u | X_{i-1} = v) = \begin{bmatrix} P(1|1) & P(1|2) & \dots & P(1|n) \\ \vdots & \vdots & \ddots & \vdots \\ P(n|1) & P(n|2) & \dots & P(n|n) \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{2} & 0 & \dots & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

Let π_t is an n -dimensional vector where

$$\pi_i = \begin{bmatrix} \Pr[X_i = 1] \\ \Pr[X_i = 2] \\ \vdots \\ \Pr[X_i = n] \end{bmatrix} \in \mathbb{R}^n$$

Then, we have

$$\pi_{i+1} = P \times \pi_i$$

for all $i \geq 1$.

$$\pi_{i+1} = \begin{bmatrix} P_{X_{i+1}}(1) \\ P_{X_{i+1}}(2) \\ \vdots \\ P_{X_{i+1}}(n) \end{bmatrix} = \begin{bmatrix} P_{X_{i+1}|X_i}(1|1) & \dots & P_{X_{i+1}|X_i}(1|n) \\ \vdots & & \\ P_{X_{i+1}|X_i}(n|1) & \dots & P_{X_{i+1}|X_i}(n|n) \end{bmatrix} \begin{bmatrix} \pi_i \\ P_{X_i}(2) \\ \vdots \\ P_{X_i}(n) \end{bmatrix}$$

$$P_{X_{i+1}}(1) = \sum_{j=1}^n P_{X_i}(j) P_{X_{i+1}|X_i}(1|j)$$

Example 02. Binary Random Process

Consider a binary random process where $p_{X_i|X_{i-1}}(0|1) = p_{X_i|X_{i-1}}(1|0) = \alpha < 1/2$ and $p_{X_i|X_{i-1}}(1|0) = p_{X_i|X_{i-1}}(0|1) = 1 - \alpha$. Then we can define a transition matrix

$$P = \begin{pmatrix} 1 - \alpha & \alpha \\ \alpha & 1 - \alpha \end{pmatrix}.$$

Suppose we have $\pi_0 = [1, 0]$, namely $X_0 = 0$ with probability 1. Then, $\pi_1 = [1 - \alpha, \alpha]$.

$$X_1 = 1 \Rightarrow \pi_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \pi_2 = \begin{pmatrix} 0 \\ \frac{1}{2} \\ \frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \end{pmatrix}$$

6.3 k-th Order Markov Process

Consider a random process $\mathbb{X}$. If $P_{X_i|X^{i-1}}(x_i|x^{i-1}) = P_{X_i|X_{i-k}^{i-1}}(x_i|x_{i-k}^{i-1})$, the sequence is said to follow the k^{th} order Markov process.

Remark

If the sequence follow the k^{th} order Markov, then

$$P_{X^n}(x^n) = \prod_{i=1}^n P_{X_i|X^{i-1}}(x_i|x^{i-1}) = \prod_{i=1}^n P_{X_i|X_{i-k}^{i-1}}(x_i|x_{i-k}^{i-1}).$$

$$P_{X_i|X_{i-1}, \dots, X_1} = P_{X_i|X_{i-1}, \dots, X_{i-k}}$$

6.4 Stationary Distribution

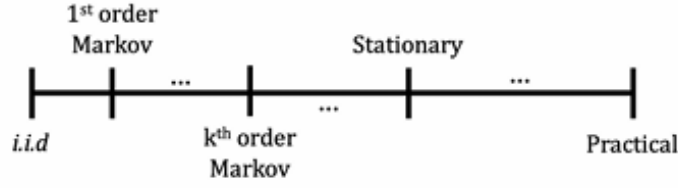


Figure 8: The assumption we've learned ordered by proximity to practical setting. The i.i.d assumption considers no correlation between the random variables in a sequence, assuming complete independence, while the practical settings have higher correlation among the sequence's random variables. A stationary distribution is a closer approximation to practical than a k^{th} order Markov process.

$\mathbb{X}$ is stationary iff

$$(X_1, X_2, \dots, X_n) \stackrel{(d)}{=} (X_{t+1}, X_{t+2}, \dots, X_{t+n}) \quad \forall t, n$$

Consider the random process $X_1, X_2, \dots, X_n$. The process is said to be *stationary* if X_i^n is *distributionally equivalent* with X_{i+1}^{i+n} for all i and n .

Note that the $P_{X_i^n}$ denotes the probability distribution of n -tuples starting from the i^{th} step, and $P_{X_{i+1}^{i+n}}$ denotes the probability distribution of n -tuples starting from the $(i+1)^{\text{th}}$ step. That is, if we take any n -tuples starting from i^{th} step and shift them to start from $(i+1)^{\text{th}}$ step, they follow the same distribution if the process is stationary. Thus, if the process is stationary, the random variable of the process does not depends on time.

Consider the random walk process:

$$X_0 = 0, \quad X_n = X_{n-1} \pm 1$$

then the possible value of X_1 and X_2 is:

$$X_1 = \begin{cases} 0 \\ 1 \end{cases}, \quad X_2 = \begin{cases} 2 \\ 0 \\ -2 \end{cases}$$

Therefore we can expect the X_n is "more random" than X_0 . Thus the process is not stationary.

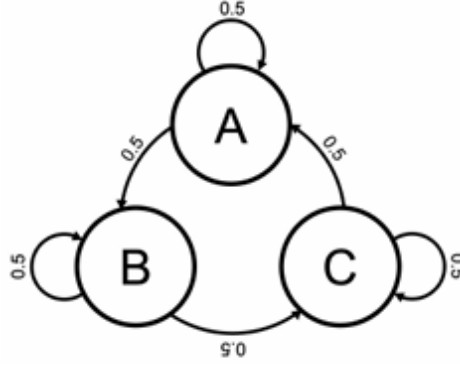
***Note** that not all Markov process is stationary. Also, stationary processes can have infinite dependencies.

6.5 Stationary Markov Process

$$1^{\text{st}} \text{ Markov} \not\Rightarrow \text{Stationary}$$

Example 03. 1^{st} order Markov Process

Consider the 1^{st} order Markov process with the initial distribution $P(A) = P(B) = P(C) = \frac{1}{3}$.

Figure 9: 1st order Markov transition.

Then we can have

$$P(X_1 = A) = P(X_1 = B) = P(X_1 = C) = \frac{1}{3}$$

$$P(X_2 = A) = P(X_1 = A) \cdot P(X_2 = A|X_1 = A) + P(X_1 = C) \cdot P(X_2 = A|X_1 = C) = \frac{1}{3}$$

When the same process is applied to $P(X_2 = B)$ and $P(X_2 = C)$, the probability consistently remains at $\frac{1}{3}$, and this holds true when extended to $X_3, X_4, \dots$ where all probabilities are $\frac{1}{3}$ as well. Thus, the probabilities at X_i always follow the initial probability of $\frac{1}{3}$, indicating that the process is stationary: if we take any n -tuples in the process starting from i^{th} step and shift to $(i+1)^{\text{th}}$ step, the distribution will be always same.

Now consider the same process described in the previous example, with the initial distribution $P(X_1 = A)$ and $P(X_2 = A) = P(X_2 = B) = \frac{1}{2}$. Given this condition, the process is no longer stationary. That is, the Markov process itself does not guarantee stationary; however, under certain assumptions, Markov process can have stationary initial distributions, as in the previous example.

***Note** that with some reasonable assumptions (*finite state, irreducible, aperiodic*), Markov process can have a *stationary initial distribution*.

Example 04. Binary Random Process 2

Consider a binary random process where $P_{X_1}(0) = 1$, $P_{X_1}(1) = 0$, $\alpha < 1/2$, and the conditional probability is as follows:

$$\begin{aligned} P_{X_i|X_{i-1}}(0|0) &= 1 - \alpha & P_{X_i|X_{i-1}}(0|1) &= \alpha \\ P_{X_i|X_{i-1}}(1|0) &= \alpha & P_{X_i|X_{i-1}}(1|1) &= 1 - \alpha \end{aligned}$$

Then we can define a transition matrix(P)

$$P = \begin{bmatrix} P(0|0) & P(0|1) \\ P(1|0) & P(1|1) \end{bmatrix} = \begin{pmatrix} 1 - \alpha & \alpha \\ \alpha & 1 - \alpha \end{pmatrix}.$$

Suppose we have $\pi_0 = [1/2, 1/2]$, then it is not hard to show that $\pi_t = [1/2, 1/2]$ for all t .

$$\pi_2 = \begin{bmatrix} 1 - \alpha & \alpha \\ \alpha & 1 - \alpha \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \Rightarrow \text{Stationary Process.}$$

The stationary distribution π^* should satisfy

$$\pi^* = P \times \pi^*.$$

Thus, the stationary distribution is an eigenvector of P corresponds to eigenvalue 1. Recall that the eigenvalue λ of matrix P is a solution of $\det(P - \lambda I) = 0$.

With some assumptions(finite state, aperiodic, irreducible), the 1st Markov process has a stationary initial distribution.

Lemma

The uniform distribution is a stationary distribution if P is symmetric.

$$\text{if } P = P^T \Rightarrow \pi^* = \begin{bmatrix} 1/k \\ 1/k \\ \vdots \\ 1/k \end{bmatrix}$$

Proof

We let π_∞ the limiting distribution (if exists), i.e.,

$$\pi_\infty = \lim_{t \rightarrow \infty} \pi_t.$$

Note that the limiting distribution may depend on initial distribution π_0

$$\pi_{i+1} = P\pi_i, \quad \pi^* = P\pi^*$$

Considering the perspective as $Av = \lambda v$, we can interpret the equation above as follows.

$$P\pi^* = \mathbb{1} \cdot \pi^* \Rightarrow P \text{ has eigenvector } \pi^* \text{ with corresponding eigenvalue } 1.$$

$$\pi_1 = \pi^* = \pi_2 = \pi_3 = \dots$$

$$\pi_{i+1} = P\pi_i$$

$$\lim_{i \rightarrow \infty} \pi_{i+1} = \lim_{i \rightarrow \infty} P\pi_i = P \lim_{i \rightarrow \infty} \pi_i$$

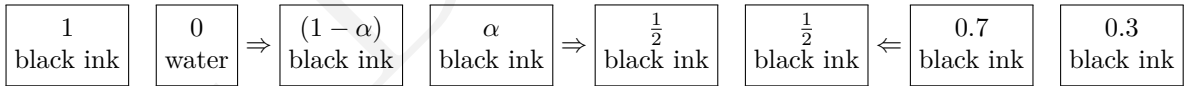
$$\pi_\infty = P\pi_\infty$$

The limiting distribution should be a stationary distribution.

Case Study 01. Equilibrium State

Consider the process of dissolving black ink into pure water

The task is to compute the concentration of the black ink (we will consider 1 if the substance consists of pure black ink, and 0 if only pure water exists.)



After many iterations, both will reach to an *equilibrium state* of concentration of $\frac{1}{2}$ regardless of the initial concentration.

Here, the equilibrium state can be denoted as a stationary process.

Case Study 02. Discrete Diffusion

The role of the Generative Model is to learn the given distribution.

Let the task as a next word prediction(π_n) process with the following corpus set X_0 .

Each word gets one-hot encoded as follows:

$$X_0 = \{\text{I, am, Iron, Man, a, the}\}$$

I	am	Iron	Man	I	am	a
$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$	$\begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \end{bmatrix}$	$\begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \end{bmatrix}$	$\begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \end{bmatrix}$
1	2	3	4	1	2	5

Given $X_0 \sim p_0$ be the random variable with data distribution p_0 .

Consider a transition matrix

$$P = \begin{bmatrix} 1-\epsilon & \epsilon/(n-1) & \dots & \epsilon/(n-1) \\ \epsilon/(n-1) & 1-\epsilon & \dots & \epsilon/(n-1) \\ \vdots & & & \\ \epsilon/(n-1) & \epsilon/(n-1) & \dots & 1-\epsilon \end{bmatrix}$$

$$\pi_2 = P\pi_1, \quad \pi_3 = P\pi_2, \quad \Rightarrow \quad \pi_{1000000} \approx \begin{bmatrix} 1/n \\ 1/n \\ \vdots \\ 1/n \end{bmatrix}$$

$$X_1 \rightarrow X_2 \xrightarrow{\leftarrow} X_3 \xrightarrow{\leftarrow} \dots \xrightarrow{\leftarrow} X_{1000000} = \begin{bmatrix} 1/n \\ \vdots \\ 1/n \end{bmatrix}$$

Let $\mathcal{X}$ be a 1st order Markov process where $\pi_{t+1} = P\pi_t$. The limiting distribution is a uniform distribution $\pi^* = [1/n, \dots, 1/n]$. For sufficiently large N , we say that $\pi_N \sim \pi^*$. On the other hand, if we can train a neural network that can recover (denoise) X_t from X_{t+1} for some t , i.e.,

$$X_t \approx f_\theta(X_{t+1}, t).$$

Then, the generation process is the following. Sample $\bar{X}_N$ according to the uniform distribution, and apply $f_\theta(\cdot, t)$ recursively to obtain $\bar{X}_0$. Then, we can expect that $\bar{X}_0$ behaves similarly to $X_0 \sim p_0$. This is a key idea of generative diffusion process.

This time consider the task with the following dictionary

$$\begin{array}{ccccccccc} \{\text{Albert}, \text{am}, \text{I}, \text{Iron}, \text{man}, \dots, \square\} & & & & & & & & \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \dots & \downarrow & & \\ 1 & 2 & 3 & 4 & 5 & & k & & \\ & & \text{I} & \text{am} & \text{Albert} & & & & \\ & & 3 & 2 & 1 & & & & \end{array}$$

The task below can be interpreted as follows:

$$\begin{array}{ccccccc} \text{I} & \text{am} & \text{Iron} & \text{man} & \dots & & \\ X_1=3 & X_2=2 & X_3=4 & X_4=5 & \dots & & \end{array}$$

Any sentence can be considered as a random process with transition matrix P as below

$$P = \begin{bmatrix} 1-\epsilon & \epsilon/(k-1) & \dots & \\ & 1-\epsilon & & \\ \vdots & & \ddots & \\ \epsilon/(k-1) & \dots & & 1-\epsilon \end{bmatrix}$$

$$\Pr(X_{n+1} = i | X_n) = 1 - \epsilon, \quad \Pr(X_{n+1} = i | X_n = j) = \frac{\epsilon}{(k-1)}$$

Training the Neural Network, a random text will be generated using this network.

$$\begin{array}{ccccccc} & X_1^{(1)} = 3 & X_2^{(1)} = 2 & X_3^{(1)} = 4 & X_4^{(1)} = 5 & & \\ f_\theta(X^{(2)}, 1) \uparrow & \downarrow P & \downarrow P & \downarrow P & \downarrow P & & \\ & X_1^{(2)} = 3 & X_2^{(2)} = 2 & X_3^{(2)} = 4 & X_4^{(2)} = 5 & & \\ f_\theta(X^{(3)}, 2) \uparrow & \downarrow P & \downarrow P & \downarrow P & \downarrow P & & \\ & X_1^{(3)} = 7 & X_2^{(3)} = 2 & X_3^{(3)} = 4 & X_4^{(3)} = 5 & & \\ f_\theta(X^{(4)}, 3) \uparrow & \downarrow P & \downarrow P & \downarrow P & \downarrow P & & \\ \vdots & & & & & & \\ f_\theta(X^{(N)}, N-1) \uparrow & \downarrow P & \downarrow P & \downarrow P & \downarrow P & & \\ & X_1^{(N)} = 7 & X_2^{(N)} = 16 & X_3^{(N)} = 29 & X_4^{(N)} = 61 & & \\ & \uparrow \pi^* & \uparrow \pi^* & \uparrow \pi^* & \uparrow \pi^* & & \end{array}$$

Here $f_\theta(X^{(i)}, i-1)$ where $i \geq 2$ denotes the denoising process.

***Note** that each process X_1, X_2, X_3, X_4 is independent each other (not pair-wise independent)

Then the denoising process from $X_1^{(N)} = 13, X_2^{(N)} = 11, X_3^{(N)} = 19, X_4^{(N)} = 23$ to I am Iron man can be interpreted as follows:

$$\begin{array}{ccccccc}
 & \downarrow \pi^* & \downarrow \pi^* & \downarrow \pi^* & \downarrow \pi^* & & \\
 & X_1^{(N)} = 13 & X_2^{(N)} = 11 & X_3^{(N)} = 19 & X_4^{(N)} = 23 & & \\
 f_\theta(X^{(N)}, N-1) \downarrow & \uparrow P^{-1} & \uparrow P^{-1} & \uparrow P^{-1} & \uparrow P^{-1} & & \\
 & X_1^{(N-1)} & X_2^{(N-1)} & X_3^{(N-1)} & X_4^{(N-1)} & & \\
 f_\theta(X^{(N-1)}, N-2) \downarrow & \uparrow P^{-1} & \uparrow P^{-1} & \uparrow P^{-1} & \uparrow P^{-1} & & \\
 \vdots & & & & & & \\
 f_\theta(X^{(3)}, 2) \downarrow & \uparrow P^{-1} & \uparrow P^{-1} & \uparrow P^{-1} & \uparrow P^{-1} & & \\
 & X_1^{(2)} & X_2^{(2)} & X_3^{(2)} & X_4^{(2)} & & \\
 f_\theta(X^{(2)}, 1) \downarrow & \uparrow P^{-1} & \uparrow P^{-1} & \uparrow P^{-1} & \uparrow P^{-1} & & \\
 \stackrel{(d)}{\approx} & X_1^{(1)} = 3 & X_2^{(1)} = 2 & X_3^{(1)} = 4 & X_4^{(1)} = 5 & & \\
 & \pi_1^{(1)} & \pi_2^{(2)} & \pi_2^{(3)} & \pi_4^{(4)} & &
 \end{array}$$

7 Estimation Theory – Expectation Review

Markov Inequality

If X is a *non-negative random variable*, then for any $\alpha > 0$

$$\Pr[X \geq \alpha \mathbb{E}[X]] \leq \frac{1}{\alpha}.$$

Chebyshev Inequality

Let X be a variable with mean μ and variance σ^2 . For any $\alpha > 0$,

$$\Pr\left[\underbrace{|X - \mathbb{E}[X]| \geq \alpha\sigma}_{\text{Probability of } X \text{ deviating from } \mu}\right] \leq \frac{1}{\alpha^2}.$$

Moment Generating Function (MGF)

For random variable X , the *moment generating function* is

$$M_X(t) = \mathbb{E}[e^{tX}].$$

The reason why it is called moment generating function is the following.

$$\begin{aligned}
 \mathbb{E}[X] &= 1^{\text{st}} \text{ moment} \\
 \mathbb{E}[X^2] &= 2^{\text{nd}} \text{ moment} \\
 \mathbb{E}[X^3] &= 3^{\text{rd}} \text{ moment} \\
 &\vdots \\
 \frac{\partial}{\partial t} M_X(t) &= \frac{\partial}{\partial t} \mathbb{E}[e^{tX}] = \mathbb{E}\left[\frac{\partial}{\partial t} e^{tX}\right] = \mathbb{E}[X e^{tX}]
 \end{aligned}$$

which implies $M'_X(0) = \mathbb{E}[X]$. Also,

$$\frac{\partial^2}{\partial t^2} M_X(t) = \mathbb{E}\left[\frac{\partial^2}{\partial t^2} e^{tX}\right] = \mathbb{E}[X \cdot X \cdot e^{tX}] = \mathbb{E}[X^2 e^{tX}]$$

which implies $M''_X(0) = \mathbb{E}[X^2]$.

***Note** that MGF completely characterises X .

$$X \text{ is a random variable with } \begin{cases} p_X & \text{p.m.f} \\ f_X & \text{p.d.f} \end{cases}$$

$$\begin{aligned}
p_X = p_Y &\Rightarrow X \stackrel{(d)}{=} Y \\
&\text{Same distribution} \\
f_X = f_Y &\Rightarrow X \stackrel{(d)}{=} Y \\
M_X(t) = M_Y(t) \quad \forall t &\Rightarrow X \stackrel{(d)}{=} Y
\end{aligned}$$

Properties of MGF

The moment generating function of Gaussian random variable $X \sim \mathcal{N}(\mu, \sigma^2)$ is given by

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad M_X(t) = \mathbb{E}[e^{tX}] = \int_{-\infty}^{\infty} e^{tX} \cdot f_X(x) dx = \exp\left(\frac{t^2\sigma^2}{2} + t\mu\right).$$

For two random variables X and Y , if $M_X(t) = M_Y(t)$ for all t , then X and Y have the same distribution. More precisely, they have the same cumulative distribution

$$F_X(x) = F_Y(x)$$

for all x .

For independent random variable X and Y , let $Z = X + Y$, then

$$M_Z(t) = \mathbb{E}[e^{t(X+Y)}] = \mathbb{E}[e^{tX} \cdot e^{tY}] = \mathbb{E}[e^{tX}] \mathbb{E}[e^{tY}] = M_X(t) M_Y(t)$$

For independent random variable $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ and $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$, let $Z = X + Y$, then

$$\begin{aligned}
M_Z(t) &= M_X(t) M_Y(t) = \exp\left(\frac{t^2\sigma_X^2}{2} + t\mu_X\right) \exp\left(\frac{t^2\sigma_Y^2}{2} + t\mu_Y\right) \\
&= \exp\left(\frac{t^2(\sigma_X^2 + \sigma_Y^2)}{2} + t(\mu_X + \mu_Y)\right) \leftarrow \text{MGF of } Z \sim \mathcal{N}(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2) \\
&\therefore Z \sim \mathcal{N}(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)
\end{aligned}$$

Central Limit Theorem

We can prove (non-rigorously) the Central Limit Theorem (CLT) using MGF. Let $X_1, \dots$ be i.i.d. random process with mean μ and variance σ^2 . Then, the CLT tells you

$$Z_n = \sum_{i=1}^n \frac{X_i - \mu}{\sqrt{n}\sigma} \xrightarrow{(d)} \mathcal{N}(0, 1).$$

Consider $Y_i = \frac{X_i - \mu}{\sqrt{n}\sigma}$, then we have

$$\mathbb{E}[Y_i] = \mathbb{E}\left[\frac{X_i - \mu}{\sqrt{n}\sigma}\right] = 0, \quad \text{Var}[Y_i] = \text{Var}\left[\frac{X_i - \mu}{\sqrt{n}\sigma}\right] = \frac{\sigma^2}{n\sigma^2} = \frac{1}{n}, \quad \mathbb{E}[Y_i^2] = \text{Var}[Y_i] + (\mathbb{E}[Y_i])^2 = \frac{1}{n}$$

This implies $M_Y'(0) = 0$, $M_Y''(0) = \frac{1}{n}$

By applying the Taylor approximation

$$M_Y(t) = \mathbb{E}[e^{tY}] = \mathbb{E}\left[1 + tY + \frac{t^2}{2!}Y^2 + \frac{t^3}{3!}Y^3 + \dots\right] \approx \mathbb{E}\left[1 + tY + \frac{t^2}{2}Y^2\right]$$

$$= 1 + t\mathbb{E}[Y] + \frac{t^2}{2}\mathbb{E}[Y^2] = 1 + \frac{t^2}{2n}$$

$$M_{Z_n}(t) = M_{Y_1}(t) M_{Y_2}(t) \dots M_{Y_n}(t) = (M_Y(t))^n = \left(1 + \frac{t^2}{2n}\right)^n$$

$$\lim_{n \rightarrow \infty} M_{Z_n}(t) = \lim_{n \rightarrow \infty} \left(1 + \frac{t^2}{2n}\right)^n = \lim_{n \rightarrow \infty} \left(1 + \frac{t^2}{2n}\right)^{\frac{2n}{t^2} \times \frac{t^2}{2}} = e^{\frac{t^2}{2}}.$$

This makes the MGF of $Z_n \xrightarrow[n \rightarrow \infty]{(d)} \mathcal{N}(0, 1)$ as $e^{t^2/2}$

Chernoff Inequality

Let X be a variable with moment-generating function $M_X(t) = \mathbb{E}[e^{tX}]$.

For any $\alpha > 0$,

$$\Pr[X \geq \alpha] \leq M_X(t) \cdot e^{-t\alpha}$$

for all $t > 0$.

Proof

$$\Pr[X \geq \alpha] = \Pr[tX \geq t\alpha] = \underbrace{\Pr[e^{tX} \geq e^{t\alpha}]}_{e^{tX} \text{ is non-negative}} \leq \frac{\mathbb{E}[e^{tX}]}{e^{t\alpha}} = M_X(t) \cdot e^{-t\alpha}$$

Law of Large Numbers (weak version)

Let $X_1, \dots, X_n$ be i.i.d. process with mean μ and variance σ^2 . The *weak law of large numbers* (LLN) tells you that

$$\frac{1}{n} \sum_{i=1}^n X_i \rightarrow \mu$$

$$\lim_{n \rightarrow \infty} \Pr\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - \mu\right| > \epsilon\right) \rightarrow 0, \quad \forall \epsilon > 0$$

when the convergence is in probability (it also converges with probability 1 in strong LLN).

Let $Z_n = \frac{1}{n} \sum_{i=1}^n X_i$, then the Z_n is a random variable with mean 0 and variance σ^2/n .

Then

$$\mathbb{E}[Z_n] = \mu, \quad \text{Var}(Z_n) = \frac{1}{n^2} (\text{Var}(X_1) + \dots + \text{Var}(X_n)) = \frac{1}{n^2} \cdot n \cdot \sigma^2 = \frac{\sigma^2}{n}.$$

By applying the Chebyshev Inequality

$$\Pr\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - \mu\right| > \epsilon\right) = \Pr\left(\left|Z_n - \mathbb{E}[Z_n]\right| > \epsilon\right) = \Pr\left(\left|Z_n - \mathbb{E}[Z_n]\right| > \frac{\sigma}{\sqrt{n}} \cdot \frac{\epsilon}{\sigma/\sqrt{n}}\right) \leq \frac{\sigma^2}{n\epsilon^2}$$

which converges to 0.

Conditional Expectation

For two random variables X and Y , the *conditional expectation* of X given $Y = y$ is

$$\mathbb{E}[X|Y = y] = \sum_x x \overbrace{p_{X|Y}(x|y)}^{Y=y} = \int x f_{X|Y}(x|y) dx.$$

Conditional expectation is an expectation when the distribution is conditional $p_{X|Y}(\cdot|y)$ or $f_{X|Y}(\cdot|y)$.

Thus, when we say $\mathbb{E}[X|Y]$ is a function Y , and also random (since Y is random).

Tower Property

$$\mathbb{E}[\mathbb{E}[X|Y]] = \mathbb{E}[X].$$

$$\mathbb{E}_Y[\mathbb{E}_{X|Y}[X|Y]] = \mathbb{E}_X[X].$$

8 Estimation

Let X be a "sample" (input), and Y be the label (output). Assume that $p_{X|Y}(x|y)$ or $f_{X|Y}(x|y)$ is known. Then, the question is to estimate y given $X = x$.

Example 01. Nationality Classification

Suppose we have two nationalities, A and B. The height distribution for individuals from nationality A follows a normal distribution $\mathcal{N}(170, 10^2)$, while for nationality B, it follows $\mathcal{N}(180, 15^2)$. In this scenario, we define Y as a binary label, where $Y = 0$ corresponds to nationality A and $Y = 1$ corresponds

to nationality B, and X represents the individual's height.

We are interested in a classification task where the goal is to predict the nationality based on the height.

For example, given a height of $X = 176$, how can we estimate the corresponding Y ?

8.1 Maximum Likelihood Estimation (MLE)

The maximum likelihood estimator finds $Y = y$ that maximises the likelihood function, i.e.,

$$\arg \max_y f_{X|Y}(x|y).$$

Specifically, it asks "on which y , the given $\{X = x\}$ is the most likely." In **Example 01**, we pick $y = 0$ if $f_{X|Y}(x|0) > f_{X|Y}(x|1)$. In other words,

$$\frac{1}{\sqrt{2\pi}10^2} \exp\left(-\frac{(x-170)^2}{2 \cdot 10^2}\right) > \frac{1}{\sqrt{2\pi}15^2} \exp\left(-\frac{(x-180)^2}{2 \cdot 15^2}\right) \Rightarrow f_{X|Y}(176|0) > f_{X|Y}(176|1) \Rightarrow \hat{y} = 0$$

8.2 Maximum A Posteriori (MAP) Estimation

Maximum a posteriori (MAP) estimator finds $Y = y$ that maximises the posterior probability $p_{Y|X}(y|x)$, i.e.,

$$\arg \max_y f_{Y|X}(y|x)$$

It asks "given $\{X = x\}$, which y is the most probable." Note that MAP requires the knowledge of $p_Y(y)$ for all y . If the Y is a continuous random variable, replace the pmf with $f_{Y|X}(y|x)$.

Applying the Bayes' rule, we have

$$f_{Y|X}(y|x) = \frac{f_{X|Y}(x|y)p_Y(y)}{f_X(x)}.$$

Since $f_X(x)$ is constant (not a function of y), MAP rule is simply

$$\arg \max_y \frac{f_{X|Y}(x|y)p_Y(y)}{f_X(x)} \Rightarrow \arg \max_y f_{X|Y}(x|y) \underbrace{p_Y(y)}_{\text{Prior}}.$$

Thus, if $p_Y(y)$ is uniform, MAP estimation is equivalent to MLE.

In **Example 01**, suppose the population from A is double i.e., $p_Y(0) = \frac{9}{10}$ and $p_Y(1) = \frac{1}{10}$. Then, MAP estimator pick $y = 0$ if

$$\frac{1}{\sqrt{2\pi} \cdot 10^2} \exp\left(-\frac{(x-170)^2}{2 \cdot 10^2}\right) \times \frac{9}{10} > \frac{1}{\sqrt{2\pi} \cdot 15^2} \exp\left(-\frac{(x-180)^2}{2 \cdot 15^2}\right) \times \frac{1}{10}.$$

Bayes Optimal

The MAP is *Bayes optimal*. In other words, it *minimises the probability of error*.

If the goal of the estimation is to minimise the probability of error $\Pr[\hat{Y} \neq Y]$, then

$$\hat{Y}_{MAP}(X) = \arg \max_y p_{X|Y}(x|y)$$

is the minimum achiever.

8.3 Fano's Inequality

Let $Y \in \mathcal{Y}$ be a true label and X be an input. Let's say like an input 28×28 image in the MNIST dataset. We want to estimate $\hat{Y}(X)$ as a function of X where we focus on hard classification namely $\hat{Y}(X)$ which is also discrete $\hat{Y}(X) \in \mathcal{Y}$. Then, we are interested in the probability of error $\Pr[Y \neq \hat{Y}(X)]$.

The *probability of error is bounded by*

$$\Pr[Y \neq \hat{Y}(X)] \geq \frac{H(Y|X) - 1}{\log |\mathcal{Y}|} = \min \Pr[\hat{Y} \neq Y(X)]$$

Proof

Let E be a binary random variable where $E = 1$ if $\hat{Y}(X) \neq Y$ and $E = 0$ if $\hat{Y}(X) = Y$. Then,

$$H(E, Y | \hat{Y}) = H(Y | \hat{Y}) + H(E | Y, \hat{Y}) = H(Y | \hat{Y}) \geq H(Y | X).$$

where the last inequality is due to data processing inequality since $\hat{Y}$ is a function of X .

$$I(X; Y) \geq I(\hat{Y}(X); Y), \quad H(Y) - H(Y | X) \geq H(Y) - H(Y | \hat{Y}(X)) \Leftrightarrow H(Y | X) \leq H(Y | \hat{Y}(X))$$

we have $H(Y | X) \leq H(E, Y | \hat{Y}(X))$

On the other hand,

$$\begin{aligned} H(E, Y | \hat{Y}) &= H(E | \hat{Y}) + H(Y | E, \hat{Y}) \\ &\leq 1 + p_E(1)H(Y | E = 1, \hat{Y}) + p_E(0)H(Y | E = 0, \hat{Y}) = 1 + p_E(1)H(Y | E = 1, \hat{Y}) \leq 1 + p_E(1) \log |\mathcal{Y}|. \end{aligned}$$

Since $p_E(1) = \Pr[Y \neq \hat{Y}(X)]$, we got

$$H(Y | X) \leq 1 + \Pr[Y \neq \hat{Y}(X)] \log |\mathcal{Y}|,$$

which concludes the proof.

Fano's inequality tells us that the probability of error cannot be small if $Y | X$ has too much randomness *i.e.*,

$H(Y | X)$ denotes the “randomness” of $Y | X$.

8.4 Parameter Estimation

Suppose $X_1, \dots, X_n$ is i.i.d. process with p_X is parameterised by θ . We are interested in estimating θ given samples $x_1, \dots, x_n$. Without assumption on distribution of θ , we can employ Maximum Likelihood Estimator. In other words, we find θ that maximises $p_{X^n | \theta}(x^n | \theta)$. Since maximising the likelihood is equivalent to maximise the log-likelihood, we often maximise

$$\sum_{i=1}^n \log p_{X_i | \theta}(x_i | \theta).$$

Example 02. COVID Test

Consider a random variable Y , the task to detect the COVID-19 disease *i.e.* what we want to know is:

$$Y = \begin{cases} 0 & \text{Not Infected} \\ 1 & \text{COVID} \end{cases}$$

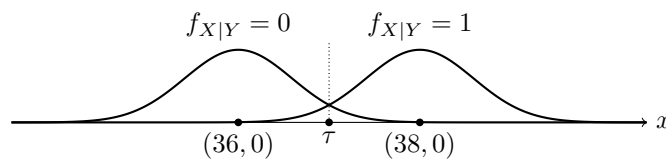
Let X denotes the body temperature of a patient. The goal is to estimate Y based on X

Assumption 1

We know that $f_{X|Y} = 0$, $f_{X|Y} = 1$.

Assume they're following a gaussian distribution, that is

$$X | Y = 0 \sim \mathcal{N}(36, 1^2), \quad X | Y = 1 \sim \mathcal{N}(38, 1^2).$$

**1. Maximum Likelihood (MLE)**

$$\begin{array}{cc} f_{X|Y=0}(x|0) & \text{vs} \quad f_{X|Y=1}(x|1) \\ f(36.5|0) & \text{vs} \quad f(36.5|1) \end{array}$$

False Positive, $Y = 0 \Rightarrow \hat{Y} = 1 \Rightarrow \text{Okay} \uparrow$
 False Negative, $Y = 1 \Rightarrow \hat{Y} = 0 \Rightarrow \text{Danger} \downarrow$

$$\text{if with } \begin{cases} \tau = 37.0 \Rightarrow \begin{cases} \text{False Positive} & 5\% \\ \text{False Negative} & 3\% \end{cases} \\ \tau = 36.5 \Rightarrow \begin{cases} \text{False Positive} & 5\% \\ \text{False Negative} & 1\% \end{cases} \end{cases}$$

2. Maximum A Posteriori (MAP)

$$\begin{aligned} p_Y(Y=0|x=x) &= \frac{f_{X|Y}(x|0) \cdot p_Y(0)}{f_X(x)} \quad \text{vs} \quad p_Y(Y=1|X=x) \\ &= \frac{f_{X|Y}(x|1)p_Y(1)}{f_X(x)} \end{aligned}$$

Consider there are total 1000000 patients with probability $p_Y(0) = 99.9\%$ and $p_Y(1) = 0.1\%$

1000000	Y	$\hat{Y}_{MLE}$	Negative Diagnosis	Positive Diagnosis
999000	0	0	30	969030
1000	1	1	970	29970

Table 1: COVID-19 Diagnosis with constraint $\tau = 37.0$

Considering this, $\Pr(Y=0|\hat{Y}=1) = \frac{29970}{29970+970}$

Example 03. Classification Task

Let X an input image of the MNIST dataset, and Y is the label.

Assume that $f_{X|Y}$ is known,

The parameter estimation denotes that $X_1, X_2, \dots, X_n$ are i.i.d. and p_X is parameterised by $\theta(\text{target})$.

Example 04. Coin Toss

Suppose $X_1, \dots, X_n$ is i.i.d. Bernoulli distribution of probability $p_X(1) = p$, like a coin toss which the variable denotes the number of "Heads". We can say that the distribution is parameterised by $\theta = p$.

Note that the Bernoulli distribution can be represented by $p_X(x) = p^x(1-p)^{1-x}$ since it returns p for $X = 1$ and $1-p$ for $X = 0$. Then,

$$\log p_{X^n}(x^n) = \sum_{i=1}^n \log p^{X_i}(1-p)^{1-X_i} = \sum_{i=1}^n X_i \log p + \sum_{i=1}^n (1-X_i) \log(1-p).$$

If we take the derivative with respect to p , we have

$$\frac{1}{p} \sum_{i=1}^n X_i = \frac{1}{1-p} \sum_{i=1}^n (1-X_i)$$

This implies $\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i$

Example 05. Continuous Task

This time suppose $X_1, \dots, X_n$ is i.i.d. Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$ We can say that the distribution is parameterised by $\theta = (\mu, \sigma^2)$. Similarly consider the log-likelihood,

$$\log f_{X^n}(x^n) = \sum_{i=1}^n -\frac{(x_i - \mu)^2}{2\sigma^2} - \frac{1}{2} \log 2\pi\sigma^2.$$

If we take the derivative with respect to μ , then

$$\sum_{i=1}^n 2 \frac{x_i - \mu}{2\sigma^2} = 0$$

which implies $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i$. Also, if we take the derivative with respect to $t = \sigma^2$, then

$$\sum_{i=1}^n \frac{(x_i - \mu)^2}{2t^2} = \frac{n}{2t}$$

Thus,

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2.$$

Remark

About Maximum A Posteriori (MAP), it can be interpreted as follows

$$\max_{\theta} f(\theta|x^n) = \frac{\overbrace{p_{X^n|\theta}(x^n|\theta)}^{\text{Known}} \overbrace{f_{\theta}(\theta)}^{\text{Assumption}}}{\underbrace{p_{X^n}(x^n)}_{\text{Known}} \quad \text{Also recognised}}$$

8.5 Naive Bayes

Overview

- X^n : observation
- Y : feature (unknown)
- Problem: $X^n|Y$ is not i.i.d.
- Naive Bayes: treat $X^n|Y$ as i.i.d.

Naive Bayes

Suppose we have $X^n = (X_1, \dots, X_n)$ is the input (image) and Y be the corresponding label. Classification is simply estimating the label Y based on the input X^n . We assume the probabilistic model $P_{X^n|Y}$ where X^n is conditionally independent given Y .

Example 06. MNIST dataset

Consider X^n be the binarised MNIST image, i.e., it is 7×7 image where each pixel is either 0 or 1. Then, the Naive Bayes assumption implies that the pixel values are independent given the label.

Case Study 01. SPAM Mail detection

Consider X^n be the collection of words, where $X_i = 1$ if the mail contains the i -th word in the dictionary (and 0 otherwise). Then, the Naive Bayes assumption is simply the occurrence of each word given $Y = 1$ (the mail is SPAM) is independent.

$$\begin{matrix} X \\ (X_1, X_2, X_3, \dots, X_n) \end{matrix} \quad \begin{matrix} Y \\ \text{Not SPAM} / \text{SPAM}, \\ 0 \quad 1 \end{matrix} \quad \hat{Y}(X^n) = \text{MLE of } \max P(X^n|Y)$$

Consider the following example about indicating SPAM whether the word "dog" is contained or not:

$$p_{X_i|Y=1}(37|1) = \Pr[X_i = \overset{\text{dog}}{37}|Y = 1] = \Pr[\text{From } i\text{-th word, the } 37^{\text{th}} \text{ word is "dog"} | \text{mail} = \text{SPAM}]$$

As the name suggests, Naive Bayes assumption is over-simplified probabilistic model, but it works reasonably in simple cases. The assumption implies

$$\log p_{X^n|Y}(x^n|y) = \sum_{i=1}^n \log p_{X_i|Y}(x_i|y), \quad p_{X^n|Y}(x^n|y) \cong \prod_{i=1}^n p_{X_i|Y}(x_i|y).$$

The MLE is

$$\hat{y}_{\text{MLE}} = \arg \max_y \sum_{i=1}^n \log p_{X_i|Y}(x_i|y).$$

In SPAM filter, each $p_{X_i|Y}(x_i|y)$ are Bernoulli distribution with parameter $p_{i|y}$. If we have samples $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(N)}, y^{(N)})\}$ where $x^{(i)}$ is the i -th mail and $y^{(i)}$ is the corresponding label, then we can estimate each parameter by

$$p_{i|y=0} = \frac{\sum_{j=1}^N \mathbf{1}\{x_i^{(j)} = 1 \text{ and } y^{(j)} = 0\}}{\sum_{j=1}^N \mathbf{1}\{y^{(j)} = 0\}}, \quad p_{i|y=1} = \frac{\sum_{j=1}^N \mathbf{1}\{x_i^{(j)} = 1 \text{ and } y^{(j)} = 1\}}{\sum_{j=1}^N \mathbf{1}\{y^{(j)} = 1\}}$$

Finally, we further apply MLE to estimate the label of input x^n .

Moreover, since Y is also a Bernoulli variable, we can approximate $p_y = \Pr[Y = 1]$ using MLE.

$$p_y = \frac{1}{N} \sum_{j=1}^N \mathbf{1}\{y^{(j)} = 1\}.$$

Then, we can apply MAP to estimate the label of input x^n .

$$\hat{y}_{\text{MAP}} = \arg \max_y p_{Y|X^n}(y|x^n).$$

where

$$p_{Y|X^n}(1|x^n) = \frac{p_{X^n|Y}(x^n|1)p_Y(1)}{p_{X^n|Y}(x^n|0)p_Y(0) + p_{X^n|Y}(x^n|1)p_Y(1)} = \frac{\prod_{i=1}^n p_{X_i|Y}(x_i|1)p_Y(1)}{\prod_{i=1}^n p_{X_i|Y}(x_i|0)p_Y(0) + \prod_{i=1}^n p_{X_i|Y}(x_i|1)p_Y(1)}$$

where the probability $p_{X_i|Y}(x_i|0)$ and $p_{X_i|Y}(x_i|1)$ is estimated via MLE.

However, in the MLE above for $p_{i|y}$, some rare word x_i may never appear in the sample, resulting in $p_{i|y=0} = p_{i|y=1} = 0$. To avoid such situation, we can use Laplace smoothing, which replace MLE with

$$p_{i|y=0} = \frac{1 + \sum_{j=1}^N \mathbf{1}\{x_i^{(j)} = 1 \text{ and } y^{(j)} = 0\}}{2 + \sum_{j=1}^N \mathbf{1}\{y^{(j)} = 0\}}, \quad p_{i|y=1} = \frac{1 + \sum_{j=1}^N \mathbf{1}\{x_i^{(j)} = 1 \text{ and } y^{(j)} = 1\}}{2 + \sum_{j=1}^N \mathbf{1}\{y^{(j)} = 1\}}$$

You can think of it as you count the occurrence of each word starting from 1 (instead of 0).

8.6 Bias-Variance

Given a sample $X_1, \dots, X_n$, consider an estimator

$$\hat{\theta}_n \equiv \hat{\theta}(X_1, \dots, X_n)$$

of a real-valued parameter $\theta \in \Omega \subset \mathbb{R}$.

The distribution of any estimator of course depends on the true parameter vector θ , i.e., more precisely,

$$\hat{\theta}_n \equiv \hat{\theta}(X_1, \dots, X_n; \theta).$$

The bias of an estimator is defined by

$$\text{Bias}(\hat{\theta}_n) = \overbrace{\mathbb{E}[\hat{\theta}_n]}^{\text{w.r.t. } p_{X_1 \dots X_n}} - \overbrace{\theta}^{(p_\theta)}.$$

Considering the bias of the MLE

$$\text{Bias}(\hat{\theta}_{\text{MLE}}) = \mathbb{E}\left[\frac{X_1 + \dots + X_n}{n}\right] - \theta = \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{p_\theta}[X_i] - \theta$$

An estimator is called *unbiased* if $\mathbb{E}[\hat{\theta}_n] = \theta$, and hence

$$\text{Bias}(\hat{\theta}_n) = 0$$

for all possible θ .

The variance of an estimator is given by

$$\text{Var}[\hat{\theta}_n] = \mathbb{E}\left((\hat{\theta}_n - \mathbb{E}(\hat{\theta}_n))^2\right).$$

***Note** that low variance (= reliability) is desirable

Performance of an estimator is most frequently evaluated with respect to the mean squared error loss (also called L_2 loss):

$$(\hat{\theta}_n - \theta)^2.$$

$$\mathbb{E}[(\hat{\theta}_n - \theta)^2] = \mathbb{E}[(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n] + \mathbb{E}[\hat{\theta}_n] - \theta)^2] = \mathbb{E}[(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n])^2] + \mathbb{E}[(\mathbb{E}[\hat{\theta}_n] - \theta)^2] + 2 \cdot \mathbb{E}[(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n])(\mathbb{E}[\hat{\theta}_n] - \theta)]$$

Since $X_1, \dots, X_n \sim p_\theta$ and $\hat{\theta}_n = \hat{\theta}(X_1, \dots, X_n)$ are random while $\mathbb{E}[\hat{\theta}_n], \theta$ are deterministic:

$$= \text{Var}[\hat{\theta}_n] + \underbrace{(\mathbb{E}[\hat{\theta}_n] - \theta)^2}_{=\text{Bias}(\hat{\theta}_n)^2} + 2 \underbrace{\mathbb{E}[(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n])]}_{=\mathbb{E}[\hat{\theta}_n] - \mathbb{E}[\hat{\theta}_n] = 0} \times \underbrace{(\mathbb{E}[\hat{\theta}_n] - \theta)}_{=\text{is not random}}$$

The corresponding risk is the Mean Squared Error (MSE):

$$\text{MSE}(\hat{\theta}_n) = \mathbb{E}((\hat{\theta}_n - \theta)^2) = \text{Bias}(\hat{\theta}_n)^2 + \text{Var}[\hat{\theta}_n].$$

For an unbiased estimator, the mean squared error is obviously equal to the variance of the estimator.

Assume an i.i.d. sample $X_1, \dots, X_n$ with mean $\mu = \mathbb{E}[X_i]$ and variance $\sigma^2 = \text{Var}[X_i] < \infty$, i.e.,

$$X_1, \dots, X_n \sim \mathcal{N}(\overset{\theta}{\downarrow} \mu, \overset{\text{Known}}{\downarrow} \sigma^2)$$

The sample mean $\hat{X}$ is an unbiased estimator of the true mean μ , since the equation

$$\hat{\mu}_{\text{MLE}} = \frac{X_1 + X_2 + \dots + X_n}{n} \Rightarrow \mathbb{E}[\hat{\mu}_{\text{MLE}}] = \mu \Rightarrow \mathbb{E}(\hat{X}) = \mu$$

holds for any possible value of the true mean μ .

The variance of the estimator $\hat{X}$ is given by

$$\text{Var}[\hat{\mu}_{\text{MLE}}] = \text{Var}[\hat{X}] = \text{Var}\left[\frac{X_1 + \dots + X_n}{n}\right] = \frac{1}{n^2} \sum_{i=1}^n \text{Var}[X_i] = \frac{\sigma^2}{n}.$$

The mean squared error of the estimator $\hat{X}$ is given by

$$\text{MSE}(\hat{X}) = \mathbb{E}[(\hat{\mu}_{\text{MLE}} - \mu)^2] = \mathbb{E}[(\hat{X} - \mu)^2] = \text{Bias}(\hat{\mu}_n)^2 + \text{Var}(\hat{\mu}_n) = 0^2 + \text{Var}(\hat{\mu}_n) = \text{Var}[\hat{X}] = \frac{\sigma^2}{n}.$$

8.7 Minimum Mean Squared Error Estimation

Let X be the source (target) and Y be the noisy measurement (observation) where we want to estimate X (sample) given Y (label). Suppose we want to find the Minimum Mean Squared Estimation $\hat{X}(Y)$ as a function of Y that minimises

$$\text{MSE}(\hat{X}(Y)) = \mathbb{E}\left[\left(\overset{\text{Random}}{\downarrow} \hat{X} - \hat{X}(Y)\right)^2\right].$$

Lemma

Conditional expectation is Bayes' optimal. In other words, $\hat{X}(Y) = \mathbb{E}[X|Y]$ minimises the MSE.

Proof

$$\begin{aligned}\mathbb{E}[(X - \hat{X}(Y))^2] &= \mathbb{E}[(X - \mathbb{E}[X|Y] + \mathbb{E}[X|Y] - \hat{X}(Y))^2] \\ &= \mathbb{E}[(X - \mathbb{E}[X|Y])^2] + \mathbb{E}[2(X - \mathbb{E}[X|Y])(\mathbb{E}[X|Y] - \hat{X}(Y))] + \mathbb{E}[(\mathbb{E}[X|Y] - \hat{X}(Y))^2]\end{aligned}$$

Consider the second term. Due to the tower property, we have

$$\begin{aligned}\mathbb{E}[2(X - \mathbb{E}[X|Y])(\mathbb{E}[X|Y] - \hat{X}(Y))] &= \mathbb{E}_Y[2\mathbb{E}_X[(X - \mathbb{E}[X|Y])(\mathbb{E}[X|Y] - \hat{X}(Y))|Y]] \\ &= \mathbb{E}_Y[2(\mathbb{E}[X|Y] - \hat{X}(Y))\mathbb{E}_X[(X - \mathbb{E}[X|Y])|Y]] = 0.\end{aligned}$$

The last inequality is due to $\mathbb{E}[X - \mathbb{E}[X|Y]|Y] = \mathbb{E}[X|Y] - \mathbb{E}[X|Y] = 0$. Thus,

$$\mathbb{E}[(X - \hat{X}(Y))^2] = \mathbb{E}[(X - \mathbb{E}[X|Y])^2] + \mathbb{E}[(\mathbb{E}[X|Y] - \hat{X}(Y))^2] \geq \mathbb{E}[(X - \mathbb{E}[X|Y])^2].$$

Which implies $\hat{X}(Y) = \mathbb{E}[X|Y]$ is the MMSE.

$$\therefore \mathbb{E}[(X - \hat{X}(Y))^2] \geq \mathbb{E}[(X - \underbrace{\mathbb{E}[X|Y]}_{\text{Bayes optimal}})^2]$$

9 Multivariate Gaussian

9.1 2D-Gaussian

Let X_1, X_2 be a *jointly Gaussian* with mean $\boldsymbol{\mu} = (\mu_1, \mu_2)$ and covariance matrix Σ

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}, \quad \sigma_1, \sigma_2 > 0, \quad -1 \leq \rho \leq 1.$$

Here, σ_1^2 denotes $\text{Var}(X_1)$, σ_2^2 represents $\text{Var}(X_2)$, and $\rho\sigma_1\sigma_2$ corresponds to $\text{Cov}(X_1, X_2)$.

We say $(X_1, X_2) \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ and the corresponding joint probability density function (pdf)

$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{\sqrt{(2\pi)^2 \det(\Sigma)}} \exp\left(-\frac{1}{2}(x^2 - \boldsymbol{\mu})\Sigma^{-1}(x^2 - \boldsymbol{\mu})\right) = \frac{1}{\sqrt{(2\pi)^2 |\Sigma|}} \exp\left(-\frac{1}{2} \begin{bmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \end{bmatrix}^\top \Sigma^{-1} \begin{bmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \end{bmatrix}\right)$$

where $x^2 = (x_1, x_2)$.

If $(X_1, X_2) \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, then $X_1|X_2$ is also Gaussian with mean $\mu_1 + \frac{\sigma_{12}}{\sigma_{22}}(x_2 - \mu_2)$ and variance $\sigma_{22} - \frac{\sigma_{12}^2}{\sigma_{22}}$ where

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix}$$

$$X_1|X_2 = x_2 \sim \mathcal{N}\left(\mu_1 + \rho \frac{\sigma_1}{\sigma_2}(x_2 - \mu_2), \sigma_1^2 - \rho^2 \sigma_1^2\right)$$

9.2 Positive Definite Matrices

Overview

$$\arg \min_x (ax^2 + bx + c), \quad a > 0, \quad x^* = -\frac{b}{2a}$$

$$\Rightarrow \arg \min_{\mathbb{X}} \mathbb{X}^T A \mathbb{X} + b\mathbb{X} + c \Leftrightarrow ax_1^2 + bx_2^2 + cx_1x_2 + dx_1 + ex_2 + f \leftarrow \text{quadratic } x_1, x_2$$

Positive Definite Matrices

Let A be a *real symmetric* $n \times n$ matrix

1. A is *positive definite*, denoted $A > 0$ if

$$x^T A x > 0, \quad \forall x \in \mathbb{R}^n, x \neq 0.$$

2. A is nonnegative definite (or *positive semi definite*), denoted $A \geq 0$, if

$$x^T A x \geq 0, \quad \forall x \in \mathbb{R}^n.$$

It is known from basic linear algebra that

1. A symmetric matrix has n real eigenvalues.
2. If $A > 0$, all eigenvalues are positive.
3. If $A \geq 0$, all eigenvalues are nonnegative.
4. Eigenvectors corresponding to distinct eigenvalues are orthogonal, and can be chosen to form an orthonormal basis of $\mathbb{R}^n$.

Every symmetric matrix A can be written as

$$A = \sum_{i=1}^n \lambda_i z_i z_i^T,$$

where λ_i are eigenvalues and z_i are orthonormal eigenvectors (which is called spectral decomposition).

For $A \geq 0$, all $\lambda_i \geq 0$, allowing us to define the symmetric square root

$$B = \sum_{i=1}^n \sqrt{\lambda_i} z_i z_i^T, \quad \text{so that} \quad B^2 = A.$$

If $A > 0$, then all $\lambda_i > 0$, and we can define the inverse as

$$A^{-1} = \sum_{i=1}^n \frac{1}{\lambda_i} z_i z_i^T.$$

For positive semidefinite matrix A , there exists a matrix B such that $A = B^T B$.

Proof

If $A = B^T B$, then for all v , we have

$$v^T A v = \|Bv\|^2 \geq 0.$$

On the other hand, let $A = Q \Lambda Q^T$ be the eigen decomposition of matrix A .

Since all its eigenvalues are nonnegative, we can let

$$B = Q \sqrt{A}$$

where $\sqrt{A}$ is a diagonal matrix with all its entries are square root of eigenvalues $\sqrt{\lambda_i}$'s.

9.3 Multivariate Gaussian

Our interest in positive definite matrices stems from their role in defining multivariate normal densities.

When $A > 0$, the quadratic form $q(x) = x^T A x$ grows unbounded as $\|x\| \rightarrow \infty$, and $e^{-q(x)}$ decays to zero, thus enabling valid probability densities.

There are three equivalent ways to define the multivariate normal distribution.

1. Density definition

A random vector $\mathbb{X} \in \mathbb{R}^n$ has a (nondegenerate) multivariate normal distribution if its joint p.d.f is

$$f_{\mathbb{X}}(\mathbb{X}) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \exp \left(-\frac{1}{2} (\mathbb{X} - \mu)^T \Sigma^{-1} (\mathbb{X} - \mu) \right),$$

for some mean vector $\mu \in \mathbb{R}^n$ and positive definite covariance matrix Σ .

2. Constructive definition

A random vector $\mathbb{X}$ is multivariate normal if

$$\mathbb{X} = D\mathbb{W} + \mu,$$

where $\mathbb{W}$ is a vector of independent standard random variables $\mathcal{N}(0, 1)$, and D is a real matrix.

3. Projection definition

A random vector $\mathbb{X}$ is multivariate normal if, for every real vector $a \in \mathbb{R}$, the scalar random variable $a^T \mathbb{X}$ is (univariate) normal.

$$a^T \mathbb{X} = a_1 X_1 + a_2 X_2 + \cdots + a_n X_n, \quad (X_1 \perp\!\!\!\perp X_2 \sim \mathcal{N} \Rightarrow X_1 + X_2 \sim \mathcal{N})$$

Degeneracy

The **density definition** requires a PDF, which only exists if Σ is positive definite. When Σ is singular (e.g., $X_2 = 0$ always), the distribution is called *degenerate*, as all mass is concentrated on a lower-dimensional subspace. **Constructive definition** and **projection definition** remain valid for degenerate normals.

9.4 Means and Covariances of Vector Random Variables

Let $X = (X_1, \dots, X_n)^T$ be a random vector. Its mean vector is

$$\mathbb{E}[X] = \begin{bmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_n] \end{bmatrix}.$$

Given random vectors $X \in \mathbb{R}^n$ and $Y \in \mathbb{R}^m$, the *covariance matrix* is

$$\begin{aligned} \text{Cov}(X, Y) &= \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])^T]. \\ \text{Cov} \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix} &\Rightarrow \begin{pmatrix} \text{Cov}(X_1, X_1) & \cdots & \text{Cov}(X_1, X_n) \\ \vdots & \ddots & \vdots \\ \text{Cov}(X_n, 1) & \cdots & \text{Cov}(X_n, X_n) \end{pmatrix} \end{aligned}$$

In particular, $\text{Cov}(X, X)$ is an $n \times n$ symmetric matrix.

$$\begin{aligned} \text{Cov} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} &\Rightarrow \begin{pmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Cov}(X_2, X_2) \end{pmatrix} = \mathbb{E} \begin{bmatrix} (X_1 - \mathbb{E}[X_1])(X_1 - \mathbb{E}[X_1]) & (X_1 - \mathbb{E}[X_1])(X_2 - \mathbb{E}[X_2]) \\ (X_2 - \mathbb{E}[X_2])(X_1 - \mathbb{E}[X_1]) & (X_2 - \mathbb{E}[X_2])(X_2 - \mathbb{E}[X_2]) \end{bmatrix} \\ &= \mathbb{E} \begin{bmatrix} (\mathbb{X} - \mathbb{E}[\mathbb{X}])_{n \times 1} (\mathbb{X} - \mathbb{E}[\mathbb{X}])_{1 \times n}^T \end{bmatrix} \end{aligned}$$

***Note** that $\text{Cov}(X, X)$ is nonnegative definite.

Proof

$$\begin{aligned} v^T \Sigma v &\geq 0 \Rightarrow \Sigma \geq 0, \\ v^T \mathbb{E}[(\mathbb{X} - \mathbb{E}[\mathbb{X}]) (\mathbb{X} - \mathbb{E}[\mathbb{X}])^T] v &= \mathbb{E} \left[\underbrace{v^T (\mathbb{X} - \mathbb{E}[\mathbb{X}]) (\mathbb{X} - \mathbb{E}[\mathbb{X}])^T v}_{((\mathbb{X} - \mathbb{E}[\mathbb{X}])^T v)^T} \right] = \mathbb{E} [||(\mathbb{X} - \mathbb{E}[\mathbb{X}])^T v||^2] \geq 0. \end{aligned}$$

Correlation Coefficient

For scalar random variables X_i and X_j , the (*Pearson*) *correlation coefficient* is defined as

$$\rho_{ij} = \frac{\text{Cov}(X_i, X_j)}{\sqrt{\text{Var}(X_i) \text{Var}(X_j)}}.$$

- $-1 \leq \rho_{12} \leq 1$
- $\rho_{12} = 1, \quad X_2 = a \cdot X_1 \quad (a > 0)$
- $\rho_{12} = -1 \quad X_2 = a \cdot X_1 \quad (a < 0)$
- $\rho_{12} = 0 : X_1 \& X_2$ uncorrelated.

$$X_1 \perp\!\!\!\perp X_2 \Rightarrow \rho_{12} = 0$$

The correlation coefficient ρ_{ij} satisfies $-1 \leq \rho_{ij} \leq 1$, and provides a normalised measure of the linear dependence between X_i and X_j . If $\rho_{ij} = 0$, then we say random variables X_i and X_j are uncorrelated.

9.5 Multivariate Normal

Properties of multivariate normal distributions

Let $\mathbb{X} = D\mathbb{W} + \mu$, where $\mathbb{W}$ has independent $\mathcal{N}(0, 1)$ components and $\mathbb{X}, \mu \in \mathbb{R}^n$. Then:

1. Each component X_i is univariate normal with mean μ_i .

Proof

Under the **projection definition**, each component X_i is a linear function of independent normal random variables, and hence it is itself normal. Since $\mathbb{E}[\mathbb{W}] = 0$, we have

$$\mathbb{E}[X_i] = \mathbb{E}[e_i^T (D\mathbb{W} + \mu)] = e_i^T \mu = \mu_i.$$

$$\mathbb{W} = \begin{bmatrix} W_1 \\ \vdots \\ W_n \end{bmatrix} \sim \text{i.i.d. } \mathcal{N}(0, 1), \quad D = \begin{bmatrix} -d_1^T & - \\ -d_2^T & - \\ \vdots & \\ -d_n^T & - \end{bmatrix} = \begin{bmatrix} d_1^T \mathbb{W} + \mu_1 \\ \vdots \\ d_n^T \mathbb{W} + \mu_n \end{bmatrix} \Rightarrow D\mathbb{W} + \mu$$

$$X_i = d_i^T \mathbb{W} + \mu_i = \underbrace{d_{i1} W_1}_{\sim \mathcal{N}(0, d_{i1}^2)} + \underbrace{d_{i2} W_2}_{\sim \mathcal{N}(0, d_{i2}^2)} + \cdots + d_{in} W_n + \mu_i \Rightarrow \mathcal{N}(\mu_i, d_{i1}^2 + \cdots + d_{in}^2) = \mathcal{N}(\mu_i, \|d_i\|^2)$$

2. $\text{Cov}(X, X) = DD^T$.

Proof

For simplicity, consider the zero-mean case ($\mu = 0$). Then

$$\text{Cov}(X, X) = \mathbb{E}[\mathbb{X}\mathbb{X}^T] = \mathbb{E}[D\mathbb{W} \cdot \mathbb{W}^T D^T] = D \cdot \underbrace{\mathbb{E}[\mathbb{W}\mathbb{W}^T]}_{\sum_{\mathbb{W}}} D^T = DD^T,$$

since the components of $\mathbb{W}$ are independent with unit variance, i.e., $\mathbb{E}[\mathbb{W}\mathbb{W}^T] = I$.

3. If $\mathbb{Y} = C\mathbb{X} + d$, where C is an $m \times n$ matrix and $d \in \mathbb{R}^m$, then

$$\mathbb{Y} \sim \mathcal{N}(C\mu + d, CDD^T C^T).$$

Proof

$$\text{Let } \mathbb{Y} = \begin{matrix} \mathbb{R}^m \\ \mathbb{R}^{m \times n} \end{matrix} C \begin{matrix} \mathbb{R}^n \\ \mathbb{R}^{n \times n} \end{matrix} \mathbb{X} + \begin{matrix} \mathbb{R}^m \\ \mathbb{R}^{m \times 1} \end{matrix} d = C \cdot (D \cdot \mathbb{W} + \mu) + d = (CD) \cdot \mathbb{W} + (C\mu + d)$$

Then $\mathbb{Y}$ is a linear function of independent $\mathcal{N}(0, 1)$ variables, hence multivariate normal. We have

$$\mathbb{E}[\mathbb{Y}] = C\mu + d, \quad \text{Cov}(\mathbb{Y}, \mathbb{Y}) = C\text{Cov}(\mathbb{X}, \mathbb{X})C^T = CDD^T C^T.$$

$\Rightarrow$ if $\mathbb{X}$ is jointly Gaussian $\Rightarrow \mathbb{Y} = C\mathbb{X} + d$ is also jointly Gaussian.

4. If D is invertible, then X is nondegenerate with covariacne $V = DD^T$, and has the density in the **Property 2** (i.e., $\mathbb{X} = D\mathbb{W} + \mu$)

Proof

if D is invertible

$$f_X(\mathbb{X}) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \exp \left(-\frac{1}{2} (\mathbb{X} - \mu)^T \Sigma^{-1} (\mathbb{X} - \mu) \right)$$

This part follows from a standard result on transformations of random variables. For simplicity, take $\mu = 0$. When $\mathbb{X} = D\mathbb{W}$ with D invertible, the change of variables formula gives

$$f_X(x) = \frac{f_W(D^{-1}x)}{|\det D|}.$$

Since the components of W are i.i.d. $\mathcal{N}(0, 1)$, we have

$$f_W(w) = \frac{1}{(2\pi)^{n/2}} \exp \left(-\frac{1}{2} w^T w \right).$$

Substituting $w = D^{-1}x$, we obtain

$$f_X(x) = \frac{1}{(2\pi)^{n/2} |DD^T|^{1/2}} \exp \left(-\frac{1}{2} x^T (D^{-1})^T D^{-1} x \right).$$

This is exactly form in the **Property 2** with $\Sigma = DD^T$, then $\text{Cov}(X, X) = \Sigma$. The same argument holds for $\mu \neq 0$ by replacing x with $(x - \mu)$.

5. The joint c.d.f. of X is completely determined by its mean μ and covariance Σ .

Proof

From **Property 4**, the joint p.d.f. of X is determined by μ and $\Sigma = DD^T$. In the degenerate case (when Σ is singular), there is no joint p.d.f, but the distribution is still determined by μ and Σ . To see this, consider the moment generating function (MGF)

$$M_X(s) = \mathbb{E}[e^{s^T X}].$$

Since $s^T X$ is a scalar normal variable with mean $s^T \mu$ and variance $s^T \Sigma s$, we have

$$M_X(s) = \exp \left(s^T \mu + \frac{1}{2} s^T \Sigma s \right).$$

By the inversion theorem for transforms, the distribution of X is completely determined by μ and Σ .

6. The components of X are uncorrelated if and only if they are independent.

Proof

$$(X_i, X_j), \mu_i, \mu_j, \sigma_i^2, \sigma_j^2, \rho_{ij} = 0$$

$$\begin{aligned} f_{X_i, X_j}(x_i, x_j) &= \frac{1}{\sqrt{(2\pi)^2 |\Sigma|}} \exp \left(-\frac{1}{2} \begin{bmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \end{bmatrix}^T \begin{bmatrix} 1/\sigma_1^2 & 0 \\ 0 & 1/\sigma_2^2 \end{bmatrix} \begin{bmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \end{bmatrix} \right) \\ &= \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp \left(-\frac{1}{2} \frac{(x_1 - \mu_1)^2}{\sigma_1^2} \right) \cdot \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp \left(-\frac{1}{2} \frac{(x_2 - \mu_2)^2}{\sigma_2^2} \right) \Rightarrow X_i \perp\!\!\!\perp X_j \end{aligned}$$

If the components of X are independent, then they are necessarily uncorrelated. For the converse, assume the components are uncorrelated so that Σ is diagonal. Let Y be a random vector whose components are independent normal variables with the same means and variances as X . Then X and Y have identical means and covariances; hence by **Property 5**, they have the same distribution. Therefore, the components of X are also independent.

Alternatively, when Σ is invertible and diagonal, the joint density factorises as

$$f_X(x) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right) = \prod_{i=1}^n f_{X_i}(x_i),$$

confirming independence.

7. If

$$\begin{bmatrix} X \\ Y \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix}, \begin{bmatrix} V_{XX} & V_{XY} \\ V_{YX} & V_{YY} \end{bmatrix} \right), \quad V_{YY} > 0,$$

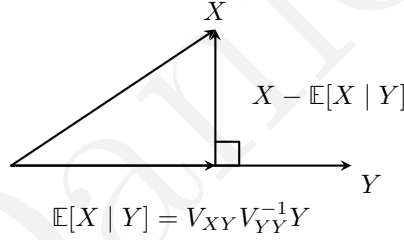
then:

- (a) $\mathbb{E}[X|Y] = \mu_X + V_{XY} V_{YY}^{-1} (Y - \mu_Y)$.
- (b) Let $\tilde{X} = X - \mathbb{E}[X|Y]$. Then $\tilde{X}$ is independent of Y and of $\mathbb{E}[X|Y]$.
- (c) $\text{Cov}(\tilde{X}, \tilde{X}) = V_{XX} - V_{XX} V_{YY}^{-1} V_{YX}$.

Proof

$$\begin{matrix} n \times 1 \\ k \times 1 \end{matrix} \sim \mathcal{N} \left(\begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix}, \begin{bmatrix} V_{XX} & V_{XY} \\ V_{YX} & V_{YY} \end{bmatrix} \right) \Rightarrow \mathbb{E} \left[\left(\begin{bmatrix} X \\ Y \end{bmatrix} - \begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix} \right) \left(\begin{bmatrix} X \\ Y \end{bmatrix} - \begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix} \right)^T \right]^{k \times k}$$

$$\begin{aligned}
\mathbb{E}[X|Y] &= \mu_X + V_{XY}V_{YY}^{-1}(Y - \mu_Y), \quad \hat{X} = \mu_X + V_{XY}V_{YY}^{-1}(Y - \mu_Y) \\
\mathbb{E}[(\hat{X} - \mu_X)(Y - \mu_Y)^T] &= V_{XY}V_{YY}^{-1} \underbrace{[(Y - \mu_Y)(Y - \mu_Y)^T]}_{V_{YY}} = V_{XY} = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)^T] \\
&\Rightarrow \mathbb{E} \left[\underbrace{(X - \hat{X})}_{\text{Jointly Gaussian}} \underbrace{(Y - \mu_Y)^T}_{\text{Gaussian}} \right] = 0 \\
\begin{bmatrix} X - \hat{X} \\ Y - \mu_Y \end{bmatrix} &= \begin{bmatrix} I & -V_{XY}V_{YY}^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} X \\ Y - \mu_Y \end{bmatrix} + \begin{bmatrix} \mu_X \\ 0 \end{bmatrix} \\
\mathbb{E}[X - \hat{X}] &= \mathbb{E}[X - \mu_X - V_{XY}V_{YY}^{-1}(Y - \mu_Y)] = V_{XY}V_{YY}^{-1}\mathbb{E}[Y - \mu_Y] = 0 \\
\mathbb{E}[(X - \hat{X})(Y - \mu_Y)^T] &= 0 \Rightarrow (X - \hat{X}) \& Y \text{ are uncorrelated} \Rightarrow \text{independent} \\
\mathbb{E}[(X - \hat{X}) \underbrace{g(Y)}_{\text{For any } g}] &= \mathbb{E}[X - \hat{X}]\mathbb{E}[g(Y)] = 0 \\
\mathbb{E}_Y[\mathbb{E}[(X - \hat{X})g(Y)|Y]] &= \mathbb{E}_Y[\underbrace{\mathbb{E}[X - \hat{X}|Y]}_{=0} \cdot g(Y)] = 0 \\
\mathbb{E}[X - \hat{X}|Y] &= 0, \quad \mathbb{E}[X|Y] = \mathbb{E}[\hat{X}|Y] = \mathbb{E}[\mu_X + V_{XY}V_{YY}^{-1}(Y - \mu_Y)|Y] = \underbrace{\mu_X + V_{XY}V_{YY}^{-1}(Y - \mu_Y)}_{\hat{X}} \\
X - \hat{X} &\perp\!\!\!\perp Y, \quad X - \mathbb{E}[X|Y] \perp\!\!\!\perp Y, \quad X \perp\!\!\!\perp Y \Leftrightarrow \langle X, Y \rangle = 0 \\
\text{For zero mean random variables } X \& Y, \quad &\mathbb{E}[XY] = \langle XY \rangle
\end{aligned}$$



$\mathbb{E} \left[\underbrace{X}_{\text{What we want to guess}} \mid \underbrace{Y}_{\text{What we know}} \right] \rightarrow \text{Best guess: Conditional expectation } (X|Y) \text{ partition of } Y$

$$\begin{aligned}
\text{Cov}(X|Y) &= \mathbb{E}[(X - \mathbb{E}[X|Y])(X - \mathbb{E}[X|Y])^T|Y] = \mathbb{E}[(X - \mathbb{E}[X|Y])(X - \mathbb{E}[X])^T] \\
&= \mathbb{E}[\underbrace{(X - \mu_X)}_{\tilde{X}} - V_{XY}V_{YY}^{-1}\underbrace{(Y - \mu_Y)}_{\tilde{Y}} \underbrace{(X - \mu_X)}_{\tilde{X}} - V_{XY}V_{YY}^{-1}\underbrace{(Y - \mu_Y)}_{\tilde{Y}}]^T \\
&\rightarrow \mathbb{E}[\tilde{X}\tilde{X}^T] - \mathbb{E}[V_{XY}V_{YY}^{-1}\tilde{Y} \cdot \tilde{X}] - \mathbb{E}[\tilde{X}\tilde{Y}^T V_{YY}^{-1}V_{YX}] + \mathbb{E}[V_{XX}V_{YY}^{-1}\tilde{Y}\tilde{Y}^T V_{YY}^{-1}V_{YX}] \\
&= V_{XX} - V_{XY}V_{YY}^{-1}V_{YX} - V_{XY}V_{YY}^{-1}V_{YX} + V_{XX}V_{YY}^{-1}V_{YX} \\
\mathbb{E}[X|Y] &= \mu_X + V_{XY}V_{YY}^{-1}(Y - \mu_Y)
\end{aligned}$$

$$\text{Cov}(X|Y) = V_{XX} - V_{XY}V_{YY}^{-1}V_{YX} \Rightarrow X|Y \sim \mathcal{N}(\mu_X + V_{XY}V_{YY}^{-1}(Y - \mu_Y), V_{XX} - V_{XY}V_{YY}^{-1}V_{YX})$$

For simplicity, assume zero mean. Define the random vector

$$\hat{X} = V_{XY}V_{YY}^{-1}Y.$$

Then

$$\mathbb{E}[XY^T] = V_{XY} = \mathbb{E}[\hat{X}Y^T],$$

implying $\mathbb{E}[(X - \hat{X})Y^T] = 0$, i.e., $X - \hat{X}$ is uncorrelated with Y . Since (X, Y) is jointly normal, $(X - \hat{X}, Y)$ is also multivariate normal, and uncorrelatedness implies independence. Therefore, $X - \hat{X}$ is independent of Y .

Now, for any measurable function g ,

$$\mathbb{E}[(X - \hat{X})g(Y)] = 0,$$

which by the definition of conditional expectation implies $\hat{X} = \mathbb{E}[X|Y]$.

Thus,

$$\mathbb{E}[X|Y] = V_{XY}V_Y^{-1}Y.$$

For part (b), define $\tilde{X} = X - \mathbb{E}[X|Y]$. Since $\tilde{X}$ is independent of Y , it is also independent of $\mathbb{E}[X|Y]$.

For part (c),

$$\begin{aligned} \text{Cov}(\tilde{X}, \tilde{X}) &= \mathbb{E}[\tilde{X}\tilde{X}^T] = \mathbb{E}[(X - \hat{X})(X - \hat{X})^T] \\ &= \mathbb{E}[XX^T] - \mathbb{E}[X\hat{X}^T] - \mathbb{E}[\hat{X}X^T] + \mathbb{E}[\hat{X}\hat{X}^T] = V_{XX} - V_{XY}V_Y^{-1}V_{YX}. \end{aligned}$$

This covariance is also equal to $\text{Cov}(\tilde{X}, \tilde{X}|Y)$ since $\tilde{X}$ and Y are independent.

9.6 Gaussian Channel

Let $X \sim \mathcal{N}(0, \sigma_x^2)$ and $Z \sim \mathcal{N}(0, \sigma_z^2)$ be independent random variables, and observe

$$X \perp\!\!\!\perp Z \Rightarrow Y = X + Z \sim \mathcal{N}(0, \sigma_x^2 + \sigma_z^2)$$

Then (X, Y) is jointly Gaussian with

$$\text{Var}(X) = \sigma_x^2, \quad \text{Var}(Y) = \sigma_x^2 + \sigma_z^2, \quad \text{Cov}(X, Y) = \sigma_x^2.$$

Because (X, Y) is jointly Gaussian,

$$X|Y = y \sim \mathcal{N}(\mu_{X|Y}(y), \sigma_{X|Y}^2),$$

where

$$\begin{aligned} \mu_{X|Y}(y) &= \frac{\text{Cov}(X, Y)}{\text{Var}(Y)}y = \frac{\sigma_x^2}{\sigma_x^2 + \sigma_z^2}y, \quad \sigma_{X|Y}^2 = \sigma_x^2 - \frac{\text{Cov}(X, Y)^2}{\text{Var}(Y)} = \frac{\sigma_x^2\sigma_z^2}{\sigma_x^2 + \sigma_z^2}. \\ X &\stackrel{(d)}{=} \underbrace{\frac{\sigma_x^2}{\sigma_x^2 + \sigma_z^2}Y}_{\text{Mean}} + \underbrace{\sqrt{\frac{\sigma_x^2\sigma_z^2}{\sigma_x^2 + \sigma_z^2}}W}_{\text{Variance}} \Rightarrow W \sim \mathcal{N}(0, 1), \quad (Y \perp\!\!\!\perp W) \end{aligned}$$

Hence, the MMSE estimator is the posterior mean:

$$\hat{X}_{\text{MMSE}}(Y) = \mathbb{E}[X|Y] = \frac{\sigma_x^2}{\sigma_x^2 + \sigma_z^2}Y.$$

Note that the following are two equivalent sampling procedures for (X, Y) .

1. **Forward (prior + noise):** Sample $X \sim \mathcal{N}(0, \sigma_x^2)$ and $Z \sim \mathcal{N}(0, \sigma_z^2)$ independently, then set

$$\overset{\text{Sample}}{X}, \overset{\text{Sample}}{Z} \Rightarrow Y = X + Z.$$

2. **Observation-first (posterior decomposition):** First sample

$$Y \sim \mathcal{N}(0, \sigma_x^2 + \sigma_z^2),$$

then, conditional on Y , sample

$$X = \frac{\sigma_x^2}{\sigma_x^2 + \sigma_z^2}Y + \sqrt{\frac{\sigma_x^2\sigma_z^2}{\sigma_x^2 + \sigma_z^2}}W, \quad W \sim \mathcal{N}(0, 1) \quad \text{independent of } Y.$$

***Note** that there is no statistical difference between two types of generations $Y = X + Z$ and

$$X = \frac{\sigma_x^2}{\sigma_x^2 + \sigma_z^2}Y + \sqrt{\frac{\sigma_x^2\sigma_z^2}{\sigma_x^2 + \sigma_z^2}}W$$

Equivalently, $X|Y$ can be viewed as adding independent Gaussian noise with variance $\frac{\sigma_x^2\sigma_z^2}{\sigma_x^2 + \sigma_z^2}$ to the MMSE estimate $\hat{X}_{\text{MMSE}}(Y)$.

We can do the simple sanity check. As $\sigma_z^2 \rightarrow 0$, we have $\mu_{X|Y} \rightarrow Y$ and $\sigma_{X|Y}^2 \rightarrow 0$. As $\sigma_z^2 \rightarrow \infty$, we have $\mu_{X|Y} \rightarrow 0$ and $\sigma_{X|Y}^2 \rightarrow \sigma_x^2$.

9.7 Gaussian Diffusion

Overview

$$p_{X,Y}, \boxed{f_{X,Y}} = \left. \begin{matrix} f_X \cdot f_{Y|X} \\ f_Y \cdot f_{X|Y} \end{matrix} \right\} \text{Main ingredients of diffusion model}$$

Let X a random variable which suffices the following constraints below $\dots (\star)$

$$X \perp\!\!\!\perp Z, \quad Z \sim \mathcal{N}(0, \sigma_z^2), \quad Y = X + Z$$

Here the random variable Y has no guarantee to follow a gaussian distribution. However, assume that $Y|X \sim \mathcal{N}(X, \sigma_z^2)$ i.e., $Y|X$ follows a gaussian distribution.

Note that $X|Y$ does **not** follow a gaussian distribution, but we can assume it approximately follows gaussian if the variance of z is extremely small. i.e.,

$$X|Y \underset{\text{approximately}}{\sim} \mathcal{N}(\mu_{X|Y}, \sigma_{X|Y}^2) \quad \text{if } \sigma_z^2 \ll 1$$

This can be proved due to the field called "*Differential Privacy*".

Suppose we can estimate $\mu_{X|Y}, \sigma_{X|Y}^2$, then we can sample Y first (instead of sampling X then Y). Then, we compute $\mu_{X|Y}$ and $\sigma_{X|Y}^2$.

***Note** that the derived output $X = \mu_{X|Y} + \sigma_{X|Y} \cdot W$ (where $W \sim \mathcal{N}(0, 1)$) is distributionally similar to X we assigned in $(\star)$.

Gaussian Diffusion

Let X a random variable where the distribution is unknown f_X , ... think of X like a "*realistic image*". The goal of the task is to sample $X \sim f_X$. However, we don't know the photorealistic distribution of f_X . In that case, the only way to sample is to estimate as follows.

$$X \xrightarrow{+Z} Y \Rightarrow \left(\begin{matrix} X_1 \rightarrow Y_1 \\ X_2 \rightarrow Y_2 \\ \vdots \\ X_N \rightarrow \underbrace{Y_N}_{\text{Noise}} \end{matrix} \right) \xrightarrow{\text{Estimate}} \mu_{X|Y}, \sigma_{X|Y}^2$$

Here, $\mu_{X|Y}$ and $\sigma_{X|Y}^2$ can be learned via some ML algorithm. Once $\mu_{X|Y}$ and $\sigma_{X|Y}^2$ are learned, we can now sample Y via computing $X = \mu_{X|Y} + \sigma_{X|Y} \cdot W$ where $W \sim \mathcal{N}(0, 1)$. As mentioned previously we can assume that the computed X is statistically similar to X that we assigned at the initial point. In other words, if X is the target, then it would be the photorealistic image. However, the problem still arises since still the distribution of Y is unknown (*=infeasible*). To solve this, we can build a Markov process of X where $X_k \sim \mathcal{N}(0, 1)$ considering the noise is very small.

$$\begin{array}{ccccccc} \xrightarrow{\text{Small Noise}} & \longrightarrow & \longrightarrow & \longrightarrow & \longrightarrow & \longrightarrow & \\ X_0 & \text{---} & X_1 & \text{---} & X_2 & \text{---} & X_3 & \text{---} & \dots & \longrightarrow & X_k \\ (=X) & & & & & & & & & & \\ \xleftarrow{\mathcal{N}(\mu_{0|1}, \sigma_{0|1}^2)} & \xleftarrow{\mathcal{N}(\mu_{1|2}, \sigma_{1|2}^2)} & \xleftarrow{\mathcal{N}(\mu_{2|3}, \sigma_{2|3}^2)} & \dots & \xleftarrow{\mathcal{N}(\mu_{k-1|k}, \sigma_{k-1|k}^2)} & & & & & & \end{array}$$

where $(X_{n-1} \perp\!\!\!\perp Z_n)$.

Let $\{X_n\}_{n \geq 0}$ be a Gaussian process defined by the recursion

$$X_n = \sqrt{1 - \beta} X_{n-1} + \sqrt{\beta} Z_n, \quad Z_n \sim \mathcal{N}(0, 1),$$

where each Z_n is an independent standard normal noise. Assume $\beta \ll 1$. Equivalently,

$$X_n|X_{n-1} \sim \mathcal{N}(\sqrt{1 - \beta} X_{n-1}, \beta).$$

Suppose the initial state $X_0 \sim f_0$ has mean 0 and variance 1. Then the first and second moments of $\{X_n\}$ satisfy

$$\begin{aligned}\mathbb{E}[X_n] &= \sqrt{1-\beta}\mathbb{E}[X_{n-1}] + \sqrt{\beta}\mathbb{E}[Z_n] = 0, \\ \text{Var}[X_n] &= (1-\beta)\text{Var}(X_{n-1}) + \beta\text{Var}(Z_n) = 1,\end{aligned}$$

for all $n \geq 1$. Hence, $\{X_n\}$ remains zero-mean and unit-variance for all steps.

Now consider the limiting distribution of X_n as $n \rightarrow \infty$.

Let $M_{X_n}(t) = \mathbb{E}[e^{tX_n}]$ denote the moment generating function (MGF) of X_n . Then

$$\begin{aligned}M_{X_n}(t) &= M_{X_{n-1}}(\sqrt{1-\beta}t)e^{\frac{\beta t^2}{2}} \\ &= M_{X_{n-2}}((\sqrt{1-\beta})^2 t)e^{\frac{(1-(1-\beta))t^2}{2}} = \dots = M_{X_0}((\sqrt{1-\beta})^n t) \exp\left(\frac{1-(1-\beta)^n}{2}t^2\right).\end{aligned}$$

As $n \rightarrow \infty$, $(\sqrt{1-\beta})^n \rightarrow 0$, so

$$M_{X_n}(t) \rightarrow e^{t^2/2},$$

which is the MGF of $\mathcal{N}(0, 1)$. Hence,

$$X_n \xrightarrow{d} \mathcal{N}(0, 1),$$

i.e., the process converges to the standard normal distribution regardless of the initial f_0 .

When β is small, the reverse conditional distribution can be locally approximated by a Gaussian:

$$X_{n-1}|X_n \approx \mathcal{N}(\mu_{n-1|n}, \sigma_{n-1|n}^2),$$

where the parameters $(\mu_{n-1|n}, \sigma_{n-1|n}^2)$ can be derived from the joint Gaussian form of (X_{n-1}, X_n) .

Let $\{\mathbf{X}_n\}_{n \geq 0}$ be a sequence of random vectors in $\mathbb{R}^d$ defined by

$$\mathbf{X}_n = \sqrt{1-\beta}\mathbf{X}_{n-1} + \sqrt{\beta}\mathbf{Z}_n, \quad \mathbf{Z}_n \sim \mathcal{N}(0, \mathbf{I}_d),$$

with $\mathbf{X}_0 \sim f_0(\mathbf{x})$ having mean $\mathbf{0}$ and covariance $\mathbf{I}_d$. Then

$$\mathbf{X}_n|\mathbf{X}_{n-1} \sim \mathcal{N}(\sqrt{1-\beta}\mathbf{X}_{n-1}, \beta\mathbf{I}_d),$$

and the marginal statistics remain stationary:

$$\mathbb{E}[\mathbf{X}_n] = \mathbf{0}, \quad \text{Cov}(\mathbf{X}_n) = \mathbf{I}_d.$$

The limiting distribution satisfies

$$\mathbf{X}_n \xrightarrow{d} \mathcal{N}(0, \mathbf{I}_d), \quad \text{as } n \rightarrow \infty.$$

Now let's consider the multidimensional case, which includes the most practical scenarios such as image, audio, and video generations. As in the scalar case, when β is small, the reverse conditional can be approximated by a Gaussian:

$$\mathbf{X}_{n-1}|\mathbf{X}_n \approx \mathcal{N}(\boldsymbol{\mu}_{n-1|n}, \boldsymbol{\Sigma}_{n-1|n}),$$

where $\boldsymbol{\mu}_{n-1|n}, \boldsymbol{\Sigma}_{n-1|n}$ can be approximated via Neural Networks. This Gaussian approximation underlies the reverse diffusion formulation used in diffusion probabilistic models. That is, each backward step can be obtained by sampling according to the above reverse process.

Based on this forward process and (approximated) reverse process, now we are ready to generate a data similar to the data distribution.

1. Initialise the terminal state from the stationary (noise) distribution (for large enough T):

$$\mathbf{X}_T \sim \mathcal{N}(0, \mathbf{I}_d).$$

2. For $n = T, T-1, \dots, 1$, sample $\mathbf{X}_{n-1}$ according to

$$\mathbf{X}_{n-1}|\mathbf{X}_n \approx \mathcal{N}(\boldsymbol{\mu}_{n-1|n}, \boldsymbol{\Sigma}_{n-1|n}),$$

3. The resulting $\mathbf{X}_0$ is a sample from the reversed process, which is (approximately) equivalent to sample according to the data distribution.

To sum up, we learn $\boldsymbol{\mu}_{n-1|n}$ and $\boldsymbol{\Sigma}_{n-1|n}$ using "diffusion model" and $X_0 = \boldsymbol{\mu}_{0|1} + W_1$ where $W_1 \sim \mathcal{N}(0, \boldsymbol{\Sigma}_{0|1})$ behaves similar to the $X \sim f_X$. Note that we still don't know the exact f_X , but we know to approximate to sample $X_0 \sim f_X$. As a result, though we do not directly parameterise the distribution, but know the way how to sample.

10 Gaussian Discriminant

Overview

Consider the following random variables

$$A : \mathcal{N}(170, 10^2) \Rightarrow f_A$$

$$B : \mathcal{N}(180, 10^2) \Rightarrow f_B$$

Let's say the given variable x is the height of the person in the group

Note that the following statistics represent the following features.

$$\text{MLE: } f_A(x) \geq f_B(x)$$

$$\text{MAP: } \underbrace{\Pr(A)f_A(x)}_{\text{from } A} \geq \underbrace{\Pr(B)f_B(x)}_{\text{from } B}$$

We can find out that these statistics require a knowledge of f_A, f_B ($\Pr(A), \Pr(B)$)

The goal is to classify x with respect to A/B .

The problem is f_A, f_B ($\Pr(A), \Pr(B)$) are unknown.

To solve this, we assume the following conditions.

$$X|A \sim \mathcal{N}(\mu_0, \Sigma_0), \quad X|B \sim \mathcal{N}(\mu_1, \Sigma_1)$$

Then,

$$f_A(x) = \frac{1}{\sqrt{(2\pi)^n |\Sigma_0|}} \exp\left(-\frac{1}{2}(x-\mu_0)^\top \Sigma_0^{-1}(x-\mu_0)\right), \quad f_B(x) = \frac{1}{\sqrt{(2\pi)^n |\Sigma_1|}} \exp\left(-\frac{1}{2}(x-\mu_1)^\top \Sigma_1^{-1}(x-\mu_1)\right)$$

Thus, the solution is to estimate the following variables

$$\hat{\mu}_0, \hat{\mu}_1, \hat{\Sigma}_0, \hat{\Sigma}_1, \quad \underbrace{\Pr(A) = \hat{p}_0, \Pr(B) = \hat{p}_1}_{\text{If we want to apply MAP instead of MLE}}$$

Parameter Estimation of Gaussian (using MLE)

Given $X^{(1)}, X^{(2)}, \dots, X^{(N)} \sim f_A$ (from A)

The empirical mean is $\hat{\mu}_0 = \frac{1}{N} \sum_{i=1}^N X^{(i)}$

Then what is the MLE of mean? (*i.e.*, $\hat{\Sigma}_0 = ?$)

$$\log \prod_{i=1}^N f_A(X^{(i)}) = \sum_{i=1}^N -\frac{1}{2}(X^{(i)} - \boldsymbol{\mu})^\top \Sigma_0^{-1}(X^{(i)} - \boldsymbol{\mu}) - \frac{N}{2} \log(2\pi)^n |\Sigma|$$

$$f_A(x) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \exp\left(-\frac{1}{2}(x - \boldsymbol{\mu})^\top \Sigma_0^{-1}(x - \boldsymbol{\mu})\right)$$

The goal is to get Σ_0 that maximises $\log \prod_{i=1}^N f_A(X^{(i)})$

$$\Phi = \Sigma_0^{-1}$$

$$\log \prod_{i=1}^N f_A(X^{(i)}) = \sum_{i=1}^N \underbrace{-\frac{1}{2}(X^{(i)} - \mu)^\top \Phi (X^{(i)} - \mu)}_{\text{linear to } \Phi} - \frac{N}{2} \log |\Phi^{-1}|$$

$$\nabla_\Phi \log \prod_{i=1}^N f_A(X^{(i)}) = \sum_{i=1}^N \underbrace{-\frac{1}{2}(X^{(i)} - \mu)^\top (X^{(i)} - \mu)}_{(*)} + \frac{N}{2} (\Phi^{-1})^\top$$

$$(*) \Rightarrow \underbrace{(x - \mu)^\top}_{(1 \times n)} \underbrace{\Phi}_{(n \times n)} \underbrace{(x - \mu)}_{(n \times 1)} = \text{tr}((x - \mu)^\top \Phi (x - \mu)) = \text{tr}((x - \mu)(x - \mu)^\top \Phi) \quad (\because \text{tr}(ABC) = \text{tr}(CAB))$$

$$\nabla_\Phi (x - \mu)^\top \Phi (x - \mu) = \nabla_\Phi \text{tr}((x - \mu)(x - \mu)^\top \Phi) = (x - \mu)(x - \mu)^\top$$

$$\nabla_\Phi \log \prod_{i=1}^N f_y(X^{(i)}) = \sum_{i=1}^N -\frac{1}{2}(X^{(i)} - \mu)^\top (X^{(i)} - \mu) + \frac{N}{2} (\Phi^{-1})^\top$$

$$\Rightarrow (\Phi^{-1})^\top = \frac{1}{N} \sum_{i=1}^N (X^{(i)} - \mu)(X^{(i)} - \mu)^\top$$

$$\Rightarrow \Sigma_0^\top = \frac{1}{N} \sum_{i=1}^N (X^{(i)} - \mu)(X^{(i)} - \mu)^\top$$

$$\Rightarrow \Sigma_0 = \frac{1}{N} \sum_{i=1}^N (X^{(i)} - \mu)(X^{(i)} - \mu)^\top$$

$$\Rightarrow \hat{\Sigma}_0 = \frac{1}{N} \sum_{i=1}^N (X^{(i)} - \hat{\mu}_0)(X^{(i)} - \hat{\mu}_0)^\top$$

$$\Sigma = \mathbb{E}[(X - \mu)(X - \mu)^\top] \approx \frac{1}{N} \sum_{i=1}^N (X^{(i)} - \mu)(X^{(i)} - \mu)^\top$$

Again consider for $X^{(1)}, \dots, X^{(N)}$, $A \rightarrow \hat{\mu}_0, \hat{\Sigma}_0$ and $B \rightarrow \hat{\mu}_1, \hat{\Sigma}_1$

$$\longrightarrow \hat{\mu}_0 = \frac{1}{N} \sum_{i=1}^N X^{(i)}, \quad \hat{\Sigma}_0 = \frac{1}{N} (X^{(i)} - \hat{\mu}_0)(X^{(i)} - \hat{\mu}_0)^\top$$

$$\longrightarrow \hat{\mu}_1 = \frac{1}{N} \sum_{i=1}^N X^{(i)}, \quad \hat{\Sigma}_1 = \frac{1}{N} (X^{(i)} - \hat{\mu}_1)(X^{(i)} - \hat{\mu}_1)^\top$$

with new sample X

$$f_A(x) \geq f_B(x)$$

$$\hat{p}_0 f_A(x) \geq \hat{p}_1 f_B(x)$$

where, $\hat{p}_0$ denotes counting A and $\hat{p}_1$ denotes counting B .

Gaussian Discriminant

We consider binary classification with class-conditional multivariate Gaussians:

$$X|Y=0 \sim \mathcal{N}(\mu_0, \Sigma_0), \quad X|Y=1 \sim \mathcal{N}(\mu_1, \Sigma_1),$$

where $X \in \mathbb{R}^n$. The MLE decision rule (equivalently, MAP with equal priors) picks $y=0$ if

$$\frac{1}{\sqrt{(2\pi)^n |\Sigma_0|}} \exp\left(-\frac{1}{2}(x - \mu_0)^\top \Sigma_0^{-1}(x - \mu_0)\right) > \frac{1}{\sqrt{(2\pi)^n |\Sigma_1|}} \exp\left(-\frac{1}{2}(x - \mu_1)^\top \Sigma_1^{-1}(x - \mu_1)\right).$$

With non-equal priors $\pi_y = \mathbb{P}(Y = y)$, the MAP rule multiplies each side by π_0 and π_1 , respectively. Both MLE and MAP requires the knowledge of parameters μ and Σ 's, which requires the MLE parameter estimation. Given i.i.d. samples $\{x^{(i)}\}_{i=1}^N$ from a single Gaussian class $\mathcal{N}(\mu, \Sigma)$, the log-likelihood is

$$\ell(\mu, \Sigma) = \sum_{i=1}^N \log p(x^{(i)}; \mu, \Sigma) = -\frac{Nn}{2} \log(2\pi) - \frac{N}{2} \log |\Sigma| - \frac{1}{2} \sum_{i=1}^N (x^{(i)} - \mu)^\top \Sigma^{-1} (x^{(i)} - \mu).$$

Taking the gradient with respect to μ :

$$\nabla_{\mu} \ell = \sum_{i=1}^N \Sigma^{-1} (x^{(i)} - \mu) = \Sigma^{-1} \left(\sum_{i=1}^N x^{(i)} - N\mu \right).$$

Setting $\nabla_{\mu} \ell = 0$ yields the MLE

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N x^{(i)}.$$

For the covariance matrix estimation, it is convenient to reparameterise with the precision matrix $\Phi := \Sigma^{-1}$. Using $|\Sigma| = |\Phi^{-1}| = |\Phi|^{-1}$, we have

$$\ell(\mu, \Phi) = -\frac{Nn}{2} \log(2\pi) + \frac{N}{2} \log |\Phi| - \frac{1}{2} \sum_{i=1}^N (x^{(i)} - \mu)^\top \Phi (x^{(i)} - \mu).$$

Define the scatter matrix

$$S := \sum_{i=1}^N (x^{(i)} - \mu)(x^{(i)} - \mu)^\top.$$

Using the trace trick,

$$(x - \mu)^\top \Phi (x - \mu) = \text{tr}((x - \mu)(x - \mu)^\top \Phi),$$

we obtain the gradients (matrix identities recalled below):

$$\nabla_{\Phi} \ell = \frac{N}{2} \Phi^{-T} - \frac{1}{2} S.$$

Setting $\nabla_{\Phi} \ell = 0$ gives

$$\Phi^{-T} = \frac{1}{N} S \quad \Rightarrow \quad \hat{\Sigma} = \Phi^{-1} = \frac{1}{N} S = \frac{1}{N} \sum_{i=1}^N (x^{(i)} - \hat{\mu})(x^{(i)} - \hat{\mu})^\top.$$

Remark

This is the MLE (biased) covariance. The unbiased sample covariance replaces $\frac{1}{N}$ with $\frac{1}{(N-1)}$. In the above derivation, the following matrix-calculus identities were used. For matrices of compatible sizes:

$$\nabla_X \text{tr}(A^\top X) = A, \quad \nabla_X \text{tr}(AX) = A^\top,$$

$$\nabla_X \log |X| = X^{-T}, \quad (X \text{ invertible}).$$

Hence, for any fixed A , $\nabla_{\Phi} \text{tr}(A\Phi) = A^\top$ and $\nabla_{\Phi} \log |\Phi| = \Phi^{-T}$.

For a matrix A and a positive semidefinite matrix Φ , $\nabla_{\Phi} \text{tr}(A\Phi) = A^\top$.

Proof

Let $C = A\Phi$, then

$$\text{tr}(C) = \sum_{i=1}^d \sum_{j=1}^d A_{i,j} \Phi_{j,i}.$$

Thus,

$$\frac{\partial}{\partial \Phi_{i,j}} \text{tr}(C) = A_{j,i}.$$

$\therefore$ Thus, $\nabla_{\Phi} \text{tr}(A\Phi) = A^T$.

Remark

For an invertible (symmetric) positive definite Φ , $\nabla_{\Phi} \log |\Phi| = \Phi^{-T} = \Phi^{-1}$.

Proof

By the scalar chain rule,

$$\frac{\partial}{\partial X_{ij}} \log |X| = \frac{1}{|X|} \frac{\partial |X|}{\partial X_{ij}} = \frac{1}{|X|} C_{ij}(X).$$

Hence, in matrix form,

$$\nabla_X \log |X| = \frac{1}{|X|} \text{adj}(X)^{\top}.$$

For any invertible X , $\text{adj}(X) = |X|X^{-1}$. Therefore,

$$\nabla_X \log |X| = \frac{1}{|X|} (|X|X^{-1})^{\top} = X^{-T}.$$

Since X is symmetric positive definite, $X^{-T} = X^{-1}$, and we conclude

$$\boxed{\nabla_X \log |X| = X^{-1}}.$$

11 Review on Linear Algebra

11.1 Rank and Range

Let $A \in \mathbb{R}^{n \times m}$ be a $n \times m$ matrix,

$$A = \begin{bmatrix} a_1 & a_2 & \dots & a_m \end{bmatrix}$$

where $a_i \in \mathbb{R}^n$.

Range / Column Space

The *range* of A is a *column space*, the span of all its column vector.

Recall that the *span* is a set of all possible linear combination of given set of vectors.

$$\text{range}(A) = \text{span}(\{a_1, \dots, a_m\}) = \{x : x = \alpha_1 a_1 + \alpha_2 a_2 + \dots + \alpha_m a_m\} = \{x = \sum_{i=1}^m \alpha_i a_i : \alpha_i \in \mathbb{R}\} \in \mathbb{R}^n.$$

The *column space* is a vector subspace. The *dimension* of vector subspace is defined by the size of basis (the maximum number of linearly independent vector).

Rank

The *rank* of matrix A is a *dimension* of $\text{range}(A)$.

$$\text{rank}(A) = \dim(\text{range}(A)).$$

Note that the dimension of a row space is also the same as the rank.

Null Space

The *null space* of A is a set of vectors that satisfies $Ax = 0$, i.e.,

$$\text{null}(A) = \{x \in \mathbb{R}^m : Ax = 0\}.$$

Rank-Nullity

$$\text{rank}(A) + \dim(\text{null}(A)) = n.$$

Consider the following examples

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 0 & 0 \end{bmatrix} \implies \begin{cases} n = 3 \\ \text{rank}(A) = 2, & \dim(\text{null}(A)) = 1 \\ \text{rank}(B) = 1, & \dim(\text{null}(B)) = 2 \end{cases}$$

Orthogonal Complement

The *orthogonal complement* of set $W \subseteq \mathbb{R}^n$ is given by

$$W^\perp = \{v \in \mathbb{R}^n : v^T w = 0, \quad \forall w \in W\}$$

If W is a subspace of $\mathbb{R}^n$, then

$$\mathbb{R}^n = W \oplus W^\perp \iff \forall x \in \mathbb{R}^n, \exists w \in W, w^\perp \in W^\perp \text{ such that } x = w + w^\perp$$

For $A \in \mathbb{R}^{n \times m}$, the *orthogonal complement* of row space is a *null space* of A .

The *orthogonal complement* of a *column space* of A is a *null space* of A^T .

$$A \in \mathbb{R}^{n \times m}, \quad \text{null}(A^T) = \text{range}(A)^\perp$$

$$x \in \text{null}(A^T) \implies A^T x = 0 \leftarrow (m \times 1)$$

$$\begin{bmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_m^T \end{bmatrix} \begin{matrix} (m \times n) \\ (n \times 1) \end{matrix} x = \begin{bmatrix} a_1^T x \\ a_2^T x \\ \vdots \\ a_m^T x \end{bmatrix} = 0 \implies \begin{cases} \alpha_1 \cdot a_1^T x + \dots + \alpha_m \cdot a_m^T x = 0 \\ (\alpha_1 a_1 + \dots + \alpha_m a_m)^T \cdot x = 0 \\ x \in \text{range}(A)^\perp \end{cases}$$

11.2 Norm and Orthogonal Matrix

Norm

We say $\|\cdot\|$ is a norm on $\mathbb{R}^n$ if $\|x\| \in [0, \infty)$ for all $x \in \mathbb{R}^n$

$$\|\cdot\| : \mathbb{R}^n \rightarrow [0, \infty)$$

$$\text{if } \begin{cases} \|x\| = 0 \iff x = 0 \\ \|\alpha \cdot x\| = |\alpha| \cdot \|x\|, & \alpha \in \mathbb{R} \\ \|x + y\| \leq \|x\| + \|y\|, & \forall x, y \in \mathbb{R}^n \end{cases}$$

Manhattan norm ($\|\cdot\|_1$, ℓ_1 -norm)

$$\|x\|_1 = |x_1| + \cdots + |x_n|$$

Euclidean norm ($\|\cdot\|_2$, ℓ_2 -norm)

$$\|x\|_2 = \sqrt{x^T \cdot x} = \sqrt{x_1^2 + \cdots + x_n^2}.$$

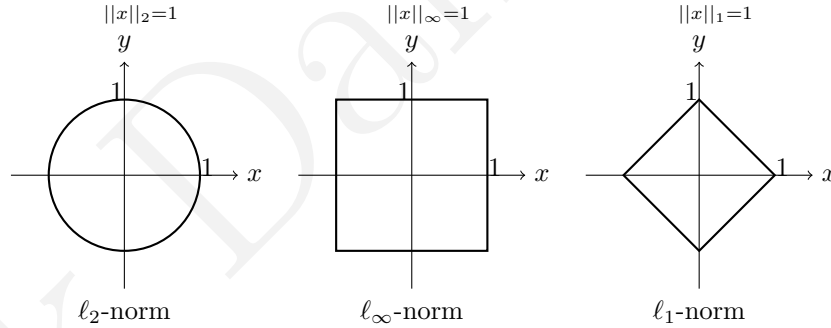
ℓ_p -norm is a generalised version for $p > 0$ ($p > 1$)

$$\|x\|_p = (|x_1|^p + \cdots + |x_n|^p)^{\frac{1}{p}}.$$

If $p \rightarrow \infty$:

$$\|x\|_\infty = \lim_{p \rightarrow \infty} (|x_1|^p + \cdots + |x_n|^p)^{\frac{1}{p}} = \max_{1 \leq i \leq n} |x_i| \lim_{p \rightarrow \infty} \left(\left| \frac{x_1}{\max_{1 \leq i \leq n} |x_i|} \right|^p + \cdots + 1 + \cdots + \left| \frac{x_n}{\max_{1 \leq i \leq n} |x_i|} \right|^p \right)^{\frac{1}{p}} = \max_{1 \leq i \leq n} |x_i|$$

For $n = 2$:



Let $U \in \mathbb{R}^{n \times n}$ is *orthogonal* (or *unitary*) if any of the following equivalent conditions hold:

$$U^T U = I \iff U U^T = I \iff \text{columns of } U : \text{orthonormal basis} \iff \text{rows of } U : \text{orthonormal basis}$$

If $U \in \mathbb{R}^{n \times n}$ is an orthogonal matrix, then $\|Ux\|_2 = \|x\|_2$ for all $x \in \mathbb{R}^n$.

Example 01. Rotation matrix

$$\text{For } n = 2, \quad U = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \implies \|Ux\|_2 = \|x\|_2.$$

Case Study 01. Quantisation

Quantisation is a technique that store data to clusters of a certain value to gain model efficiency.

Consider the data given is $\mathcal{D} = \{-3, -1.5, 2.7, 1.1\}$

Note that error $\propto$ length (= max - min)

The range we can think of it is $\mathcal{D} \subset [-3, 3]$

Considering this, we can determine the quantisation classification as follows:

$$\text{For } d_i \in \mathcal{D} \text{ where } 1 \leq i \leq 4, \quad \text{Quantise}(d_i) = \begin{cases} -2 & d_i \in [-3, -1) \\ 0 & d_i \in [-1, 1) \\ 2 & d_i \in [1, 3) \end{cases}$$

This makes the data quantised as follows:

$$\mathcal{D} = \{-3, -1.5, 2.7, 1.1\} \implies \text{Quantise}(\mathcal{D}) = \{-2, -2, 2, 2\}$$

This will reduce memory storage which gains model efficiency.

Similarly if there exist some arbitrary data $x = (-100, +1, -1, 0, +2, -2, \dots)$:

Quantisation matrix can be interpreted as an orthogonal matrix U ,

where the quantised data can be viewed as Ux

$$x = (-100, +1, -1, 0, +2, -2, \dots) \xrightarrow{\times U} Ux = (+3, -3, -2, +4, \dots)$$

Since the total amount of data should be preserved, quantisation matrix also suffices $\|Ux\|_2 = \|x\|_2$.

11.3 Singular Value Decomposition

Let $A \in \mathbb{R}^{m \times n}$. The singular value decomposition (SVD) of A has the form

$$A = U\Sigma V^T$$

Components of SVD

- A : $m \times n$ matrix
- U : $m \times m$ orthogonal matrix
- Σ : $m \times n$ diagonal covariance matrix with the form

$$\Sigma_{ij} = \begin{cases} \sigma_i & \text{where } 1 \leq i = j \leq r = \text{rank}(A) \\ 0 & \text{otherwise} \end{cases}, \quad \text{for } \sigma_1 \geq \sigma_2 \geq \dots \sigma_r \geq 0$$

- V^T : $n \times n$ orthogonal matrix

This makes the SVD of A as follows:

$$\begin{aligned} A = U\Sigma V^T &= \begin{bmatrix} u_1 & u_2 & \dots & u_m \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 & \dots & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 & \dots & 0 \\ \vdots & & & & & \\ 0 & 0 & \dots & \sigma_r & \dots & 0 \\ \vdots & & & & & \\ 0 & 0 & \dots & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_r^T \\ \vdots \\ v_n^T \end{bmatrix} \\ &= u_1 \sigma_1 v_1^T + u_2 \sigma_2 v_2^T + \dots + u_r \sigma_r v_r^T = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots + \sigma_r u_r v_r^T = \sum_{i=1}^r \sigma_i u_i v_i^T \end{aligned}$$

where $u_1, \dots, u_m \in \mathbb{R}^m$ are orthonormal basis of $\mathbb{R}^m$ and $v_1, \dots, v_n \in \mathbb{R}^n$ are orthonormal basis of $\mathbb{R}^n$

The SVD always exists and can be computed. Note that we have compact version of SVD:

$$A = U_c \Sigma_c V_c^T = \underbrace{\begin{bmatrix} u_1 & u_2 & \dots & u_r \end{bmatrix}}_{U_c} \underbrace{\begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & & & \\ 0 & 0 & \dots & \sigma_r \end{bmatrix}}_{\Sigma_c} \underbrace{\begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_r^T \\ \vdots \\ v_n^T \end{bmatrix}}_{V_c^T}$$

Note that U_c and V_c contain orthonormal columns, but the matrices are not orthogonal matrices:

$$\begin{aligned} U_c^T U_c &= I \quad \text{but} \quad U_c U_c^T \neq I \\ V_c^T V_c &= I \quad \text{but} \quad V_c V_c^T \neq I. \end{aligned}$$

(Moore-Penrose) Pseudo-Inverse

The (Moore-Penrose) *pseudo-inverse* is as defined as follows:

$$A^\dagger = V_c \Sigma_c^{-1} U_c^T = \begin{bmatrix} v_1 & v_2 & \dots & v_r \end{bmatrix} \begin{bmatrix} \sigma_1^{-1} & 0 & \dots & 0 \\ 0 & \sigma_2^{-1} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_r^{-1} \end{bmatrix} \begin{bmatrix} u_1^T \\ u_2^T \\ \vdots \\ u_r^T \end{bmatrix}$$

12 Matrix Norms, Nuclear Norm, LoRA

12.1 Matrix Norms

Vector p -norm

For a vector $x \in \mathbb{R}^n$, the ℓ_p -norm is

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{\frac{1}{p}}, \quad \|x\|_\infty = \max_i |x_i|.$$

Matrix induced p -norm

For a matrix $A \in \mathbb{R}^{m \times n}$, the induced (or operator) p -norm is defined by

$$\|A\|_p = \max_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p}.$$

This ensures the inequality

$$\|Ax\|_p \leq \|A\|_p \cdot \|x\|_p \quad \forall x,$$

which is an essential property of operator norms.

Frobenius Norm

$$\|A\|_F = \sqrt{\sum_i \sum_j a_{ij}^2}$$

Spectral Norm ($p = 2$)

For $p = 2$, the induced matrix norm is the *spectral norm*:

$$\|A\|_2 = \max_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2}.$$

Components of the spectral norm

- $\|A\|_2$: Matrix $\mathbb{R}^{m \times n}$ norm
- $\|x\|_2$: Vector $\mathbb{R}^n$ norm
- $\|Ax\|_2$: Vector $\mathbb{R}^m$ norm

Let $A = U \Sigma V^T$ be the SVD of A , with singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$, then:

$$\|A\|_2 = \sigma_1$$

Here, $\|A\|_2$ denotes the largest singular value.

For any matrix norm intended to measure "how much A can stretch a vector," it is essential that

$$\|Ax\|_2 \leq \|A\|_2 \cdot \|x\|_2$$

holds for every x . If we defined the norm in any other arbitrary way, it might fail to satisfy this inequality, making it useless for conditioning analysis, stability bounds, and perturbation theory. The quantity

$$\sup_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2}$$

is exactly the minimum number that makes the inequality hold for all x ,

hence giving a principal and mathematically tight definition of the operator norm.

Case Study 01. Low-Rank Approximation (LoRA)

Note that the *rank* of A denotes the number of nonzero singular values.

Let $A \in \mathbb{R}^{m \times n}$, with rank r .

Using the Singular Value Decomposition of $A = U\Sigma V^T$:

$$A = U_c \Sigma_c V_c^T = \begin{pmatrix} U_c & \Sigma_c & V_c^T \end{pmatrix} = \begin{pmatrix} (m \times r) & (r \times r) & (r \times n) \\ (L) & (R) \end{pmatrix}$$

LoRA is a technique to convert A into LR via SVD to reduce the number of parameters.

Let suppose $m = n = 4096$, and $r = 16$

The original model A requires $(4096)^2$ number of parameters,

By applying $A = LR$, A require $2 \times 16 \times 4096$ number of parameters. ($2 \times 16 \times 4096 \ll (4096)^2$)

$\therefore$ The number of parameters surely reduces by applying LoRA which triggers model efficiency.

This is usually applied when we store parameters in weight matrices.

Consider model having weight $W \in \mathbb{R}^{m \times n}$ which the size is often 4096×4096

Applying LoRA, $W \approx W_r = \begin{pmatrix} (m \times r) \\ L \end{pmatrix} \cdot \begin{pmatrix} (r \times n) \\ R \end{pmatrix}$

This makes the number of parameters decrease from mn to $mr + rn$

Case Study 02. Model Compression

When it comes to model compression, it handles the balance between rate vs. quality

- Rate: we want small r ($r \leq k$)
- Quality: we want the Frobenius norm $\|W - W_r\|_F$ to be minimised.

One of the solution is for fixed r , compute the $\min \|W - W_r\|_F$.

This can be handled as an *optimisation problem* where $\text{rank}(W_r) = r$.

$$W = U\Sigma V^T = \begin{bmatrix} u_1 & \dots & u_r & \dots \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 & \dots & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 & \dots & 0 \\ \vdots & & & & & \\ 0 & 0 & \dots & \sigma_k & \dots & 0 \\ \vdots & & & & & \\ 0 & 0 & \dots & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} v_1^T \\ \vdots \\ v_r^T \end{bmatrix}, \quad \text{where } \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r \geq \sigma_k$$

By adapting LoRA, the optimised weight W_r^* is as follows:

$$W_r^* = \begin{bmatrix} u_1 & \dots & u_r \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & & & \\ 0 & 0 & \dots & \sigma_r \end{bmatrix} \begin{bmatrix} v_1^T \\ \vdots \\ v_r^T \end{bmatrix}$$

12.2 Nuclear Norm (Trace Norm)

Let A have a singular values $\sigma_1, \dots, \sigma_r$. The *nuclear norm* (also called the *trace norm*) is defined by

$$\|A\|_* = \sum_{i=1}^r \sigma_i.$$

Interpretation as a measure of low rank

The nuclear norm is the convex envelope of the matrix rank on the unit spectral-norm ball:

$$\text{rank}(A) \approx \|A\|_* \quad (\text{convex surrogate}).$$

Low-rank matrices have rapidly decaying singular values, making $\|A\|_*$ small.

Thus the nuclear norm encourages low-rank solutions in convex optimisation problems.

$$A : \Omega = \{(i, j) : A_{ij} \text{ is known}\}$$

- $A^* \approx A$
- A^* : low rank

$$\min_{\underbrace{\sum_{(i,j) \in \Omega} (A_{ij} - A_{ij}^*)^2}_{A \approx A^*}} + \lambda \underbrace{\|A^*\|_{\text{tr}}}_{\text{low rank}}$$

$$\|A^*\|_{\text{tr}} = \text{trace (nuclear) norm} = \sum_{i=1}^k \sigma_i$$

$$A = U \begin{bmatrix} \sigma_1 & 0 & \dots & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 & \dots & 0 \\ \vdots & & & & & \\ 0 & 0 & \dots & \sigma_k & \dots & 0 \\ \vdots & & & & & \\ 0 & 0 & \dots & 0 & \dots & 0 \end{bmatrix} V^T$$

- $\text{rank}(A) = k$ (relax): ideally count nonzero σ 's $\xRightarrow{\text{replace}} \sum \sigma_i$

12.3 Matrix Completion and the Netflix Problem

In the Netflix challenge, the data consists of a partially observed user-movie rating matrix. Let

$$\Omega \subseteq \{1, \dots, m\} \times \{1, \dots, n\}$$

denote the set of observed (user, movie) pairs, and let $M \in \mathbb{R}^{m \times n}$ be the unknown full rating matrix.

The task is to infer the missing entries assuming that M is well-approximated by a low-rank structure (latent user and movie factors).

Netflix prize (Matrix Completion)

user	movie 1	movie 2	3	4	5	6	7	...	M
user 1	+1	?	?	+1	-1	+1	-1	...	?
user 2	+1	?	?	?	-1	+1	-1	...	?
3	?	?	?	...					
4	?	?	?	...	...				
5	?	?	?	...	...	...			
$\vdots$									
N	?	?	?	...	...	...	...	...	...

To predict the value of ?, our natural guess is +1 since the rows of the matrix behave similarly.

Ideal Low-Rank Formulation

A natural formulation is to search for a rank- k matrix that best fits the observed entries:

$$\min_X \sum_{(i,j) \in \Omega} (X_{ij} - M_{ij})^2 \quad \text{s.t.} \quad \text{rank}(X) \leq k.$$

This is a direct analogue of the classical low-rank factor model used in collaborative filtering.

However, the rank constraint makes the problem nonconvex and, in general, NP-hard.

Convex Relaxation via Nuclear Norm

To obtain a tractable optimisation problem, the rank constraint is replaced with its tightest convex surrogate, the nuclear norm:

$$\text{rank}(X) \longrightarrow \|X\|_* = \sum_i \sigma_i(X),$$

leading to the convex program

$$\min_X \sum_{(i,j) \in \Omega} (X_{ij} - M_{ij})^2 + \lambda \|X\|_*.$$

This relaxation enables efficient algorithms such as singular value thresholding, and under suitable incoherence and sampling assumptions, it can recover the true low-rank matrix exactly.

12.4 Best Low-Rank Approximation Theorem

Let $A = U\Sigma V^\top$ be the SVD of A , and let

$$A_k = U\Sigma_k V^\top$$

denote the rank- k truncation keeping only the top k singular values.

Eckart-Young-Mirsky

For any rank- k matrix B ,

$$\begin{aligned} \|A - A_k\|_2 &= \min_{\text{rank}(B)=k} \|A - B\|_2 = \sigma_{k+1}, \\ \|A - A_k\|_F &= \min_{\text{rank}(B)=k} \|A - B\|_F = \left(\sum_{i>k} \sigma_i^2 \right)^{\frac{1}{2}}. \end{aligned}$$

Thus A_k is the unique best rank- k approximation of A in both spectral and Frobenius norms.

12.5 Proof of the Best Rank- k Approximation Theorem (2-Norm Case)

Let $A \in \mathbb{R}^{n \times p}$ have SVD

$$A = U\Sigma V^\top, \quad \Sigma = \text{diag}(\sigma_1, \dots, \sigma_p), \quad \sigma_1 \geq \dots \geq \sigma_p \geq 0.$$

Let B be any matrix with $\text{rank}(B) = k$. We aim to show

$$\|A - B\|_2 \geq \sigma_{k+1}, \quad \text{with equality achieved by } A_k.$$

Step 1: Use the null space of B . Since $\text{rank}(B) = k$, the rank-nullity theorem implies

$$\dim N(B) = p - k.$$

Step 2: Use the span of the first $k + 1$ right-singular vectors. Consider the matrix

$$V_{k+1} = [v_1 \dots v_{k+1}] \in \mathbb{R}^{p \times (k+1)},$$

whose columns are the first $k + 1$ right singular vectors of A . This matrix has rank $k + 1$, so its column space $C(V_{k+1})$ is $(k + 1)$ -dimensional.

Step 3: Dimension argument to find a shared vector.

Both $N(B)$ and $C(V_{k+1})$ lie in the p -dimensional space $\mathbb{R}^p$. But

$$\dim N(B) + \dim C(V_{k+1}) = (p - k) + (k + 1) = p + 1 > p.$$

Thus they must intersect:

$$N(B) \cap C(V_{k+1}) \neq \{0\}.$$

Choose any vector w in this intersection:

$$w \in N(B), \quad w \in C(V_{k+1}), \quad \|w\|_2 = 1.$$

Since $w \in C(V_{k+1})$, we can write

$$w = \sum_{i=1}^{k+1} w_i v_i, \quad \sum_{i=1}^{k+1} w_i^2 = 1.$$

Step 4: Lower bound the error using the chosen vector.

Using the definition of the spectral norm,

$$\|A - B\|_2^2 \geq \|(A - B)w\|_2^2.$$

But $w \in N(B)$, so $Bw = 0$. Hence

$$\|(A - B)w\|_2^2 = \|Aw\|_2^2 = w^\top V \Sigma^2 V^\top w.$$

Substituting the expansion of w in the v_i basis,

$$w^\top V \Sigma^2 V^\top w = \sum_{i=1}^{k+1} \sigma_i^2 w_i^2 \geq \sigma_{k+1}^2 \sum_{i=1}^{k+1} w_i^2 = \sigma_{k+1}^2.$$

Thus

$$\|A - B\|_2^2 \geq \sigma_{k+1}^2, \quad \text{i.e.,} \quad \|A - B\|_2 \geq \sigma_{k+1}.$$

Step 5: Tightness. For the truncated SVD matrix

$$A_k = U \begin{bmatrix} \text{diag}(\sigma_1, \dots, \sigma_k) & 0 \\ 0 & 0 \end{bmatrix} V^\top,$$

whose largest singular value is exactly σ_{k+1} .

Thus for every rank- k matrix B ,

$$\|A - A_k\|_2 = \sigma_{k+1} \leq \|A - B\|_2.$$

This completes the proof.

Frobenius norm case. Let B be any rank- k matrix. Write $A - B$ in the singular basis of A :

$$A = \sum_{i=1}^r \sigma_i u_i v_i^\top, \quad A_k = \sum_{i \leq k} \sigma_i u_i v_i^\top.$$

Since $\text{rank}(B) \leq k$, it can match at most k singular directions. Thus the residual Frobenius energy is at least the energy of the truncated tail:

$$\|A - B\|_F^2 \geq \sum_{i > k} \sigma_i^2.$$

Again A_k achieves equality.

13 Linear Regression

Given a data

$$(\mathbb{X}^{(1)}, \mathbb{y}^{(1)}), \dots, (\mathbb{X}^{(N)}, \mathbb{y}^{(N)})$$

where $\mathbb{X}^{(i)} \in \mathbb{R}^n$ and $\mathbb{y}^{(i)} \in \mathbb{R}$, we want to find a linear model that approximate

$$\mathbb{y} \approx f_\theta(\mathbb{X}) = \theta^T \mathbb{X} = \theta_1 \mathbb{X}_1 + \dots + \theta_N \mathbb{X}_N$$

for new data-label pair $(\mathbb{X}, \mathbb{y})$.

Consider $N = 1$, where $\mathbb{X} \in \mathbb{R}$ and $\mathbb{y} \approx f_\theta(\mathbb{X})$.

To model the linearity relationship, we can define as:

$$\mathbb{y} = \theta_1 \mathbb{X}.$$

This linear model does not have a bias term though.

In that case, adding a dummy feature as a bias term may generalise the situation.

Let $\tilde{\mathcal{X}} = (1, \mathcal{X}_1, \dots, \mathcal{X}_n)$, and formulate the model to find $\tilde{\theta} = (b, \theta_1, \dots, \theta_n)$ that fits:

$$y \approx \theta^T \mathcal{X} + b = \theta_1 \mathcal{X}_1 + \dots + \theta_n \mathcal{X}_n + b = \underbrace{\begin{bmatrix} b & \theta_1 & \dots & \theta_n \end{bmatrix}}_{\tilde{\theta}} \underbrace{\begin{bmatrix} 1 \\ \mathcal{X}_1 \\ \vdots \\ \mathcal{X}_n \end{bmatrix}}_{\tilde{\mathcal{X}}}$$

Linear regression is interpretable by considering the θ_k term which denotes the “importance” of $\mathcal{X}_k$:

- if θ_k is small, then $\mathcal{X}_k$ has less influence on prediction $f_\theta(\mathcal{X})$.
- if $\theta_k = 0$ means the prediction does not depend on $\mathcal{X}_k$.
- θ_k is the amount prediction $f_\theta(\mathcal{X})$ increases when $\mathcal{X}_k$ increments by 1.

We need to formally define the goodness (*i.e.*, how well it approximates y).

In linear regression, Mean Squared Error (MSE) or equivalently ℓ_2 loss is normally used to minimise *i.e.*,

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^N \left\| y^{(i)} - \theta^T \mathcal{X}^{(i)} \right\|^2 = \min_{\theta} \frac{1}{N} \sum_{i=1}^N \ell_2(f_\theta(\mathcal{X}^{(i)}), y^{(i)})$$

To minimise the MSE, the task is to find θ^* , the optimal solution:

$$\theta^* = \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N \ell_2(f_\theta(\mathcal{X}^{(i)}), y^{(i)}) = \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N (f_\theta(\mathcal{X}^{(i)}) - y^{(i)})^2 = \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N (\theta^T \mathcal{X}^{(i)} - y^{(i)})^2$$

Note that $\frac{1}{N}$ does not affect the solution. We can further simplify the above summation by defining

$$\begin{aligned} A &= \begin{bmatrix} (\mathcal{X}^{(1)})^T \\ (\mathcal{X}^{(2)})^T \\ \vdots \\ (\mathcal{X}^{(N)})^T \end{bmatrix}, \quad B = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(N)} \end{bmatrix} \\ y &= \tilde{\theta}^T \tilde{\mathcal{X}} = \frac{1}{N} \sum_{i=1}^N \left\| \theta^T \mathcal{X}^{(i)} - y^{(i)} \right\|^2 = \frac{1}{N} \left\| \begin{pmatrix} \theta^T \mathcal{X}^{(1)} - y^{(1)} \\ \vdots \\ \theta^T \mathcal{X}^{(N)} - y^{(N)} \end{pmatrix} \right\|_2^2 \\ &\rightarrow \frac{1}{N} \left\| \begin{pmatrix} -\mathcal{X}^{(1)T} \\ \vdots \\ -\mathcal{X}^{(N)T} \end{pmatrix} \theta - \begin{pmatrix} y^{(1)} \\ \vdots \\ y^{(N)} \end{pmatrix} \right\|_2^2 = \frac{1}{N} \left\| \begin{matrix} A & \cdot & \theta \\ (N \times n) & & (n \times 1) \end{matrix} - \begin{matrix} B \\ (N \times 1) \end{matrix} \right\|_2^2 \cdots \Rightarrow \min_{\theta} \frac{1}{N} \|A\theta - B\|_2^2 \end{aligned}$$

where $A \in \mathbb{R}^{N \times n}$ and $B \in \mathbb{R}^N$. We normally assume $N > n$ and the matrix A is tall,

which means $A\theta = B$ is over-determined (have more equations than variables).

For most cases, there is no exact solution θ such that $A\theta = B$.

Suppose we are interested in solving linear equation $A\theta = B$. If A is a square invertible matrix,

then the solution is $\theta = A^{-1}B$. If A is a tall matrix, then we have more equations than the variables,

which is over-determined setting, which usually means no solution satisfies $A\theta = B$. If A is a fat matrix,

then we have less equations than the variables, which is under-determined setting,

which usually have more than 1 solutions that satisfies $Ax = B$ (if A is full rank).

Then, we have

$$\theta^* = \arg \min_{\theta} \frac{1}{2} \|A\theta - B\|_2^2$$

Again $\frac{1}{2}$ does not affect the solution.

The linear least squares optimisation problem

$$\min_{\theta \in \mathbb{R}^n} \frac{1}{2} \|A\theta - B\|_2^2.$$

Then, the optimal solution is given by

$$\theta^* = A^\dagger B$$

The solution is unique if $n = r \leq N$. Otherwise, θ^* is not unique but it has the *least-norm solution* that achieves minimum $\frac{1}{2} \|A\theta - B\|_2^2$ while having smallest $\|\theta\|_2^2$.

Proof

$$\text{Let } A = \begin{matrix} U_c & \Sigma_c & V_c^T \\ (N \times r) & (r \times r) & (r \times n) \end{matrix}, \quad U = \begin{bmatrix} U_c & \tilde{U}_c \\ (N \times r) & (N \times (N-r)) \end{bmatrix}$$

Then,

$$\begin{aligned} \frac{1}{2} \|A\theta - B\|_2^2 &= \frac{1}{2} \|U_c \Sigma_c V_c^T \theta - B\|_2^2 = \frac{1}{2} \left\| \begin{bmatrix} U_c^T \\ \tilde{U}_c^T \end{bmatrix} (U_c \Sigma_c V_c^T \theta - B) \right\|_2^2 \\ &= \frac{1}{2} \left\| \begin{bmatrix} U_c^T U_c \Sigma_c V_c^T \theta & - U_c^T B \\ 0 & - \tilde{U}_c^T B \end{bmatrix} \right\|_2^2 = \frac{1}{2} \|\Sigma_c V_c^T \theta - U_c^T B\|_2^2 + \frac{1}{2} \|\tilde{U}_c^T B\|_2^2 = \frac{1}{2} \|\Sigma_c V_c^T \theta - U_c^T B\|_2^2 \end{aligned}$$

$$\text{Let } \theta = \theta_1 + \theta_2, \text{ where } \theta_1 \in \text{null}(V_c^T)^\perp \text{ and } \theta_2 \in \text{null}(V_c^T)$$

In turn, this is equivalent to

$$\frac{1}{2} \|A\theta - B\|_2^2 = \frac{1}{2} \|\Sigma_c V_c^T (\theta_1 + \theta_2) - U_c^T B\|_2^2 = \frac{1}{2} \|\Sigma_c V_c^T \theta_1 - U_c^T B\|_2^2$$

Since $V_c^T V_c = I$, the above minimum is achieved by

$$\theta^* = (\Sigma_c V_c^T)^{-1} U_c^T B = V_c \Sigma_c^{-1} U_c^T B = A^\dagger B.$$

Thus, $\theta^* = \theta_1^* + \theta_2$ is a solution that achieves the minimum for all $\theta_2 \in \text{null}(V_c^T)$.

However, among them, the minimum norm solution is $\theta^* = \theta_1^*$ since

$$\|\theta_1^* + \theta_2\|_2^2 = \|\theta_1^*\|_2^2 + \|\theta_2\|_2^2 \geq \|\theta_1^*\|_2^2.$$

***Note** that this is a common setup in Machine Learning. Given a dataset, we first define a function class $\mathcal{F}$ (in above case, $\mathcal{F} = \{f_\theta : f_\theta(\mathbf{x}) = \theta^T \mathbf{x}, \theta \in \mathbb{R}^n\}$), and define the loss function $\ell(f_\theta(\mathbf{x}), y)$.

Then, minimises the "empirical loss":

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \ell(f_\theta(\mathbf{x}^{(i)}), y^{(i)}).$$

This is also the case in neural network training where one can set $\mathcal{F}$ be the class of neural network with appropriate loss ℓ (like Cross-Entropy loss). The training is simply to minimise the empirical loss $\mathcal{L}(\theta)$ with an appropriate optimisation algorithm.

Case Study 01. Feature Engineering

We can generalise this approach to a broader class of functions. For instance, in the $n = 1$ case, we might consider a more complex model such as $f_\theta(x) = \theta_1 \log x + \theta_2 x + \theta_3 x^2$. Although this model incorporates nonlinear transformations of x , it remains linear with respect to θ , so we can apply the same method to find the best model fit.

This approach introduces the problem of selecting the most useful features (like $\log x$, x^2 , etc.), a process

known as *feature engineering*. In later courses, we will explore feature selection techniques in more depth. In the deep learning era, however, feature engineering is largely automated, with neural networks learning the relevant features directly from data.

More generally, the output dimension can be larger than 1.

$$(\mathbf{x}^{(1)}, \mathbf{y}^{(1)}), (\mathbf{x}^{(2)}, \mathbf{y}^{(2)}), \dots, (\mathbf{x}^{(N)}, \mathbf{y}^{(N)})$$

where $\mathbf{x}^{(i)} \in \mathbb{R}^n$ and $\mathbf{y}^{(i)} \in \mathbb{R}^m$.

We want to find a linear model that approximate

$$f_{\Theta}(\mathbf{x}) = \Theta \mathbf{x} \approx \mathbf{y}$$

for new data-label pair $(\mathbf{x}, \mathbf{y})$ where Θ is $n \times m$ matrices.

If we let

$$\Theta = \begin{bmatrix} \theta_1^T \\ \theta_2^T \\ \vdots \\ \theta_m^T \end{bmatrix}.$$

Then, the problem is simply to find $\theta_k \in \mathbb{R}^n$ that satisfies $\theta_k \mathbf{x} \approx \mathbf{y}_k$ for all $1 \leq k \leq m$. This is because

$$\sum_{i=1}^N \|\Theta \mathbf{x}^{(i)} - \mathbf{y}^{(i)}\|_2^2 = \sum_{i=1}^N \sum_{k=1}^m \|\theta_k^T \mathbf{x}^{(i)} - \mathbf{y}_k^{(i)}\|_2^2 = \sum_{k=1}^m \sum_{i=1}^N \|\theta_k^T \mathbf{x}^{(i)} - \mathbf{y}_k^{(i)}\|_2^2.$$

We can solve the same linear regression problem for each k that minimises

$$\sum_{i=1}^N \|\theta_k^T \mathbf{x}^{(i)} - \mathbf{y}_k^{(i)}\|_2^2.$$

then merge them to form Θ .

Eckart-Young-Mirsky

$A : m \times n$ matrix

$$A = U \begin{bmatrix} \sigma_1 & 0 & \dots & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 & \dots & 0 \\ \vdots & & & & & \\ 0 & 0 & \dots & \sigma_k & \dots & 0 \\ \vdots & & & & & \\ 0 & 0 & \dots & 0 & \dots & 0 \end{bmatrix} V^T$$

Assume $B : \text{rank}(B) \leq r$

$$\min \|A - B\|_2 \implies B^* = Ar = [u_1 \dots u_r] \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & & & \\ 0 & 0 & \dots & \sigma_r \end{bmatrix} \begin{bmatrix} v_1^T \\ \vdots \\ v_r^T \end{bmatrix}$$

Since $B : m \times n$ matrix, $\text{rank}(B) \leq r$

$\dim(\text{null}(B)) \geq n - r$ ($\because \dim(\text{null}(B)) + \text{rank}(B) = n$)

$$V_{r+1} = \underbrace{[v_1 \dots v_{r+1}]}_{(n \times (r+1))} \longrightarrow \text{rank}(V_{r+1}) = r + 1$$

$$\begin{aligned} \dim(\text{null}(B)) &\geq n - r \\ \dim(\text{range}(V_{r+1})) &= r + 1 \geq n + 1, \quad \text{null}(B) \cap \text{range}(V_{r+1}) \ni w \neq 0 \end{aligned}$$

$$w \in \text{range}(V_{r+1}) \Rightarrow w = a_1 v_1 + \cdots + a_{r+1} v_{r+1}$$

$$\|w\|_2 = 1 \text{ (we can always normalise), } V^T w = \begin{bmatrix} v_1^T \\ \vdots \\ v_n^T \end{bmatrix} (a_1 v_1 + \cdots + a_{r+1} v_{r+1}) = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_{r+1} \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\begin{aligned} \|A - B\|_2^2 &\geq \|(A - B)w\|_2^2 = \|Aw\|_2^2 = w^T A^T \cdot Aw = w^T V \Sigma^T \Sigma V^T w \\ &= a_1^2 \sigma_1^2 + a_2^2 \sigma_2^2 + \cdots + a_{r+1}^2 \sigma_{r+1}^2 \quad (\text{where } \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{r+1}) \\ &\geq (a_1^2 + \cdots + a_{r+1}^2) \sigma_{r+1}^2 = \|w\|^2 \cdot \sigma_{r+1}^2 = \sigma_{r+1}^2 \end{aligned}$$

For any rank r matrix $B \Rightarrow \|A - B\|_2^2 \geq \sigma_{r+1}^2 = \|A - Ar\|_2^2$

$$\begin{aligned} \|A - Ar\|_2^2 &= U \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 & \cdots & 0 \\ \vdots & & & & & \\ 0 & 0 & \cdots & \sigma_k & \cdots & 0 \\ \vdots & & & & & \\ 0 & 0 & \cdots & 0 & \cdots & 0 \end{bmatrix} V^T - U \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 & \cdots & 0 \\ \vdots & & & & & \\ 0 & 0 & \cdots & \sigma_r & \cdots & 0 \\ \vdots & & & & & \\ 0 & 0 & \cdots & 0 & \cdots & 0 \end{bmatrix} V^T \\ &= U \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & & & & & & & \\ 0 & 0 & \cdots & \boxed{\sigma_{r+1}} & \cdots & 0 & \cdots & 0 \\ \vdots & & & & & & & \\ 0 & 0 & \cdots & 0 & \cdots & \sigma_k & \cdots & 0 \\ \vdots & & & & & & & \\ 0 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 0 \end{bmatrix} V^T \\ &\quad \underbrace{\hspace{15em}}_{\text{SVD of } A - Ar} \end{aligned}$$

This makes $\|A - Ar\|_2 = \boxed{\sigma_{r+1}}$

14 Empirical Risk Minimisation

14.1 Empirical Risk

Given a data

$$(\mathbf{x}^{(1)}, \mathbf{y}^{(1)}), \dots, (\mathbf{x}^{(N)}, \mathbf{y}^{(N)})$$

that are sampled from P in i.i.d. manner where $\mathbf{x}^{(i)} \in \mathcal{X}$ and $\mathbf{y}^{(i)} \in \mathcal{Y}$.

Classifier

The classifier $f : \mathcal{X} \rightarrow \hat{\mathcal{Y}}$ is to predict Y given X for new data-label pair $(X, Y) \sim P$ so that $f_\theta \approx y$.

Loss Function

A loss function $\ell : \hat{\mathcal{Y}} \times \mathcal{Y} \rightarrow \mathbb{R}$ quantifies how well $\hat{y}$ approximates y .

Smaller $\ell(\hat{y}, y)$ indicates $\hat{y}$ is a good approximation of y .

Typically, the loss function is nonnegative (but not always).

– Examples of loss functions ($\ell(f_\theta(x), y)$)

- Quadratic Loss (Mean Squared Error)

$$\ell(f_\theta(x), y) = \|f_\theta(x) - y\|_2^2.$$

- Absolute Loss (ℓ_1 -loss)

$$\ell(f_\theta(x), y) = \|f_\theta(x) - y\|_1.$$

- Indicator Function

$$\mathbb{1}(f_\theta(x) \neq y)$$

Expected Risk

The *expected risk* (or *true risk* or *population risk*) is

$$\mathcal{R}[f_\theta] = \mathbb{E}_P[\ell(f_\theta(X), Y)]$$

Our goal is to solve

$$\mathcal{R}^* = \min_{f_\theta} \mathcal{R}[f_\theta]$$

Bayes Risk

We call

$$\mathcal{R}^* = \inf_{f_\theta} \mathcal{R}[f_\theta^*]$$

the *Bayes risk* or the *optimal risk*, where the infimum (minimum) is over all possible functions.

The optimal achieving $f^* : \mathcal{X} \rightarrow \mathcal{Y}$ is called *Bayes predictor* (*Bayes optimal*). I.e.,

$$\mathcal{R}^* = \mathcal{R}[f_\theta^*]$$

Empirical Risk

Although our goal is to minimise the true risk $\mathcal{R}[f_\theta]$, it is often impossible to compute since the true data distribution is unknown. Instead we use approximate

$$\hat{\mathcal{R}}[f_\theta] = \frac{1}{N} \sum_{i=1}^N \ell(f_\theta(\mathcal{X}^{(i)}, \mathcal{Y}^{(i)}))$$

We call $\hat{\mathcal{R}}[f_\theta]$ the **empirical risk** which is used to train/optimize the model.

Though $\hat{\mathcal{R}}(f_\theta) \neq \mathcal{R}(f_\theta)$,

$$\hat{\mathcal{R}}(f_\theta) \xrightarrow{N \rightarrow \infty} \mathbb{E}[\ell(f_\theta(X), Y)] = \mathcal{R}[f_\theta]$$

As N goes to infinity, $\hat{\mathcal{R}}[f_\theta] \rightarrow \mathcal{R}[f_\theta]$ by the law of large numbers (LLN).

15 Gradient Descent

Consider the **Linear Regression** problem again, *i.e.*,

$$(\mathcal{X}^{(1)}, \mathcal{Y}^{(1)}), \dots, (\mathcal{X}^{(N)}, \mathcal{Y}^{(N)})$$

$f_\theta(x) \approx y$, ℓ : loss function

$$\text{Loss} = \frac{1}{N} \sum_{i=1}^N \ell(f_\theta(\mathcal{X}^{(i)}, \mathcal{Y}^{(i)}))$$

Suppose the task is to find the best θ that minimises the loss function.

Let's say... like $\min_{\theta} \frac{1}{N} \sum_{i=1}^N (\theta^T \mathcal{X}^{(i)} - \mathcal{Y}^{(i)})^2 = \min_{\theta} L(\theta)$

One of the metrics we can find such θ is *Gradient Descent* (**GD**)

From now on, we will focus on the constraints that will guarantee Gradient Descent.

15.1 Convexity

A function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is *convex* if

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$$

for all $x, y \in \mathbb{R}^d$, and $t \in [0, 1]$.

Differentiable & Convex

If f is differentiable & convex, then every tangent line lies below the function graph, i.e.,

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle = f(x) + \nabla f(x)^T (y - x)$$

This is clear if $d = 1$, where

$$f(y) \geq f(x) + f'(x)(y - x),$$

which implies

$$\begin{aligned} \frac{f(y) - f(x)}{y - x} &\leq f'(x), \text{ if } y < x \\ \frac{f(y) - f(x)}{y - x} &\geq f'(x), \text{ if } y > x \end{aligned}$$

Note that ∇f is a gradient of f , i.e.,

$$\nabla f(x) = \begin{bmatrix} \frac{\partial}{\partial x_1} f(x) \\ \frac{\partial}{\partial x_2} f(x) \\ \vdots \\ \frac{\partial}{\partial x_d} f(x) \end{bmatrix}$$

The gradient of f is d -dimensional generalisation of derivative. A directional derivative measures the rate at which a function $f(x_1, x_2, x_3, \dots, x_d) = \square$ changes in a particular direction v at a given point x , i.e., $x \rightarrow x + \epsilon \cdot v$

$$\begin{aligned} \nabla f \cdot v &= v_1 \frac{\partial f}{\partial x_1} + \dots + v_d \frac{\partial f}{\partial x_d} \\ \langle \nabla f(x), v \rangle &= \nabla f(x)^T v. \end{aligned}$$

Twice Differentiable & Convex

This time consider $f'' \geq 0$ where f is a convex function like $f(x) = x^2$

For a twice differentiable function f , let $\nabla^2 f$ is a Hessian of f , where

$$\nabla^2 f(x) = \begin{bmatrix} \frac{\partial^2}{\partial x_1 \partial x_1} f(x) & \frac{\partial^2}{\partial x_1 \partial x_2} f(x) & \dots & \frac{\partial^2}{\partial x_1 \partial x_d} f(x) \\ \frac{\partial^2}{\partial x_2 \partial x_1} f(x) & \frac{\partial^2}{\partial x_2 \partial x_2} f(x) & \dots & \frac{\partial^2}{\partial x_2 \partial x_d} f(x) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2}{\partial x_d \partial x_1} f(x) & \frac{\partial^2}{\partial x_d \partial x_2} f(x) & \dots & \frac{\partial^2}{\partial x_d \partial x_d} f(x) \end{bmatrix}.$$

We have alternative definition for convex function f .

If $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is *twice-differentiable*, the function f is convex if and only if

$$\nabla^2 f \geq 0 \text{ (i.e., } \nabla^2 f \text{ is positive semidefinite)}$$

In other words $\nabla^2 f(x)$ is positive semi-definite for every x , and equivalently, $v^T \nabla^2 f(x) v \geq 0$ for all vector $v \in \mathbb{R}^d$.

$$f \text{ is convex} \iff \nabla^2 f \geq 0, \quad v^T \nabla^2 f v \geq 0.$$

15.2 Smoothness

L-Lipshitz Continuous

A function f is *L-Lipshitz continuous* if

$$|f(x) - f(y)| \leq L \cdot \|x - y\|_2$$

for all x, y .

A Lipschitz continuous function is limited in how fast it can change.

L-smooth

A differentiable function f is said to be *L-smooth* if its gradients are *L-Lipshitz continuous*, i.e.,

$$\|\nabla f(x) - \nabla f(y)\|_2 \leq L \cdot \|x - y\|_2$$

for all x, y

This means an *L-smooth* function is limited in how fast its gradient can change.

(i.e., f' should not increase too fast)

Twice differentiable & *L-smooth*

If f is twice-differentiable and *L-smooth* (i.e., the smaller L is, the smoother f becomes), then

$$\nabla^2 f(x) \leq LI$$

i.e., $LI - \nabla^2 f(x)$ is positive semidefinite matrix.

Proof

If f is twice differentiable, then the Taylor expansion gives

$$\nabla f(x + tv) - \nabla f(x) = \int_{s=0}^t \nabla^2 f(x + sv) v ds$$

for all vector $v \in \mathbb{R}^d$. Since

$$\left\| \int_{s=0}^t \nabla^2 f(x + sv) v ds \right\|_2 \leq Lt \|v\|_2$$

and therefore,

$$\frac{\left\| \int_{s=0}^t \nabla^2 f(x + sv) v ds \right\|_2}{t \|v\|} \leq L.$$

Taking the limit as $t \rightarrow 0$, then

$$\nabla^2 f(x) \leq LI.$$

Lemma

For all $\delta \in \mathbb{R}^d$

W.T.S is below:

$$f(x + \delta) \leq f(x) + \langle \nabla f(x), \delta \rangle + \frac{L}{2} \|\delta\|^2.$$

Proof

Let $g(t) = f(x + t\delta) \Rightarrow g(0) = f(x), \quad f(1) = f(x + \delta)$.

Then, $g'(t) = \nabla f(x + t\delta)^T \delta$, and

$$g'(t) - g'(0) = (\nabla f(x + t\delta) - \nabla f(x))^T \cdot \delta \leq \|\nabla f(x + t\delta) - \nabla f(x)\| \cdot \|\delta\| \leq L \cdot \|t\delta\| \cdot \|\delta\| = L \cdot t \cdot \|\delta\|^2$$

$$\begin{aligned} f(x + \delta) = g(1) &= g(0) + \int_0^1 g'(t) dt \leq g(0) + \int_0^1 g'(0) + Lt\|\delta\|^2 dt \leq g(0) + g'(0) + \frac{L}{2}\|\delta\|^2 \\ &= f(x) + \langle \nabla f(x), \delta \rangle + \frac{L}{2}\|\delta\|^2. \end{aligned}$$

$$\therefore f(x + \delta) \leq f(x) + \langle \nabla f(x), \delta \rangle + \frac{L}{2}\|\delta\|^2.$$

***Note** that the **Lemma** above does not require a convexity of function f , and it only requires the L -smootheness.

This time, if f is L -smooth and has a minimiser x^* , where $f(x) \geq f(x^*)$ for all x , then

$$\frac{1}{2L} \|\nabla f(z)\|^2 \leq f(z) - f(x^*) \leq \frac{L}{2} \|z - x^*\|^2$$

for all $z \in \mathbb{R}^d$.

Proof

Considering the right-hand inequality:

If z and x^* are close $\Rightarrow f(z) - f(x^*)$ is small

$$f(z) \leq f(x^*) + \nabla f(x^*)^T (z - x^*) + \frac{L}{2} \|z - x^*\|^2 = f(x^*) + \frac{L}{2} \|z - x^*\|^2$$

since $\nabla f(x^*) = 0$ at the optimal point

$$\therefore f(z) - f(x^*) \leq \frac{L}{2} \|z - x^*\|^2$$

This time, considering the left-hand inequality:

$$f(x^*) = \min_y f(y) \leq \min_y f(z) + \nabla f(z)^T (y - z) + \frac{L}{2} \|y - z\|^2$$

Plugging in $y = z + t \cdot u$ and $\|u\| = 1$:

$$= f(z) + \min_{t, \|u\|=1} \nabla f(z)^T \cdot t \cdot u + \frac{L}{2} t^2 = f(z) + \min_{\|u\|=1} \min_t \nabla f(z)^T \cdot t \cdot u + \frac{L}{2} t^2$$

Plugging in $t^* = -\frac{1}{L} \nabla f(z)^T u$:

$$= f(z) + \min_{\|u\|=1} \left(-\frac{1}{2L} (\nabla f(z)^T u)^2 \right) = f(z) - \frac{1}{2L} \|\nabla f(z)\|^2$$

$$\therefore f(x^*) \leq f(z) - \frac{1}{2L} \|\nabla f(z)\|^2 \Rightarrow \frac{1}{2L} \|\nabla f(z)\|^2 \leq f(z) - f(x^*)$$

***Note** that the following expressions in the examples represents the equivalent meaning.

“Small L ” = “Easy to Optimise” = “Easy to Predict” = “More Smooth”

15.3 Smooth and Convex

In many optimisation problems, objective function is both smooth and convex.

Lemma

If f is *convex* and *L-smooth*, then:

1. L-Smoothness

$$f(y) - f(x) \leq \langle \nabla f(y), y - x \rangle - \frac{1}{2L} \|\nabla f(y) - \nabla f(x)\|^2$$

2. Co-Coercivity

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq \frac{1}{L} \|\nabla f(x) - \nabla f(y)\|^2$$

Co-coercivity means the gradient of a smooth convex function cannot twist or oscillate: its change is aligned with the displacement, making gradient descent (GD) stable, predictable, and convergent.

Proof

First, from the *L-smoothness*

$$f(x + z) \leq f(x) + \langle \nabla f(x), z \rangle + \frac{L}{2} \|z\|^2$$

and from convexity

$$\underbrace{f(x + z)}_{f(a) \geq f(b) + \langle f(b), a - b \rangle} \geq f(y) + \langle \nabla f(y), x + z - y \rangle$$

By adding the above two and rearrange terms, we have

$$f(y) + \langle \nabla f(y), x + z - y \rangle \leq f(x) + \langle \nabla f(x), z \rangle + \frac{L}{2} \|z\|^2.$$

By plugging $z = \frac{1}{L} (\nabla f(y) - \nabla f(x))$

$$f(y) + \langle \nabla f(y), x - y \rangle + \underbrace{\langle \nabla f(y) - \nabla f(x), z \rangle}_{L \cdot \|z\|^2} \leq f(x) + \frac{L}{2} \|z\|^2$$

$$\therefore f(y) - f(x) \leq \langle \nabla f(y), y - x \rangle - \frac{1}{2L} \|\nabla f(y) - \nabla f(x)\|^2$$

Next, from the **Co-coercivity**

Let $g_x(x) = \nabla_z (f(z) - \nabla f(x)^T z) = \nabla_z f(z) - \nabla f(x)$. Then we have

$$g_x(y) - g_x(x) \geq \frac{1}{2L} \|\nabla_x g_x(y)\|^2 = \frac{1}{2L} \|\nabla f(y) - \nabla f(x)\|^2$$

If we similarly define $g_y(z) = f(z) - \nabla f(y)^T z$, then

$$g_y(x) - g_y(y) \geq \frac{1}{2L} \|\nabla f(x) - \nabla f(y)\|^2.$$

By adding two inequality, we have

$$\begin{aligned} (g_x(y) - g_x(x)) + (g_y(x) - g_y(y)) &= (f(y) - \nabla f(x)^T y - f(x) + \nabla f(x)^T x) + (f(x) - \nabla f(y)^T x - f(y) + \nabla f(y)^T y) \\ &= (\nabla f(x) - \nabla f(y))^T (x - y). \end{aligned}$$

and

$$(g_x(y) - g_x(x)) + (g_y(x) - g_y(y)) \geq \frac{1}{2L} \|\nabla f(y) - \nabla f(x)\|^2 + \frac{1}{2L} \|\nabla f(x) - \nabla f(y)\|^2 \geq \frac{1}{L} \|\nabla f(x) - \nabla f(y)\|^2.$$

Finally,

$$\begin{aligned} (\nabla f(x) - \nabla f(y))^T (x - y) &\geq \frac{1}{L} \|\nabla f(x) - \nabla f(y)\|^2. \\ \therefore \langle \nabla f(y) - \nabla f(x), y - x \rangle &\geq \frac{1}{L} \|\nabla f(x) - \nabla f(y)\|^2 \end{aligned}$$

The proof of Co-coercivity can also be interpreted as if f is convex & L -smooth then, $\langle \nabla f(y) - \nabla f(x), y - x \rangle$ cannot be too small since it has a lower-bound $\geq \frac{1}{L} \|\nabla f(y) - \nabla f(x)\|^2$

This inequality shows $\nabla f(y) - \nabla f(x)$ and $y - x$ is aligned.

In the case “*gradient f is predictable*”, let

$$\begin{aligned} f_x(z) &= f(z) - \nabla f(x)^T z \\ f_y(z) &= f(z) - \nabla f(y)^T z \end{aligned}$$

Note that $f_x(z)$ can be viewed as a fixed variable $f_n \cdot f_z$

$$\nabla_z f_x(z) = \nabla f(z) - \nabla f(x), \quad \nabla f_x(x) = 0 \Rightarrow f_x(z) \geq f_x(x)$$

$$f(y) - f(x) - \nabla f(x)^T (y - x) = f_x(y) - f_x(x) \geq \frac{1}{2L} \|\nabla f_x(y)\|^2 = \frac{1}{2L} \|\nabla f(y) - \nabla f(x)\|^2$$

From the **Lemma**

$$f(y) - f(x) \leq \langle \nabla f(y), y - x \rangle - \frac{1}{2L} \|\nabla f(y) - \nabla f(x)\|^2 \Rightarrow \langle \nabla f(y), -\nabla f(x), y - x \rangle \geq \frac{1}{L} \|\nabla f(y) - \nabla f(x)\|^2$$

$$f(x) - f(y) - \nabla f(y)^T (x - y) \geq \frac{1}{2L} \|\nabla f_y(x)\|^2 = \frac{1}{2L} \|\nabla f(x) - \nabla f(y)\|^2$$

The function that both suffices Convex & L -smooth, such as a quadratic function ($L \cdot \|y - x\|^2$), leads to the concept of "Strong Convexity" which will be discussed soon.

15.4 Strong Convexity

μ -Strongly Convex

A convex function f is called μ -strongly convex if

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2} \|y - x\|^2$$

for all $x, y \in \mathbb{R}^d$.

It implies the function is approximately behave like quadratic function near x .

Note that the existence of a function even for a very small value μ actually differs from without it. *i.e.*, there exist difference between $\mu = 1e - 8$ and $\mu = 0$

If a *convex, twice differentiable function* f is μ -strongly convex, then

$$\nabla^2 f(x) \geq \mu I.$$

Lemma

If f is differentiable and convex, then

$$(\nabla f(x) - \nabla f(y))^T (x - y) \geq 0$$

for all x, y .

Proof

We have

$$\begin{aligned} f(x) &\geq f(y) + \nabla f(y)(x - y) \\ f(y) &\geq f(x) + \nabla f(x)(y - x). \end{aligned}$$

By adding two inequalities, we have

$$(\nabla f(x) - \nabla f(y))^T (x - y) \geq 0.$$

This time if f is differentiable and μ -strongly convex,

$$(\nabla f(x) - \nabla f(y))^T (x - y) \geq \mu \|x - y\|^2$$

Proof

We have

$$\begin{aligned} f(x) &\geq f(y) + \nabla f(y)(x - y) + \frac{\mu}{2} \|x - y\|^2 \\ f(y) &\geq f(x) + \nabla f(x)(y - x) + \frac{\mu}{2} \|y - x\|^2. \end{aligned}$$

By adding two inequalities, we have

$$(\nabla f(x) - \nabla f(y))^T (x - y) \geq \mu \|x - y\|^2.$$

Lemma

If f is μ -strongly convex, then it satisfies the *Polyak-Lojasiewicz (PL)* condition, which is

$$\|\nabla f(x)\| \geq 2\mu(f(x) - f(x^*)).$$

Proof

$$f(x) - f(x^*) \leq \langle \nabla f(x), x - x^* \rangle - \frac{\mu}{2} \|x^* - x\|^2 = -\frac{1}{2} \|\sqrt{\mu}(x - x^*) - \frac{1}{\sqrt{\mu}} \nabla f(x)\|^2 + \frac{1}{2\mu} \|\nabla f(x)\|^2 \leq \frac{1}{2\mu} \|\nabla f(x)\|^2.$$

Note that strong convexity is sufficient condition for Polyak-Lojasiewicz (PL) condition.

The PL condition can also be interpreted as follows:

$$\underbrace{\|\nabla f(x)\|^2}_{\text{Norm is smallest}} \geq \underbrace{\left(\underbrace{f(x)}_{\text{What you have}} - \underbrace{f(x^*)}_{\text{Ultimate Goal}} \right)}_{\text{function value difference}}$$

That is, small $\|\nabla f(x)\|^2$ guarantees small $f(x) - f(x^*)$

PL condition is an interesting condition by itself, which will guarantee the linear convergence of GD.

Compared to μ strongly convex, PL condition is a more “*generalised*” condition, *i.e.*, a weaker constraint.

15.5 Gradient Descent

Overview

Recall the “**Empirical Loss**” *i.e.*,

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \ell(f_{\theta}(\mathcal{X}^{(i)}), \mathcal{Y}^{(i)})$$

What we want to find is θ that minimises the function above.

In the linear case where $\theta_1 x_1 + \dots + \theta_d x_d$ are parameters and we just need to compute the corresponding formula.

When it comes to Large Language Models, however, there are too much parameters *i.e.*, about parameters of 10^{11} , so there is no typical analytic formula ($\nabla \mathcal{L}(\theta) = 0$), thus a heuristic approach is necessary.

Though there is no analytic $\nabla \mathcal{L}(\theta)$, but we can compute $\nabla \mathcal{L}(\theta)$, and the idea is “*Gradient Descent*”.

Gradient Descent (**GD**) is determined as follows:

- $\nabla \mathcal{L}(\theta)$ is the fastest ascending direction
- $-\nabla \mathcal{L}(\theta)$ is the fastest descending direction

Gradient Descent

Consider a differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, and we want to solve

$$\text{minimise}_{x \in \mathbb{R}^d} f(x).$$

Here, d denotes the number of parameters, and $f(=\mathcal{L}(\theta))$ is what we want to minimise.

***Note** that we will use the term f instead of $\mathcal{L}(\theta)$ in this course.

Without any assumption, we are not able to find a global minimum x^* (*i.e.* target we want to achieve).

However, if f is convex, then x is the solution if and only if $\nabla f(x) = 0$

Let $x^{(0)}$ an initial point(= initialisation) which is picked randomly (maybe your favourite value)

A gradient descent algorithm is

$$x^{(k+1)} = x^{(k)} - \alpha_k \nabla f(x^{(k)})$$

where $\alpha_k > 0$ is learning rate at k -th iteration. The intuition of the algorithm is to update x toward the maximally decreasing direction (since $\nabla f(x)$ is the maximally increasing direction of $f(x)$).

Remark

If $d = 1$, the motivation is clear. We are looking for a point $f'(x) = 0$. Starting from a random point x_0 , compute $f'(x_0)$. If $f'(x_0) > 0$, the function is increasing at $x = x_0$, and therefore search the region $x < x_0$, where the next iteration would be $x_1 = x_0 - \alpha f'(x_0)$. If $f'(x_0) < 0$, the function is decreasing at $x = x_0$, and therefore search the region $x > x_0$ where the next iteration would be $x_1 = x_0 - \alpha f'(x_0)$ (here we have $f'(x_0) < 0$). The learning rate is a "step-size", how far your next iteration would be.

We further assume that ∇f is L -Lipshitz continuous. Then, the following theorem guarantees that the Gradient descent always converges with appropriate learning rate.

Lemma

If f has a global minimum x^* and f is L -smooth, then Gradient Descent algorithm converges to the local minimum (where $\nabla f(x) = 0$).

Proof

Then,

$$\begin{aligned} f(x^{(k+1)}) &= f(x^{(k)} - \alpha_k \nabla f(x^{(k)})) \leq f(x^{(k)}) + \langle \nabla f(x^{(k)}), -\alpha_k \nabla f(x^{(k)}) \rangle + \frac{L}{2} \|\alpha_k \nabla f(x^{(k)})\|^2 \\ &\leq f(x^{(k)}) + \left(\frac{L}{2} \alpha_k^2 - \alpha_k \right) \|\nabla f(x^{(k)})\|^2 \end{aligned}$$

If we pick $\frac{L}{2} \alpha_k^2 - \alpha_k < 0$ (e.g., $\alpha_k < 2/L$), then

$$\begin{aligned} f(x^{(k+1)}) &\leq f(x^{(k)}) \\ f(x^{(0)}) &\geq f(x^{(1)}) \geq \dots \geq f(x^{(k)}) \geq \dots \geq f(x^*) \end{aligned}$$

Actually, we don't know exactly what value the lower bound $f(x^*)$ is,

but we know there exist x^* that reaches the lower bound.

Since $f(x^{(k)})$ is lower bounded by $f(x^*)$ (where x^* is a global minimum) and decreasing, thus it converges to "something". If $f(x^{(k)})$ converges to "something", then $\|\nabla f(x^{(k)})\|^2$ should converge to 0.

Remark

Do you think this theoretical result is too weak? or seems useless? (since we do not know whether

the function is Lipschitz or not) Indeed this theorem tells you the most important fact. If your function (and its gradient) varies too quickly, you need to be careful (adjust step size). Intuitively, if your function is varying crazily, no algorithm can guarantee the convergence. This beautiful theorem basically shows the relation between steepness of the function and the right step size of the algorithm.

How step size matters

Considering the previous problem, if $\alpha_k < 2/L$, it will definitely converge.

This time consider $\alpha_k = 1/L$, then

$$\underbrace{f(x^{(k+1)})}_{\text{Converge}} \leq \underbrace{f(x^{(k)})}_{\text{Converge}} - \frac{1}{2L} \|\nabla f(x^{(k)})\|^2$$

$$\text{if } k \rightarrow \infty \quad f(x^{(k+1)}) = f(x^{(k)}) \Rightarrow \|\nabla f(x)\|^2 = 0$$

The result concludes that it converges to a “stationary” point *i.e.*, $\|\nabla f(x)\|^2 = 0$

What we really need to know during Gradient Descent is to find the reasonable step size α_k

The correlation between the magnitude of step size and GD is as follows

- Small α_k : the algorithm is too slow to update, likely to converge but too slow
- Huge α_k : α_k tends to get out of the scope which would never get to the optimal point (diverge behaviour)

Note that the step size α_k corresponds to the concept of “learning rate” which will be discussed soon.

For $\alpha < 2/L$:

$$\text{if } \begin{cases} L \text{ is large} \\ L \text{ is small} \end{cases} \Rightarrow \begin{cases} \alpha \text{ should be small} \\ \alpha \text{ can be large} \end{cases}$$

From the feature above, it can be inferred that it is good to select a small α_k for a noisy function, whereas larger α_k is good for a smooth function.

Interpretation of the meaning of $\alpha_k < L/2$.

We know that if f is L -smooth, Gradient Descent will converge.

What if f is convex and L -smooth?

Consider the basic gradient step when f is a convex function.

$$x^+ = x - \alpha \nabla f(x).$$

Then,

$$f(x^+) = f(x - \alpha \nabla f(x)) \leq f(x) + \nabla f(x)^T (x - \alpha \nabla f(x) - x) + \frac{L}{2} \|x - \alpha \nabla f(x) - x\|^2 = f(x) - \alpha(1 - \alpha L/2) \|\nabla f(x)\|^2.$$

Let $0 < \alpha \leq 1/L$, then

$$f(x^+) \leq f(x) - \frac{\alpha}{2} \|\nabla f(x)\|^2.$$

From the above inequality, we can say that the single gradient step decreases a function value

since $f(x^+) \leq f(x)$.

From convexity,

$$\nabla f(x)^T (x^* - x) \leq f(x^*) - f(x).$$

By adding two inequalities,

$$f(x^+) - f(x^*) \leq \nabla f(x)^T (x - x^*) - \frac{\alpha}{2} \|\nabla f(x)\|^2 = \frac{1}{2\alpha} (\|x - x^*\|^2 - \|x - x^* - \alpha \nabla f(x)\|^2) = \frac{1}{2\alpha} (\|x - x^*\|^2 - \|x^+ - x^*\|^2).$$

Since $f(x^*) \leq f(x^+)$, we have

$$\|x - x^*\|^2 \geq \|x^+ - x^*\|^2.$$

In other words, the distance to the optimal point also decreases in the gradient method.

If we iteratively apply the gradient method with initial point x_0 and constant step size α , we have

$$\sum_{i=1}^T f(x_i) - f(x^*) \leq \frac{1}{2\alpha} \sum_{i=1}^T (\|x_{i-1} - x^*\|^2 - \|x_i - x^*\|^2) = \frac{1}{2\alpha} (\|x_0 - x^*\|^2 - \|x_T - x^*\|^2) \leq \frac{1}{2\alpha} \|x_0 - x^*\|^2.$$

Note that $\frac{1}{2\alpha} \|x_0 - x^*\|^2$ is some upper bound which is not a function of T .

Since $f(x_i)$ is non-increasing (i.e., $f(x_i) \leq f(x_{i-1})$), we have

$$f(x_t) - f(x^*) \leq \frac{1}{T} \sum_{i=1}^T f(x_i) - f(x^*) \leq \frac{1}{2\alpha T} \|x_0 - x^*\|^2.$$

To be more specific

$$(f(x_1) - f(x^*)) + (f(x_2) - f(x^*)) + \cdots + \underbrace{(f(x_{T+1}) - f(x^*))}_{\text{Converges to zero}} + \cdots \leq \blacksquare$$

Using the fact below

$$\sum_{k=1}^n a_k \leq \blacksquare \quad \& \quad \alpha_k \geq 0 \quad \Rightarrow \quad \lim_{k \rightarrow \infty} a_k = 0$$

we know

$$\lim_{k \rightarrow \infty} f(x_{k+1}) = f(x^*)$$

Due to the fact that Gradient Descent converges and it achieves the minimum function value

$$f(x_T) - f(x^*) \leq \frac{1}{T} \sum_{k=0}^{T-1} f(x_{k+1}) - f(x^*) \leq \frac{1}{T} \frac{L}{2} \|x_0 - x^*\|^2 \longrightarrow 0$$

$\underbrace{\hspace{10em}}_{\text{Since } f(x_T) \leq f(x_{k+1})}$

As a matter of fact, it decreases as a function of $\mathcal{O}(\frac{1}{T})$

$$f(x_T) - f(x^*) \in \mathcal{O}\left(\frac{1}{T}\right) \Rightarrow \text{this denotes the speed of convergence}$$

Thus, the function value decreases with a rate of $\mathcal{O}(1/T)$.

Corollary

If you want to achieve $f(x_T) - f(x^*) \leq \epsilon$, then

the number of iteration should be $\mathcal{O}(1/\epsilon)$ (i.e., $T \in \mathcal{O}(\frac{1}{\epsilon})$)

15.6 μ -strongly Convex Function

We can achieve (guarantee) the faster convergence if f is strongly convex.

If f is μ -strongly convex and L -smooth (twice differentiable), then h is $L - \mu$ -smooth.

$$h(x) = f(x) - \frac{\mu}{2} \|x\|^2$$

Proof

Since

$$\nabla^2 f = \nabla^2 h + \mu I,$$

we have

$$\nabla^2 h = \nabla^2 f - \mu I \leq (L - \mu)I.$$

Since h is convex and $\nabla h(x) = \nabla f(x) - \mu x$

$$0 \leq (\nabla h(x) - \nabla h(y))^T (x - y) = (\nabla f(x) - \nabla f(y))^T (x - y) - \mu \|x - y\|^2 \leq (L - \mu) \|x - y\|^2.$$

From co-coercivity of h , we also have

$$(\nabla h(x) - \nabla h(y))^T(x - y) \geq \frac{1}{L - \mu} \|\nabla h(x) - \nabla h(y)\|^2.$$

Equivalently,

$$\begin{aligned} & (\nabla f(x) - \nabla f(y) - \mu x + \mu y)^T(x - y) \geq \frac{1}{L - \mu} \|\nabla f(x) - \nabla f(y) - \mu x + \mu y\|^2 \\ \Leftrightarrow & (\nabla f(x) - \nabla f(y))^T(x - y) - \mu \|x - y\|^2 \geq \frac{1}{L - \mu} \|\nabla f(x) - \nabla f(y)\|^2 + \frac{\mu^2}{L - \mu} \|x - y\|^2 - 2 \frac{\mu}{L - \mu} (\nabla f(x) - \nabla f(y))^T(x - y) \\ \Leftrightarrow & \frac{L + \mu}{L - \mu} (\nabla f(x) - \nabla f(y))^T(x - y) \geq \frac{1}{L - \mu} \|\nabla f(x) - \nabla f(y)\|^2 + \frac{L\mu}{L - \mu} \|x - y\|^2 \\ \Leftrightarrow & (\nabla f(x) - \nabla f(y))^T(x - y) \geq \frac{1}{L + \mu} \|\nabla f(x) - \nabla f(y)\|^2 + \frac{L\mu}{L + \mu} \|x - y\|^2. \end{aligned}$$

Consider the gradient descent method with μ -strongly convex assumption of f , where

$$x^+ = x - \alpha \nabla f(x).$$

Let the step size $\alpha \leq \frac{2}{\mu + L}$, then from the above co-coercivity:

$$\begin{aligned} \|x^+ - x^*\|^2 &= \|x - \alpha \nabla f(x) - x^*\|^2 \\ &= \|x - x^*\|^2 - \underbrace{2\alpha \nabla f(x)^T(x - x^*)}_{=2\alpha(\nabla f(x) - \nabla f(x^*))^T(x - x^*)=0} + \alpha^2 \|\nabla f(x)\|^2 \\ &\leq \|x - x^*\|^2 - 2\alpha \cdot \frac{1}{L + \mu} \|\nabla f(x)\|^2 - 2\alpha \cdot \frac{L\mu}{L + \mu} \|x - x^*\|^2 + \alpha^2 \|\nabla f(x)\|^2 \\ &\leq \left(1 - 2\alpha \frac{L\mu}{L + \mu}\right) \|x - x^*\|^2 + \left(\alpha^2 - \frac{2\alpha}{L + \mu}\right) \|\nabla f(x)\|^2 \leq \left(1 - 2\alpha \frac{L\mu}{L + \mu}\right) \|x - x^*\|^2. \end{aligned}$$

which is because $\nabla f(x^*) = 0$.

Let $c = 1 - 2\alpha \frac{L\mu}{L + \mu} < 1$. Then, the gradient descent is a "contractive" operation

$$\begin{aligned} \|x^+ - x^*\|^2 &\leq \underbrace{1 - 2\alpha \cdot \frac{L\mu}{L + \mu}}_c \|x - x^*\|^2 = c \|x - x^*\|^2 \quad (c < 1) \\ \|x^{(k+1)} - x^*\|^2 &\leq c \cdot \|x^{(k)} - x^*\|^2 \leq \dots \leq \underbrace{c^{(k+1)} \|x^{(0)} - x^*\|^2}_{\text{Converges to 0 as } k \text{ increases "exponentially"}} \end{aligned}$$

Thus, now $x^{(k+1)}$ converges to x^* exponentially fast. Furthermore

$$f(x^{(k+1)}) - f(x^*) \leq \frac{L}{2} \|x^{(k+1)} - x^*\|^2 \leq \underbrace{c^{(k+1)} \cdot \frac{L}{2} \|x^{(0)} - x^*\|^2}_{\text{Converges to 0}}.$$

Thus, the function value also converges to $f(x^*)$ exponentially fast.

Remark

If f is

- Convex

$$\begin{aligned} f(x^{(k+1)}) - f(x^*) &\approx 1/k \\ k &\approx \mathcal{O}\left(\frac{1}{\epsilon}\right) \end{aligned}$$

- μ -strongly convex

$$\begin{aligned} f(x^{(k+1)}) - f(x^*) &\approx c^k \\ k &\approx \mathcal{O}\left(\log\left(\frac{1}{\epsilon}\right)\right) \end{aligned}$$

This denotes that smaller number of Gradient Descent iterations can be applied.

16 Stochastic Gradient Descent

In many deep learning applications, the loss function is of form

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \ell(f_{\theta}(x^{(i)}), y^{(i)}) = \frac{1}{N} \sum_{i=1}^N \mathcal{L}_i(\theta) \quad (N \gg 1)$$

in principle

$$\theta^+ = \theta - \alpha \cdot \nabla \mathcal{L}(\theta) = \theta - \alpha \cdot \nabla \sum_{i=1}^N \mathcal{L}_i(\theta)$$

For example, in linear regression problem, we have N data points $\{(x^{(1)}, y^{(1)}), \dots, (x^{(N)}, y^{(N)})\}$ with model h_{θ} and loss function $\ell(\cdot, \cdot)$. The ERM is to minimise the following risk:

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \ell(h_{\theta}(x^{(i)}), y^{(i)}),$$

where h_{θ} is a model, which is of form of $\mathcal{L}(\theta)$. In this section, we assume that the N is extremely large (number of data is huge, for instance, LLM training). Since computing the gradient of N functions are challenging, we often apply stochastic gradient descent (SGD) in such case.

In full gradient descent, one should compute

$$\theta_{k+1} = \theta_k - \alpha \frac{1}{N} \sum_{i=1}^N \nabla \ell(h_{\theta}(x^{(i)}), y^{(i)})$$

at k -th iteration. The idea of stochastic gradient descent (SGD) is to approximate the true gradient with a subsample, i.e.,

$$\theta_{k+1} = \theta_k - \alpha \nabla \ell(h_{\theta}(x^{(i_k)}), y^{(i_k)})$$

where i_k is randomly chosen index form $\{1, 2, \dots, N\}$. Note that

$$\mathbb{E}[\nabla \ell(h_{\theta}(x^{(i_k)}), y^{(i_k)})] = \frac{1}{N} \sum_{i=1}^N \nabla \ell(h_{\theta}(x^{(i)}), y^{(i)})$$

and the estimate is unbiased. In practice, the indices are usually chosen without replacement until we complete one full cycle through the entire dataset.

16.1 Mini-Batching

A common technique employed with SGD is mini-batching, where we choose a random subset

$I_k \in \{1, \dots, N\}$ with size $|I_k| = b \ll N$. We then repeat

$$\theta_{k+1} = \theta_k - \alpha \cdot \frac{1}{b} \sum_{i \in I_k} \nabla \mathcal{L}_i(\theta) = \theta_k - \alpha \cdot \frac{1}{b} \sum_{i \in I_k} \nabla \ell(h_{\theta}(x^{(i)}), y^{(i)})$$

Since

$$\mathbb{E} \left[\frac{1}{b} \sum_{i \in I_k} \nabla \ell(h_{\theta}(x^{(i)}), y^{(i)}) \right] = \frac{1}{N} \sum_{i=1}^N \nabla \ell(h_{\theta}(x^{(i)}), y^{(i)})$$

we still have an unbiased estimate of the full gradient. Furthermore, we not reduced the variance of our gradient estimate by $1/b$, but we have also incurred b times more computational cost. In practice, the batch-update can be parallelised and we then indeed gain computational savings.

16.2 SGD Analysis

If we pick I_k uniform at random

$$\mathbb{E}_{I_k} \left[\underbrace{\frac{1}{b} \sum_{i \in I_k} \nabla \mathcal{L}_i(\theta)}_{g_k} \right] = \frac{1}{N} \sum_{i=1}^N \nabla \mathcal{L}_i(\theta)$$

SGD is

$$\theta_{k+1} = \theta - \alpha \cdot g_k \leftarrow \text{stochastic gradient}$$

Now let g_k be a stochastic gradient of $\mathcal{L}$ at θ_k . We assume that it is unbiased

$$\mathbb{E}_k[g_k] = \mathbb{E}[g_k | \theta_0, \dots, \theta_k] = \nabla \mathcal{L}(\theta_k)$$

where the expectation is conditioned on the iterates up to θ_k . We also assume that the conditional variance is bounded:

$$\mathbb{E}_k[(g_k - \mathbb{E}_k[g_k])^2] \leq \sigma^2.$$

This implies that the stochastic gradient is not too noisy (or close to the true gradient).

***Note** that the following expressions in the examples represents the equivalent meaning.

“ g_k has small variance” = “variance is reasonable” = “gradient is not too noisy” = “close to the true gradient”

Lemma

Let $\mathcal{L}(\theta)$ be L -Lipschitz continuous convex function where the minimiser θ^* exists. Let θ_0 be a starting point and $K > 0$ be the total iteration count. Then, Stochastic Gradient Descent (SGD) with the constant learning rate

$$\alpha_k = \alpha = \frac{\|\theta_0 - \theta^*\|}{\sqrt{L^2 + \sigma^2} \sqrt{K+1}}, \quad (\exists \alpha)$$

exhibits the rate

$$\mathbb{E}[\mathcal{L}(\bar{\theta}_K) - \mathcal{L}(\theta^*)] \leq \underbrace{\frac{\sqrt{L^2 + \sigma^2} \|\theta_0 - \theta^*\|}{\sqrt{K+1}}}_{\mathcal{O}(1/\sqrt{k})}$$

where

$$\bar{\theta}_K = \frac{1}{K+1} \sum_{k=0}^K \theta_k \quad (\text{i.e., } \theta_0 \rightarrow \theta_1 \rightarrow \theta_2 \rightarrow \dots \rightarrow \theta_K)$$

Proof

First,

$$\begin{aligned} \mathbb{E}[\|\theta_{k+1} - \theta^*\|^2 | \theta_0 - \theta_k] &= \|\theta_k - \theta^*\|^2 - 2\alpha \langle \mathbb{E}[g_k], \theta_k - \theta^* \rangle + \alpha^2 \mathbb{E}[\|g_k\|^2] \\ &= \|\theta_k - \theta^*\|^2 - 2\alpha \langle \nabla \mathcal{L}(\theta_k), \theta_k - \theta^* \rangle + \alpha^2 \mathbb{E}[\|g_k\|^2] \\ &\leq \|\theta_k - \theta^*\|^2 - 2\alpha (\mathcal{L}(\theta_k) - \mathcal{L}(\theta^*)) + \alpha^2 (L^2 + \sigma^2). \\ \mathbb{E}[\|\theta_{k+1} - \theta^*\|^2] &\leq \|\theta_k - \theta^*\|^2 - 2\alpha (\mathcal{L}(\theta_k) - \mathcal{L}(\theta^*)) + \alpha^2 (L^2 + \sigma^2). \end{aligned}$$

We take the total expectation on both sides to get

$$\mathbb{E}[\|\theta_{k+1} - \theta^*\|^2] \leq \mathbb{E}[\|\theta_k - \theta^*\|^2] - 2\alpha \mathbb{E}[\mathcal{L}(\theta_k) - \mathcal{L}(\theta^*)] + \alpha^2 (L^2 + \sigma^2).$$

By summing up from $k = 0$ to K , we have

$$\mathbb{E}[\|\theta_K - \theta^*\|^2] \leq \mathbb{E}[\|\theta_0 - \theta^*\|^2] - \sum_{k=0}^K 2\alpha \mathbb{E}[\mathcal{L}(\theta_k) - \mathcal{L}(\theta^*)] + (K+1)\alpha^2 (L^2 + \sigma^2).$$

I.e.,

$$\begin{aligned} \frac{1}{K+1} \sum_{k=0}^K \mathbb{E}[\mathcal{L}(\theta_k) - \mathcal{L}(\theta^*)] &\leq \frac{1}{(K+1)2\alpha} \mathbb{E}[||\theta_0 - \theta^*||^2] - \frac{1}{(K+1)2\alpha} \mathbb{E}[||\theta_{K+1} - \theta^*||^2] + \frac{\alpha}{2}(L^2 + \sigma^2) \\ &\leq \frac{1}{(K+1)2\alpha} ||\theta_0 - \theta^*||^2 + \frac{\alpha}{2}(L^2 + \sigma^2) = \frac{1}{\sqrt{K+1}\sqrt{L^2 + \sigma^2}} ||\theta_0 - \theta^*|| \quad (\text{for the predefined } \alpha) \end{aligned}$$

On the other hand, due to the convexity, we have

$$\frac{1}{K+1} \sum_{k=0}^K \mathbb{E}[\mathcal{L}(\theta_k) - \mathcal{L}(\theta^*)] \geq \mathbb{E}[\mathcal{L}(\bar{\theta}_K) - \mathcal{L}(\theta^*)].$$

Finally, we have

$$\mathbb{E}[\mathcal{L}(\bar{\theta}_K) - \mathcal{L}(\theta^*)] \leq \frac{1}{\sqrt{K+1}\sqrt{L^2 + \sigma^2}} ||\theta_0 - \theta^*||$$

Remark

$$\underbrace{\frac{1}{\sqrt{K+1}\sqrt{L^2 + \sigma^2}} ||\theta_0 - \theta^*||^2}_{\text{Converges to 0 as } k \rightarrow \infty} \geq \frac{1}{K+1} \sum_{k=0}^K \mathbb{E}[\mathcal{L}(\theta_k) - \mathcal{L}(\theta^*)] \geq \underbrace{\mathbb{E}[\mathcal{L}(\bar{\theta}_K) - \mathcal{L}(\theta^*)]}_{\left(\sum_{k=0}^K \mathcal{L}(\theta_k) \geq \mathcal{L}\left(\frac{1}{K+1} \sum_{k=0}^K \theta_k\right)\right)}$$

$$\mathcal{L}(\bar{\theta}_K) - \mathcal{L}(\theta^*) \approx \mathcal{O}\left(\frac{1}{\sqrt{K}}\right)$$

The convergence rate of Stochastic Gradient Descent of convex function is $\mathcal{O}(1/\sqrt{K})$.

- GD & Convex $\Rightarrow \mathcal{O}(1/K)$
- GD & Strongly Convex $\Rightarrow \mathcal{O}(c^K)$

***Note** that in stochastic gradient descent (SGD), the theoretical proof guarantees on $\bar{\theta}_K$ rather than θ_K . This means SGD is applied over K iterations, and the average of θ_k values from $k = 0$ to K is considered. However, in practice, it is common to use only the final iterate, θ_K .

17 Regularised Method

Overview

For data $(\mathcal{X}^{(1)}, \mathcal{Y}^{(1)}), \dots, (\mathcal{X}^{(N)}, \mathcal{Y}^{(N)})$

- Model: $f_\theta(x) \approx y$
- Loss: $\frac{1}{N} \sum_{i=1}^N \ell(f_\theta(\mathcal{X}^{(i)}), \mathcal{Y}^{(i)}) = \mathcal{L}(\theta)$
- Goal: $\min_{\theta} \mathcal{L}(\theta)$
 - $f_\theta(\boxed{x}) = y$
 - $f_\theta(\boxed{x'}) = y$
 - if $x \approx x' \Rightarrow f_\theta(x) \approx f_\theta(x')$

Regularised Method

Consider the predictor (or classifier) $f_\theta(x)$ with parameter θ . An important attribute of predictor f is "sensitivity" or "continuity". We say f_θ is insensitive if for x near $\tilde{x}$, the prediction $f_\theta(x)$ is near $f_\theta(\tilde{x})$. Insensitive predictors often generalise well, especially when we do not have a lot of training data.

This is because there is a gap between the population risk

$$\mathbb{E}[\ell(f_\theta(X), Y)]$$

versus the empirical risk

$$\sum_{i=1}^N \ell(f_\theta(x^{(i)}), y^{(i)}).$$

More precisely, the predictor shows a good performance in empirical risk, but not in population risk. Thus, the insensitive predictor is desirable.

A regulariser is a function $r : \mathbb{R}^d \rightarrow \mathbb{R}$ that measures the sensitivity of f_θ . In other words, if $r(\theta)$ is small then f_θ is insensitive.

Consider the linear regression $f_\theta(x) = \theta^T x$, small sensitivity is associated with small θ .

By Cauchy-Schwartz inequality

$$\|f_\theta(x) - f_\theta(\tilde{x})\|_2 \leq \|\theta\|_2 \cdot \|x - \tilde{x}\|_2.$$

This suggest a regulariser $r(\theta) = \|\theta\|_2$

The common choice of regulariser is ℓ_2 or square or ridge regularisation:

$$r(\theta) = \sum_{i=1}^d \theta_i^2.$$

Another possible choice is the ℓ_1 -regulariser where

$$r(\theta) = \sum_{i=1}^d \|\theta_i\|.$$

With the above regulariser, we have the ridge empirical risk minimisation (RERM) problem that minimises

$$\min_{\theta} \underbrace{\frac{1}{N} \sum_{i=1}^N \ell(f_\theta(x^{(i)}, y^{(i)}))}_{\mathcal{L}(\theta)} + \lambda \|\theta\|^2.$$

As λ increases, we have more insensitive solution.

***Note** the following relation between the regularisation and λ

- λ is huge: Robustness matters, *i.e.*, $\lambda \|\theta\|^2$ is more important
- λ is small or $\lambda \approx 0$: Sensitivity matters, *i.e.*, $\frac{1}{N} \sum \ell f_\theta(\cdot)$ is more important

This time the least square and ridge regression objective function is given by the task below

$$\min_{\theta} \|X\theta - Y\|^2 + \lambda \|\theta\|^2.$$

If $\lambda = 0 \Rightarrow \theta^* = X^\dagger Y = (X^T X)^{-1} X^T Y$

If X is a full rank matrix, the minimum achiever is

$$\theta^* = (X^T X + \lambda I)^{-1} X^T Y.$$

Proof

$$\|X\theta - Y\|^2 + \lambda \|\theta\|^2 = \left\| \begin{bmatrix} X \\ \sqrt{\lambda} I \end{bmatrix} \theta - \begin{bmatrix} Y \\ 0 \end{bmatrix} \right\|^2$$

Thus, the minimum achieving θ is

$$\begin{aligned} \theta^* &= \begin{bmatrix} X \\ \sqrt{\lambda} I \end{bmatrix}^\dagger \begin{bmatrix} Y \\ 0 \end{bmatrix} \\ &= \left(\begin{bmatrix} X \\ \sqrt{\lambda} I \end{bmatrix}^T \begin{bmatrix} X \\ \sqrt{\lambda} I \end{bmatrix} \right)^{-1} \begin{bmatrix} X \\ \sqrt{\lambda} I \end{bmatrix}^T \begin{bmatrix} Y \\ 0 \end{bmatrix} \\ &= \left([X^T \quad \sqrt{\lambda} I] \begin{bmatrix} X \\ \sqrt{\lambda} I \end{bmatrix} \right)^{-1} [X^T \quad \sqrt{\lambda} I] \begin{bmatrix} Y \\ 0 \end{bmatrix} \\ &= (X^T X + \lambda I)^{-1} X^T Y. \end{aligned}$$

17.1 Strongly Convex SGD

Overview

Consider $\tilde{\mathcal{L}}(\theta) = \mathcal{L}(\theta) + \lambda \|\theta\|^2$

if $\mathcal{L}$ is convex $\Rightarrow \tilde{\mathcal{L}}(\theta)$ is 2λ -strongly convex

Also, recall the following characteristics

$$\begin{cases} \text{GD} \begin{cases} \Rightarrow \text{Convex} \in \mathcal{O}(1/T) \\ \Rightarrow \text{Strongly Convex} \in \mathcal{O}(c^T) \end{cases} \\ \text{SGD} \Rightarrow \text{Convex} \in \mathcal{O}(1/\sqrt{T}) \end{cases}$$

If $f(x)$ is convex, then $\tilde{f}(x) = f(x) + \frac{\mu}{2}\|x\|^2$ is μ -strongly convex.

$$\min_x f(x) + \frac{\mu}{2}\|x\|^2 = f(x^*) + \frac{\mu}{2}\|x^*\|^2.$$

SGD

- $\mathbb{E}[g_k] = \nabla f(x_k)$
- $\mathbb{E}[\|g_k - \nabla f(x_k)\|^2] \leq \sigma^2$

Strongly convex SGD

Consider the stochastic optimisation problem

$$\min f(x) + \frac{\mu}{2}\|x\|^2.$$

Assume that F is Lipschitz continuous. Consider the stochastic gradient descent (SGD)

$$x_{k+1} = x_k - \alpha_k(g_k + \mu x_k)$$

where g_k is stochastic gradient of f at x_k .

Lemma

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a G -Lipschitz continuous function and x^* be a minimiser of $f(x) + \mu/2\|x\|^2$.

Suppose $\mathbb{E}_k[g_k] = \nabla f(x_k)$ and $\text{Var}_k(g_k) \leq \sigma^2$, then SGD with step size

$$\alpha_k = \frac{1}{\mu(k+1)}$$

exhibits the rate

$$f(\bar{x}_k) - f(x^*) \leq \underbrace{\frac{2(G^2 + \sigma^2)}{\mu}}_c \cdot \underbrace{\frac{1 + \log(k+1)}{k+1}}_{\rightarrow \frac{\log k}{k}} \Rightarrow \mathcal{O}\left(\frac{\log k}{k}\right)$$

where

$$\bar{x}_k = \frac{1}{k+1} \sum_{i=0}^k x_i. \Rightarrow \tilde{\mathcal{O}}\left(\frac{1}{k}\right)$$

***Note** the following properties may help predictability

$$\begin{array}{lll} \text{Predictability} & \Rightarrow & \left. \begin{array}{ll} \text{Curvature} & \Rightarrow \mu \uparrow \\ \text{Concavity} & \Rightarrow G \downarrow \\ \text{Noise of gradient} & \Rightarrow \sigma \downarrow \end{array} \right\} \text{Helps} \end{array}$$

Proof

For simplicity, let assume $x_0 = 0$.

$$x_{k+1} = x_k - \nabla \tilde{f}(x_k) = x_k - \alpha_k \left(\underbrace{g_k}_{\text{Noisy version of } \nabla f(x_k)} + \underbrace{\mu x_k}_{\frac{\mu}{2}\|x_k\|^2} \right) = (1 - \alpha_k \mu) x_k + \alpha_k \mu \left(-\frac{1}{\mu} g_k \right) = \sum_{i=0}^k \theta_i \left(-\frac{1}{\mu} g_i \right)$$

where $\theta_i = \alpha_i \mu \prod_{s=i+1}^k (1 - \alpha_s \mu)$ and $\sum_{i=0}^k \theta_i = 1$.

Using Jensen's inequality,

$$\begin{aligned} \mathbb{E}[\|g_k + \mu x_k\|^2] &\leq 2\mathbb{E}[\|g_k\|^2] + 2\mu^2 \mathbb{E}[\|x_k\|^2] \\ &\leq 2(G^2 + \sigma^2) + 2\mu^2 \sum_{i=0}^k \theta_i \mathbb{E}[\|(-1/\mu)g_i\|^2] = 2(G^2 + \sigma^2) + 2(G^2 + \sigma^2) = 4(G^2 + \sigma^2) \quad \dots\dots (*) \end{aligned}$$

Finally,

$$\begin{aligned} \mathbb{E}[\| \underbrace{x_{k+1} - x^*}_{=x_k - \alpha_k(g_k + \mu x_k)} \|^2 | x_k] &= \|x_k - x^*\|^2 - 2\alpha_k \langle \mathbb{E}_k[g_k] + \mu x_k, x_k - x^* \rangle + \alpha_k^2 \mathbb{E}_k[\|g_k + \mu x_k\|^2] \\ &\leq \|x_k - x^*\|^2 - 2\alpha_k \underbrace{\left(f(x_k) - f(x^*) + \frac{\mu}{2} \|x_k - x^*\|^2 \right)}_{\text{Strong convexity of } \bar{f}} + \underbrace{\alpha_k^2 4(G^2 + \sigma^2)}_{(*)} \\ &\leq (1 - \alpha_k \mu) \|x_k - x^*\|^2 - 2\alpha_k (f(x_k) - f(x^*)) + \alpha_k^2 4(G^2 + \sigma^2). \\ &= \frac{k}{k+1} \|x_k - x^*\|^2 - \frac{2}{\mu(k+1)} (f(x_k) - f(x^*)) + \frac{4(G^2 + \sigma^2)}{\mu^2(k+1)^2} \end{aligned}$$

Rearranging the terms and plugging in $\alpha_k = \frac{1}{\mu(k+1)}$,

$$f(x_k) - f(x^*) \leq \frac{\mu k}{2} \|x_k - x^*\|^2 - \frac{\mu(k+1)}{2} \mathbb{E}_k[\|x_{k+1} - x^*\|^2] + \frac{2(G^2 + \sigma^2)}{\mu(k+1)}.$$

Then, from the telescoping sum argument, we have

$$\begin{aligned} \sum_{i=0}^k f(x_i) - f(x^*) &\leq \underbrace{\frac{\mu \cdot 0}{2} \|x_0 - x^*\|^2}_{\rightarrow 0} - \frac{\mu(k+1)}{2} \|x_{k+1} - x^*\|^2 + \sum_{i=0}^k \frac{2(G^2 + \sigma^2)}{\mu(i+1)} \\ &\leq \frac{2(G^2 + \sigma^2)}{\mu} \underbrace{\sum_{i=0}^k \frac{1}{i+1}}_{\leq 1 + \log(k+1)} \leq \frac{2(G^2 + \sigma^2)}{\mu} (1 + \log(k+1)) \end{aligned}$$

Finally,

$$f(\bar{x}_k) - f(x^*) \leq \frac{1}{k} \sum_{i=0}^k f(x_i) - f(x^*) \leq \frac{2(G^2 + \sigma^2)}{\mu k} (1 + \log(k+1))$$

The convergence rate is

$$\mathcal{O}\left(\frac{\log k}{k}\right).$$

This is faster than SGD convergence rate

$$\mathcal{O}\left(\frac{1}{\sqrt{k}}\right).$$

Remark

The optimal rate (can be achieved and cannot do better than) is

$$\mathcal{O}\left(\frac{1}{k}\right).$$

The above SGD is slightly suboptimal with factor of $\log k$.

18 Operator Theory

Let $\mathbb{T} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be an operator.

Nonexpansive

We say an operator $\mathbb{T}$ is *nonexpansive* if

$$\|\mathbb{T}x - \mathbb{T}y\| \leq \|x - y\|$$

for all x, y .

Note that $\mathbb{T}$ is nonexpansive if it is “1-Lipshitz”.

Furthermore, we say $\mathbb{T}$ is a contraction if it is L -Lipshitz with $L < 1$.

Contractive

We say an operator $\mathbb{T}$ is *contractive* if there exists $L < 1$ such that

$$\|\mathbb{T}x - \mathbb{T}y\| \leq L\|x - y\|$$

for all x, y .

θ -Averaged

For $\theta \in (0, 1)$, we say operator $\mathbb{T}$ is *θ -averaged* if

$$\mathbb{T} = (1 - \theta)\mathbb{I} + \theta\mathbb{S}$$

for an identity operator $\mathbb{I}$ and some nonexpansive operator $\mathbb{S}$.

Lemma

For θ -averaged operator $\mathbb{T}$:

$$\mathbb{T} = (1 - \theta)\mathbb{I} + \theta\mathbb{S}$$

is also nonexpansive.

Proof

For $0 < L < \infty$, define the class of L -Lipshitz operators as

$$\mathcal{L}_L = \{\mathbb{A} : \mathbb{R}^n \rightarrow \mathbb{R}^n \mid \|\mathbb{A}x - \mathbb{A}y\|^2 \leq L^2\|x - y\|^2, \text{ for all } x, y\}.$$

For $0 < \beta < \infty$, define the class of β -coercive operators as

$$\mathcal{C}_\beta = \{\mathbb{A} : \mathbb{R}^n \rightarrow \mathbb{R}^n \mid \langle \mathbb{A}x - \mathbb{A}y, x - y \rangle \geq \beta\|\mathbb{A}x - \mathbb{A}y\|^2, \text{ for all } x, y\}.$$

For $0 < \theta < 1$, define the class of θ -averaged operators as

$$\mathcal{N}_\theta = \{(1 - \theta)\mathbb{I} + \theta\mathbb{A} \mid \mathbb{A} \in \mathcal{L}_1\}.$$

Let

$$\mathbb{A} = (1 - \theta)\mathbb{I} + \theta\mathbb{B} \in \mathcal{N}_\theta$$

for some $\mathbb{B} \in \mathcal{L}_1$. Then,

$$\langle (\mathbb{I} - \mathbb{A})x - (\mathbb{I} - \mathbb{A})y, x - y \rangle = \langle \theta x - \theta\mathbb{B}x - \theta y + \theta\mathbb{B}y, x - y \rangle = \theta\|x - y\|^2 - \theta \langle \mathbb{B}x - \mathbb{B}y, x - y \rangle.$$

On the other hand, from $\mathbb{B} \in \mathcal{L}_1$, we have

$$\begin{aligned} \frac{1}{2\theta} \|(\mathbb{I} - \mathbb{A})x - (\mathbb{I} - \mathbb{A})y\|^2 &= \frac{1}{2\theta} \|\theta x - \theta\mathbb{B}x - \theta y + \theta\mathbb{B}y\|^2 = \frac{1}{2} \theta \|x - y\|^2 + \theta \langle \mathbb{B}x - \mathbb{B}y, x - y \rangle + \frac{1}{2} \theta \|\mathbb{B}x - \mathbb{B}y\|^2 \\ &\leq \frac{1}{2} \theta \|x - y\|^2 - \theta \langle \mathbb{B}x - \mathbb{B}y, x - y \rangle + \frac{1}{2} \theta \|x - y\|^2 = \theta \|x - y\|^2 - \theta \langle \mathbb{B}x - \mathbb{B}y, x - y \rangle = \langle (\mathbb{I} - \mathbb{A})x - (\mathbb{I} - \mathbb{A})y, x - y \rangle. \end{aligned}$$

This implies $\mathbb{A} \in \mathcal{N}_\theta \Rightarrow \mathbb{I} - \mathbb{A} \in \mathcal{C}_{1/(2\theta)}$.

Now let's show the converse. Suppose $\mathbb{I} - \mathbb{A} \in \mathcal{C}_{1/(2\theta)}$, which implies

$$\langle (\mathbb{I} - \mathbb{A})x - (\mathbb{I} - \mathbb{A})y, x - y \rangle \geq \frac{1}{2\theta} \|(\mathbb{I} - \mathbb{A})x - (\mathbb{I} - \mathbb{A})y\|^2.$$

If we decompose $\mathbb{A} = (1 - \theta)\mathbb{I} + \theta\mathbb{B}$, then we have equivalent form

$$\begin{aligned}
 &\Leftrightarrow \langle \theta x - \theta \mathbb{B}x - \theta y + \theta \mathbb{B}y, x - y \rangle \geq \frac{1}{2\theta} \|\theta x - \theta \mathbb{B}x - \theta y + \theta \mathbb{B}y\|^2 \\
 &\Leftrightarrow \theta \|x - y\|^2 - \theta \langle \mathbb{B}x - \mathbb{B}y, x - y \rangle \geq \frac{1}{2} \theta \|x - y\|^2 + \frac{1}{2} \theta \|\mathbb{B}x - \mathbb{B}y\|^2 - \theta \langle \mathbb{B}x - \mathbb{B}y, x - y \rangle \\
 &\Leftrightarrow \theta \|x - y\|^2 - \theta \langle \mathbb{B}x - \mathbb{B}y, x - y \rangle \geq \frac{1}{2} \theta \|x - y\|^2 + \frac{1}{2} \theta \|\mathbb{B}x - \mathbb{B}y\|^2 - \theta \langle \mathbb{B}x - \mathbb{B}y, x - y \rangle \\
 &\Leftrightarrow \frac{1}{2} \theta \|x - y\|^2 \geq \frac{1}{2} \theta \|\mathbb{B}x - \mathbb{B}y\|^2 \\
 &\Leftrightarrow \|x - y\|^2 \geq \|\mathbb{B}x - \mathbb{B}y\|^2
 \end{aligned}$$

which implies $\mathbb{B} \in \mathcal{L}_1$, and finally $\mathbb{I} - \mathbb{A} \in \mathcal{C}_{1/(2\theta)} \Rightarrow \mathbb{A} \in \mathcal{N}_\theta$.

Fixed Point

We say x is a *fixed point* of $\mathbb{T}$ if $\mathbb{T}x = x$.

The algorithm "fixed-point iteration", also called the *Picard iteration* is

$$x^{(k+1)} = \mathbb{T}x^{(k)} \implies x^* = \mathbb{T}x^*$$

for $k \geq 0$ with some initial starting point $x^{(0)}$.

Remark

If $\mathbb{T}$ is a contraction with Lipschitz constant $L < 1$, then

$$\|x^{(k+1)} - x^*\| = \|\mathbb{T}x^{(k)} - \mathbb{T}x^*\| \leq L\|x^{(k)} - x^*\| \leq L^k \|x^{(0)} - x^*\|$$

for fixed point x^* .

Gradient Descent (f is L -smooth)

- x^0 : initial point
- $x^{(k+1)} = x^{(k)} - \alpha \nabla f(x^{(k)})$, $(0 < \alpha < \frac{2}{L})$
- $\mathbb{S}x = x - \frac{2}{L} \nabla f(x) \Rightarrow$ nonexpansive
- $\mathbb{T}x = \theta \mathbb{I} + (1 - \theta)\mathbb{S} \rightarrow \theta$ -averaged

Lemma

Suppose $\mathbb{T}$ is θ -averaged with $\theta \in (0, 1)$, and there exists a fixed point of $\mathbb{T}$.

Then the fixed point iteration converge with any starting point converges to some fixed point x^* , and

$$\|x^{(k+1)} - x^{(k)}\|^2 \leq \frac{1}{k+1} \frac{\theta}{1-\theta} \|x^{(0)} - x^*\|^2.$$

***Note** that this does not guarantee $x^{(k)} \rightarrow x^*$

Proof

We have

$$x^{(k+1)} = \mathbb{T}x^{(k)} = (1 - \theta)x^{(k)} + \theta \mathbb{S}x^{(k)}.$$

Note that for all x, y , the following identity holds

$$\|(1 - \theta)x + \theta y\|^2 = (1 - \theta)\|x\|^2 + \theta\|y\|^2 - \theta(1 - \theta)\|x - y\|^2.$$

Then, for a fixed point x^* , we have

$$\begin{aligned}
 \|x^{(k+1)} - x^*\|^2 &= (1 - \theta)\|x^{(k)} - x^*\|^2 + \theta\|\mathbb{S}x^{(k)} - x^*\|^2 - \theta(1 - \theta)\|\mathbb{S}x^{(k)} - x^{(k)}\|^2 \\
 &\leq (1 - \theta)\|x^{(k)} - x^*\|^2 + \theta\|x^{(k)} - x^*\|^2 - \theta(1 - \theta)\|\mathbb{S}x^{(k)} - x^{(k)}\|^2 \\
 &= \|x^{(k)} - x^*\|^2 - \theta(1 - \theta)\|\mathbb{S}x^{(k)} - x^{(k)}\|^2. \quad \dots (*)
 \end{aligned}$$

This implies the monotone decreases $\|x^{(k+1)} - x^*\| \leq \|x^{(k)} - x^*\|$.

Consider the fixed point residual $\mathbb{T}x^{(k)} - x^{(k)}$. Since $\mathbb{T}$ is nonexpansive,

$$\|x^{(k+1)} - x^{(k)}\| \leq \|x^{(k)} - x^{(k-1)}\|.$$

By summing (*),

$$\|x^{(k+1)} - x^*\|^2 \leq \|x^{(0)} - x^*\|^2 - \frac{1-\theta}{\theta} \sum_{j=0}^k \|\mathbb{T}x^{(j)} - x^{(j)}\|^2.$$

This is equivalent to

$$\sum_{j=0}^k \|\mathbb{T}x^{(j)} - x^{(j)}\|^2 \leq \frac{\theta}{1-\theta} \|x^{(0)} - x^*\|^2 - \frac{\theta}{1-\theta} \|x^{(k+1)} - x^*\|^2.$$

Since $\|\mathbb{T}x^{(j)} - x^{(j)}\|$ monotonically decreases, we have

$$(k+1) \|x^{(k+1)} - x^{(k)}\|^2 \leq \sum_{j=0}^k \|\mathbb{T}x^{(j)} - x^{(j)}\|^2 \leq \frac{\theta}{1-\theta} \|x^{(0)} - x^*\|^2.$$

The gradient descent algorithm is

$$x^{(k+1)} = \mathbb{T}x^{(k)}$$

where

$$\mathbb{T} = \mathbb{I} - \alpha \nabla f.$$

for a learning rate α .

Consider an operator $\mathbb{S}$

$$\mathbb{S} = \mathbb{I} - \frac{2}{L} \nabla f.$$

From L -smoothness of f ,

$$\begin{aligned} \|\mathbb{S}x - \mathbb{S}y\|^2 &= \|x - \left(\frac{2}{L}\right) \nabla f(x) - (y - \left(\frac{2}{L}\right) \nabla f(y))\|^2 \\ &= \|x - y\|^2 - \frac{4}{L} \langle x - y, \nabla f(x) - \nabla f(y) \rangle + \frac{4}{L^2} \|\nabla f(x) - \nabla f(y)\|^2 \\ &= \|x - y\|^2 - \frac{4}{L} \underbrace{\left(\langle x - y, \nabla f(x) - \nabla f(y) \rangle - \frac{1}{L} \|\nabla f(x) - \nabla f(y)\|^2 \right)}_{\geq 0} \leq \|x - y\|^2. \end{aligned}$$

Thus, $\mathbb{S}$ is nonexpansive, and $\mathbb{T}$ is θ -averaged operator for $\alpha \in (0, \frac{2}{L})$ with $\theta = \frac{\alpha L}{2} < 1$.

Consequently $x^{(k)}$ converges to a fixed point x^* , where the fixed point x^* satisfies

$$\nabla f(x^*) = 0.$$

If there exists a fixed point, then

$$\begin{aligned} x^{(k+1)} &= x^{(k)} - \alpha \cdot \nabla f(x^{(k)}) \\ \|\nabla f(x^{(k)})\|^2 &= O\left(\frac{1}{k}\right). \end{aligned}$$

Further, if f is μ -strongly convex and ∇f is L -Lipschitz, we can show the iteration is a contraction.

By the fundamental theorem of calculus,

$$\begin{aligned} \mathbb{T}x - \mathbb{T}y &= (x - y) - \alpha(\nabla f(x) - \nabla f(y)) \\ &= (x - y) - \alpha \int_0^1 \nabla^2 f(tx + (1-t)y)(x - y) dt \\ &= \int_0^1 [\mathbb{I} - \alpha \nabla^2 f(tx + (1-t)y)](x - y) dt. \end{aligned}$$

Then,

$$\begin{aligned} \|\mathbb{T}x - \mathbb{T}y\| &= \left\| \int_0^1 [\mathbb{I} - \alpha \nabla^2 f(tx + (1-t)y)](x-y) dt \right\| \\ &\leq \int_0^1 \left\| [\mathbb{I} - \alpha \nabla^2 f(tx + (1-t)y)](x-y) \right\| dt \\ &\leq \int_0^1 \left\| \mathbb{I} - \alpha \nabla^2 f(tx + (1-t)y) \right\| dt \|x-y\|. \end{aligned}$$

Since $\mu I \leq \nabla^2 f(z) \leq LI$ for all z , every eigenvalue λ of $\nabla^2 f(z)$ lies in $[\mu, L]$, so the eigenvalues of $\mathbb{I} - \alpha \nabla^2 f(z)$ lie in $\{1 - \alpha\lambda : \lambda \in [\mu, L]\}$. Therefore

$$\left\| \mathbb{I} - \alpha \nabla^2 f(z) \right\| \leq \max_{\lambda \in [\mu, L]} |1 - \alpha\lambda| = \max\{|1 - \alpha\mu|, |1 - \alpha L|\}.$$

Hence

$$\|\mathbb{T}x - \mathbb{T}y\| \leq \max\{|1 - \alpha\mu|, |1 - \alpha L|\} \|x - y\|.$$

In particular, if $0 < \alpha < \frac{2}{L}$, then $\max\{|1 - \alpha\mu|, |1 - \alpha L|\} < 1$ and $\mathbb{T}$ is a contraction.

19 Acceleration Methods

Overview

- f : Convex and L -smooth function
- Gradient Descent Method: $\mathcal{O}(\frac{1}{k})$ rate
- AGM: $\mathcal{O}(\frac{1}{k^2})$

Acceleration Methods

For convex and L -smooth function f , the gradient descent method provides an $\mathcal{O}(1/k)$ rate. It is natural to ask whether we can increase the speed of convergence (*i.e.*, the worst-case rate), and we can.

In optimisation, an *acceleration* is a modification of a base method to improve the convergence rate.

Considering the problem

$$\min_{x \in \mathbb{R}^n} f(x)$$

where f is convex and L -smooth. The method $\dots(\star)$

$$x_{k+1} = y_k - \frac{1}{L} \nabla f(y_k)$$

$$y_{k+1} = x_{k+1} + \frac{k-1}{k+2} (x_{k+1} - x_k)$$

where $x_0 = y_0$, is called *Nesterov's accelerated gradient method (AGM)*.

***Note** that the following formula is equivalent to AGM: $\dots(\star\star)$

$$x_{k+1} = y_k - \frac{1}{L} \nabla f(y_k)$$

$$z_{k+1} = z_k - \frac{k+1}{2L} \nabla f(y_k)$$

$$y_{k+1} = \left(1 - \frac{2}{k+2}\right) x_{k+1} + \frac{2}{k+2} z_{k+1}.$$

where $x_0 = y_0 = z_0$.

Proof

Suppose that the sequences x_k and y_k from both methods match up to iteration k .

From $(\star\star)$, we have

$$z_{k+1} = z_k + \frac{k+1}{2}(x_{k+1} - y_k).$$

From $(\star\star)$, we have

$$\begin{aligned} y_{k+1} &= \left(1 - \frac{2}{k+2}\right)x_{k+1} + \frac{2}{k+2}z_{k+1} = \left(1 - \frac{2}{k+2}\right)x_{k+1} + \frac{2}{k+2}\left(z_k + \frac{k+1}{2}(x_{k+1} - y_k)\right) \\ &= x_{k+1} + \frac{k-1}{k+2}(x_{k+1} - x_k) + \frac{k+1}{k+2}\left(\frac{k-1}{k+1}x_k + \frac{2}{k+1}z_k - y_k\right) = x_{k+1} + \frac{k-1}{k+2}(x_{k+1} - x_k). \end{aligned}$$

where the last inequality is due to the definition of y_k in $(\star\star)$.

This coincide with the formula in $(\star)$.

Lemma

Assume the convex, L -smoothe function f has a global minimiser x^* .

Then AGM converges with the rate:

$$f(x_k) - f(x^*) \leq \frac{2L}{k^2} \|x_0 - x^*\|^2.$$

Proof

Let $\theta_k = \frac{k+1}{2}$, then it satisfies

$$\theta_k^2 - \theta_k \leq \theta_{k-1}^2.$$

That $f(x)$ is L -smooth implies $f(x) - \frac{L}{2}\|x - y_k\|^2$ is concave due to the fact that $\nabla^2 f - LI \leq 0$.

This concavity implies:

$$f(x) - \frac{L}{2}\|x - y_k\|^2 \leq f(y_k) + \nabla f(y_k)^T(x - y_k).$$

Instead of x , we plug in $x = x_{k+1} = -y_k - \frac{1}{L}\nabla f(y_k)$, then we have

$$f(x_{k+1}) - f(y_k) + \frac{1}{2L}\|\nabla f(y_k)\|^2 \leq 0.$$

From convexity of f , we have

$$\begin{aligned} f(y_k) - f(x_k) &\leq \nabla f(y_k)^T(y_k - x_k) \\ f(y_k) - f(x^*) &\leq \nabla f(y_k)^T(y_k - x^*). \end{aligned}$$

Note that

$$z_{k+1} = x_k - \frac{\theta_k}{L}\nabla f(y_k)$$

and

$$\|z_{k+1} - x^*\| = \left\| z_k - \frac{\theta_k}{L}\nabla f(y_k) - x^* \right\|^2 = \|z_k - x^*\|^2 - 2\frac{\theta_k}{L}\nabla f(y_k)^T(z_k - x^*) + \frac{\theta_k^2}{L^2}\|\nabla f(y_k)\|^2.$$

Define

$$V_k = \theta_{k-1}^2(f(x_k) - f(x^*)) + \frac{L}{2}\|z_k - x^*\|^2.$$

Then,

$$\begin{aligned} V_{k+1} - V_k &= \theta_k^2(f(x_{k+1}) - f(x^*)) + \frac{L}{2}\|z_{k+1} - x^*\|^2 - \theta_{k-1}^2(f(x_k) - f(x^*)) - \frac{L}{2}\|z_k - x^*\|^2 \\ &= \theta_k^2\left(f(x_{k+1}) - f(x^*) + \frac{1}{2L}\|\nabla f(y_k)\|^2\right) - \theta_k\nabla f(y_k)^T(z_k - x^*) - \theta_{k-1}^2(f(x_k) - f(x^*)) \\ &\leq \theta_k^2(f(y_k) - f(x^*)) - \theta_k\nabla f(y_k)^T(z_k - x^*) - \theta_{k-1}^2(f(x_k) - f(x^*)) \end{aligned}$$

$$\begin{aligned}
&= (\theta_k^2 - \theta_k)(f(y_k) - f(x^*)) + \theta_k(f(y_k) - f(x_k)) + (\theta_k^2 - \theta_{k-1}^2)(f(x_k) - f(x^*)) - \theta_k \nabla f(y_k)^T (z_k - x^*) \\
&\leq (\theta_k^2 - \theta_k)(f(y_k) - f(x_k)) + \theta_k(f(y_k) - f(x^*)) - \theta_k \nabla f(y_k)^T (z_k - x^*) \\
&\leq (\theta_k^2 - \theta_k) \nabla f(y_k)^T (y_k - x_k) + \theta_k \nabla f(y_k)^T (y_k - x^*) - \theta_k \nabla f(y_k)^T (z_k - x^*) \\
&\leq \theta_k \nabla f(y_k)^T ((1 - \theta_k)x_k + \theta_k y_k - z_k) = 0,
\end{aligned}$$

where the last equality is from the definition of z_k .

$$y_k = \left(1 - \frac{1}{\theta_k}\right)x_k + \frac{1}{\theta_k}z_k.$$

Thus, V_k is a non-increasing sequence

$$V_k \leq V_{k-1} \leq \dots \leq V_0.$$

Since $\theta_{-1} = 0$, we have

$$\theta_{k-1}^2(f(x_k) - f(x^*)) \leq V_k = f(x_k) - f(x^*) + \frac{L}{2}\|z_k - x^*\|^2 \leq \frac{L}{2}\|z_0 - x^*\|^2.$$

Since $z_0 = x_0$, we finally get

$$f(x_k) - f(x^*) \leq \frac{2L}{k^2}\|x_0 - x^*\|^2.$$

First Order Method

We call “*First-order method*” be any iterative algorithm that selects x_{k+1} in the set

$$x_0 + \text{span}\{\nabla f(x_0), \nabla f(x_1), \dots, \nabla f(x_k)\}.$$

Consider the function class f that satisfies the followings.

1. f is convex and differentiable
2. f is L -smooth
3. The optimal x^* exists.

Then, the following results shows the converse (impossibility).

For every integer $k \leq \frac{d-1}{2}$ and every x_0 , there exists a function in the class that for any *first-order method*

$$f(x_k) - f(x^*) \geq \frac{2}{32} \frac{L\|x_0 - x^*\|^2}{(k+1)^2}$$

This implies that you cannot optimise (meaningfully) better than the lower bound represented above.

20 Variants of Gradient Descent

20.1 Gradient Descent with Momentum (GDM)

Gradient Descent with Momentum accelerates convergence by adding a fraction of the previous update to the current update. It is particularly effective on loss surfaces with steep gradients or oscillations.

Key Components

- $f(\mathbf{w})$: what to minimise
- $v_0 = 0, \mathbf{w}_0$: initialisation

Update Rule:

$$\begin{aligned}
\mathbf{v}_t &= \beta \mathbf{v}_{t-1} + \eta \nabla_{\mathbf{w}} \mathcal{L}(\mathbf{w}_t), \\
\mathbf{w}_{t+1} &= \mathbf{w}_t - \mathbf{v}_t,
\end{aligned}$$

where:

- $\mathbf{v}_t$ is the velocity (momentum) term.
- $\beta \in [0, 1)$ controls the contribution of past gradients.

- η is the learning rate.

Role of the Momentum

Momentum helps the optimiser navigate through ravines and accelerates convergence in directions of consistent gradients while dampening oscillations.

It also helps solve problems like *obstacles* or *saddle-points*.

20.2 RMSProp

RMSprop (Root Mean Square Propagation) adjusts the learning rate for each parameter based on the magnitude of recent gradients. It normalises gradients by their running average, preventing excessively large updates and stabilising training.

Update Rule:

$$\begin{aligned} \mathbf{s}_t &= \gamma \mathbf{s}_{t-1} + (1 - \gamma) (\nabla_{\mathbf{w}} \mathcal{L}(\mathbf{w}))^2, \\ \mathbf{w}_{t+1} &= \mathbf{w}_t - \frac{\eta}{\sqrt{\mathbf{s}_t + \epsilon}} \nabla_{\mathbf{w}} \mathcal{L}(\mathbf{w}), \end{aligned}$$

where:

- The square of Gradient is an element-wise operation
- Square root and division $\sqrt{\mathbf{s}_t + \epsilon}$ is also an element-wise operation.
- $\mathbf{s}_t$ is the exponentially decaying average of squared gradients.
- γ is the decay rate, typically set to 0.9
- ϵ is a small constant added for numerical stability.

RMSprop works well for non-stationary objects and is particularly effective in deep learning tasks.

20.3 Adam

Adam (Adaptive Moment Estimation) combines the benefits of RMSProp and Momentum. It maintains a moving average of both the gradients and their squared values, providing an adaptive learning rate that stabilises updates.

Update Rule:

$$\begin{aligned} \mathbf{v}_t &= \beta_1 \mathbf{v}_{t-1} + (1 - \beta_1) \nabla_{\mathbf{w}} \mathcal{L}(\mathbf{w}), \\ \mathbf{s}_t &= \beta_2 \mathbf{s}_{t-1} + (1 - \beta_2) (\nabla_{\mathbf{w}} \mathcal{L}(\mathbf{w}))^2, \\ \hat{\mathbf{v}}_t &= \frac{\mathbf{v}_t}{1 - \beta_1^t}, \quad \hat{\mathbf{s}}_t = \frac{\mathbf{s}_t}{1 - \beta_2^t}, \\ \mathbf{w}_{t+1} &= \mathbf{w}_t - \frac{\eta}{\sqrt{\hat{\mathbf{s}}_t + \epsilon}} \hat{\mathbf{v}}_t, \end{aligned}$$

where:

- $\mathbf{v}_t$ and $\mathbf{s}_t$ are the first and second moment estimates, respectively.
- β_1 and β_2 control the decay rates for the moments, typically set to 0.9 and 0.999, respectively.
- ϵ ensures numerical stability.

Adam is highly effective in practice, offering robust convergence across a variety of tasks and settings.

20.4 Key Differences

- **Gradient Adaption:** RMSprop adapt learning rates for each parameter, while GDM applies a single global learning rate with added momentum.
- **Efficiency:** Adam combines the strengths of RMSprop and Momentum, making it a default choice in many deep learning applications.
- **Decay Rates:** RMSprop and Adam rely on exponential moving averages, while Adagrad accumulates gradients over time, leading to different behaviours during training.

21 Sharpness-Aware Minimisation (SAM)

Overview

$$\mathbb{E}[\ell(f_\theta(x), y)], \quad (x, y) \sim \mathcal{D}$$

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \ell(f_\theta(x_i), y_i), \quad \theta^* = \min \mathcal{L}(\theta)$$

$$\theta^* \text{ works well on } ((x_1, y_1), \dots, (x_N, y_N))$$

$$\implies \theta^* \text{ will work well, for generic } (x, y) \sim \mathcal{D} \quad (\text{generalisation})$$

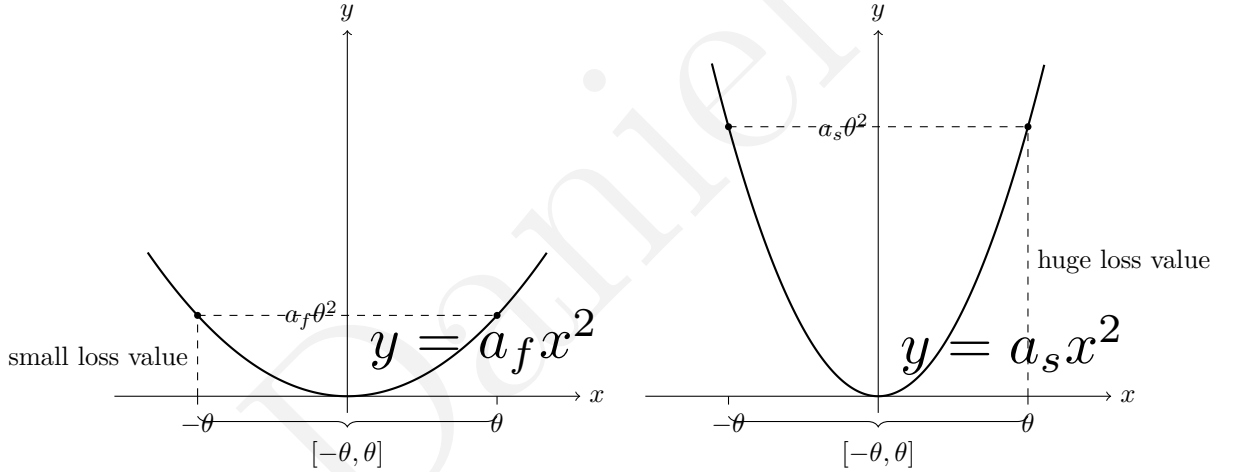
$$\implies \text{Sharpness matters.}$$

For small loss value (i.e., almost approximate to zero), we want “flatness”.

Considering these, the constraints are as follows:

We want θ that minimises $\mathcal{L}(\theta)$ & we want θ in flat region.

$$\min_{\theta} \left[\underbrace{\left(\max_{\|\epsilon\| \leq \rho} \mathcal{L}(\theta + \epsilon) - \mathcal{L}(\theta) \right)}_{\text{Sharpness}} + \underbrace{\mathcal{L}(\theta)}_{\text{Loss}} \right] \rightarrow \begin{pmatrix} \min_{\theta} \max_{\|\epsilon\| \leq \rho} \mathcal{L}(\theta + \epsilon) \cong \min_{\theta} \mathcal{L}(\theta + \epsilon^*) \\ \theta_{t+1} = \theta_t - \eta \nabla \mathcal{L}(\theta + \epsilon^*) \\ \epsilon^* = \rho \frac{\nabla \mathcal{L}(\theta)}{\|\nabla \mathcal{L}(\theta)\|} \end{pmatrix}$$



Sharpness-Aware Minimisation (SAM)

Deep learning models often exhibit overfitting and sensitivity to perturbations in input data or model parameters, primarily due to the sharpness of the loss landscape. Sharp minima, characterised by steep gradients around the optimal parameter values, make models sensitive to small changes, leading to poor generalisation on unseen data. This issue is especially pronounced in over-parameterised models, such as deep neural networks, where numerous solutions may perform equally well on the training set but differ significantly in generalisation.

To address this, **Sharpness-Aware Minimisation (SAM)** focuses on finding flat minima, defined as regions in the loss landscape where the loss remains relatively constant under small perturbations of the model parameters. Flat minima are empirically correlated with better generalisation properties, making SAM an effective method for robust optimisation.

Given a loss function $\mathcal{L}(\mathbf{w}, \mathcal{D})$ for model parameters $\mathbf{w}$ over a dataset $\mathcal{D}$, SAM seeks to optimise a surrogate objective that incorporates worst-case perturbations in the parameter space. The objective is defined as:

$$\min_{\mathbf{w}} \max_{\|\epsilon\|_p \leq \rho} \mathcal{L}(\mathbf{w} + \epsilon, \mathcal{D}),$$

where $\|\epsilon\|_p \leq \rho$ constrains the perturbation ϵ to a p -norm ball of radius ρ . This formulation ensures that

the learned model is robust to small perturbations in parameter space, favouring flatter minima.

Theorem (Informal): *Minimising the sharpness-aware objective* leads to a flatter solution in the loss landscape, which empirically correlates with better generalisation performance.

This theorem encapsulates the core intuition of SAM. By optimising for a worst-case adversarial perturbation, the solution lies in a flat region of the loss landscape. Flatness, a geometric property of the loss landscape, is strongly associated with model robustness and improved generalisation.

To solve the min-max optimisation problem efficiently, SAM introduces a two-step iterative process:

1. **Gradient Ascent (Inner Maximisation):** Compute the adversarial perturbation ϵ^* that maximises the loss in a neighborhood of the current parameters $\mathbf{w}$. Using a first-order approximation and the ℓ_2 -norm, this is given by:

$$\epsilon^* = \rho \frac{\nabla_{\mathbf{w}} \mathcal{L}(\mathbf{w}, \mathcal{D})}{\|\nabla_{\mathbf{w}} \mathcal{L}(\mathbf{w}, \mathcal{D})\|_2}.$$

2. **Gradient Descent (Outer Minimisation):** Update the model parameters $\mathbf{w}$ to minimise the adversarially perturbed loss:

$$\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla_{\mathbf{w}} \mathcal{L}(\mathbf{w} + \epsilon^*, \mathcal{D}),$$

where η is the learning rate.

The additional adversarial perturbation introduced by SAM ensures that the optimisation process favours flatter minima in the loss landscape, which are empirically linked to improved generalisation. The computational overhead of SAM is manageable, requiring only one additional gradient computation per iteration. This makes SAM a practical and effective approach for robust deep learning optimisation.

Epilogue

As we conclude the course, I hope you found the experience intellectually demanding and ultimately worthwhile. I would like to sincerely acknowledge the level of effort and commitment you demonstrated throughout the semester.

The material we covered was intentionally rigorous; a significant portion aligns with graduate-level coursework offered at leading research universities in the United States, and you rose to that challenge. The concepts and ways of thinking you have developed over the semester matter far more than any single numerical outcome. What you have accomplished (engaging seriously with advanced theoretical foundations) is itself a meaningful achievement that will continue to pay dividends as you progress.

For those of you aspiring to become deep learning engineers or researchers, a strong theoretical grounding is not optional; it is along-term asset. The capacity to reason carefully about models, assumptions, and limitations (and to question them when necessary) is what ultimately enables genuine innovation rather than superficial application. I strongly encourage you to keep cultivating this habit of critical inquiry as you move forward in your studies and careers.

Thank you again for your sustained effort and engagement throughout the semester. It has been a genuine pleasure to guide you through this material, and I wish you continued success in your future academic and professional endeavors.

Regards,
Albert